

Neurons and glia

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The five worked problems from the slides are written out in full here. Silverthorn chapter 9.

COMPETENCIES 7 TO 11 OF WEEK 4

23 SLIDES · 5 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each problem yourself before you read the answer.

The problem for today

SLIDES 1 TO 2 OF THE DECK

A neuron has one job, which is to carry a signal from the place where it receives information to the place where it passes that information on. This lecture looks at how the cell is built to do that, and at the other cells in nervous tissue, the glia, that keep it working. It is a physiology lecture, so for every structure the question is the same: what does this part do to move a signal from input to output?

Here is the problem we solve by the end. A 28 year old graduate student wakes up with blurred vision in her right eye, and the eye hurts when she moves it. Over several weeks her vision comes back. Months later she notices that in a hot shower, or on a run on a hot day, the vision in that eye blurs again, and within an hour of cooling down it clears. Her eye is healthy, and every neuron that carries the signal from that eye to her brain is still there.

Her results at the clinic, on a cool day, are these. Plasma potassium is 4.1 mM (range 3.5 to 5.0) and plasma sodium is 140 mM (range 135 to 145), so the resting potential you built in the last lecture is normal. Her temperature at rest is 36.9 °C (98.4 °F). The informative result is a timing test called a visual evoked potential. She watches a checkerboard pattern that flips back and forth, one eye at a time, and electrodes on the scalp over the visual cortex record how long the signal takes to arrive. Normal is about 100 ms, up to about 115 ms, and the limit varies a little between laboratories. From her left eye the signal takes 102 ms. From her right eye it takes 142 ms. The signal arrives, and it is close to its normal size, but it is about 40 ms late. So the questions are what sets the speed of a signal along an axon, what could change that speed in one optic nerve and not the other, and why warming her body would make it worse.

See it move: send cargo along an axon

SLIDE 19 OF THE DECK

This is a working model of axonal transport, which is covered in the second half of these notes. It is placed here so that you can find it easily. Choose a cargo, 1 to 4, and watch which motor protein carries it, which way it goes and how far it has traveled. The table beside the drawing converts what you see into real time, using typical textbook speeds. You can change the length of the axon with the slider, from 1 mm for a short interneuron in the brain to 1 m for a motor neuron that reaches the foot, and you can remove ATP. The animation is sped up enormously and is not to scale, so read the real times from the table.

Moving model: run it on the course site, drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-neurons-glia-notes.html

THREE THINGS TO TRY

- 1 Set the axon length to 1 m. Send cargo 1, the synaptic vesicle protein, and then cargo 2, the cytoskeleton protein. Write down the real time for each whole trip and compare them with worked problem 4.
- 2 Choose cargo 4, the virus. Notice that it starts at the axon terminal and that the motor is dynein. Then slide the axon length from 1 m down to 100 mm and see how the time to reach the cell body changes. That is the difference between a bite on the foot and a bite much closer to the spinal cord.
- 3 While any cargo is moving, press the ATP button to remove ATP. Every motor stops. Put the ATP back and they walk again.

Two kinds of cells, and neurons sorted by shape

SLIDES 3 TO 4 OF THE DECK

Nervous tissue has two kinds of cells. Neurons carry electrical signals. Glia, also called glial cells or neuroglia, keep the neurons working, and most of them do not carry signals. The word glia comes from the Greek word for glue, because early anatomists thought these cells only held the tissue together. Two abbreviations are used all through this unit. The central nervous system, or CNS, is the brain and the spinal cord. The peripheral nervous system, or PNS, is every nerve and every ganglion outside them. A ganglion is a cluster of neuron cell bodies outside the CNS.

A neuron has a cell body, which holds the nucleus and makes the cell's proteins, and it has processes, which are the extensions that leave the cell body. Dendrites are processes that receive signals. The axon is the process that carries the signal away from the cell body to the axon terminal. The structural classes are named by counting the processes that leave the cell body.

The functional classes describe the direction in which a neuron carries its signal. Sensory neurons, also called afferent neurons, carry signals from receptors toward the CNS. Interneurons lie entirely inside the CNS and connect neurons to each other. Motor neurons, also called efferent neurons, carry signals away from the CNS to muscles and glands. Afferent means carrying toward, and efferent means carrying away. The efferent group also includes the autonomic neurons that control smooth muscle, heart muscle and glands, which come later in the course.

Shape and job are two separate labels, but they pair up in a regular way. Most sensory neurons are pseudounipolar, and the sensory neurons for smell, vision and hearing are bipolar. Interneurons and motor neurons are multipolar, and a few interneurons are anaxonic. Interneurons outnumber the other two classes enormously. A common textbook estimate is that about 99 of every 100 neurons are interneurons, which tells you that most of what the CNS does is process information between the input and the output.

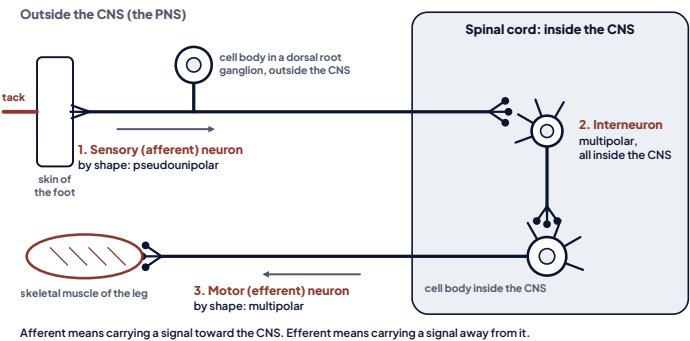


Figure 2. You step on a tack and your leg pulls back. Three neurons in a row carry the signal, and each one is labeled by job and by shape.

The pseudounipolar shape has a functional advantage. With the cell body off to the side, the signal can travel from the receptor in the skin to the spinal cord along one continuous axon, and it never has to cross the cell body. In a multipolar neuron, signals from the dendrites have to spread across the cell body before a new action potential can start, and signals that spread that way fade with distance. A touch neuron from the toe is about a meter long, and its action potential starts near the receptor and runs the whole way without interruption. The cell body still does its other job, which is to make the proteins that keep the cell alive.

Signals travel from the top of each drawing to the bottom

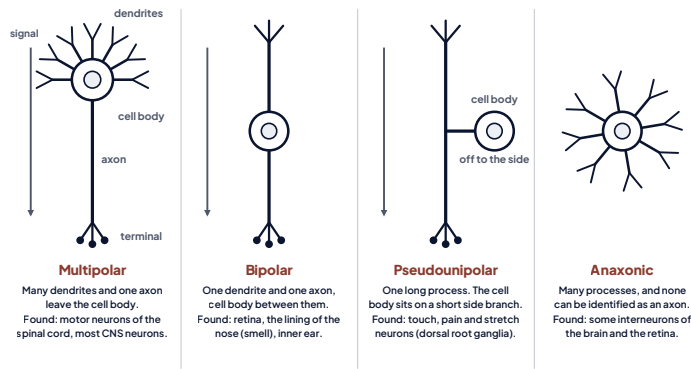


Figure 1. The four structural classes, with the signal traveling from the top of each drawing to the bottom.

Process

Any extension that leaves the cell body of a neuron: a dendrite or an axon.

Dendrite

A process that receives incoming signals.

Axon

The single process that carries the signal away from the cell body to the axon terminal.

Ganglion

A cluster of neuron cell bodies outside the CNS.

Multipolar, bipolar, pseudounipolar, anaxonic

Many dendrites and one axon; one dendrite and one axon; one fused process with the cell body on a side branch; many processes with no identifiable axon.

Neurons sorted by job

WORKED PROBLEM 1

You step on a tack and your leg pulls back. Label each of the three neurons in the pathway twice, once by shape and once by job, and say whether its cell body is inside or outside the CNS.

Use Figure 2 with the labels covered.

THE WORK

- 1** **Neuron 1** runs from the skin to the spinal cord, so it carries the signal toward the CNS. By job it is a **sensory (afferent)** neuron. By shape it is **pseudounipolar**. Its cell body is in a dorsal root ganglion, which is **outside** the CNS.
- 2** **Neuron 2** begins and ends inside the spinal cord. By job it is an **interneuron**. By shape it is **multipolar**. Its cell body is **inside** the CNS, as is every other part of it.
- 3** **Neuron 3** runs from the spinal cord to the muscle, so it carries the signal away from the CNS. By job it is a **motor (efferent)** neuron. By shape it is **multipolar**. Its cell body is **inside** the CNS, in the ventral horn of the spinal cord, and only its axon leaves.
- 4** Check the pattern. Shape and job are two separate labels, but they pair up in a regular way. Most sensory neurons are pseudounipolar, and the sensory neurons for smell, vision and hearing are bipolar. Interneurons and motor neurons are multipolar, and a few interneurons are anaxonic.
- 5** Interneurons outnumber the other two classes enormously. A common textbook estimate is that about 99 of every 100 neurons are interneurons. That tells you most of what the CNS does is process information between the input and the output.

Sensory (afferent) neuron

Carries signals from a receptor toward the CNS. Its cell body is in a ganglion outside the CNS.

Interneuron

A neuron that lies entirely inside the CNS and connects other neurons. By far the most numerous class.

Motor (efferent) neuron

Carries signals away from the CNS to a muscle or gland. Its cell body is inside the CNS.

Two kinds of electrical signal are involved, and the next lecture explains how both are produced. A graded potential is a small change in membrane potential whose size depends on the strength of the stimulus, and it fades as it spreads. An action potential is always the same size and does not fade, which is what all or none means. For now you need to know where in the cell each one happens, and the answer follows from which ion channels are in the membrane of each region. These are the four channel types from the last lecture.

The input zone has ligand gated channels, which open when a neurotransmitter binds, and the receptor endings of sensory neurons have mechanically gated channels and other channels that respond to a physical stimulus. These channels produce graded potentials. The trigger zone has voltage gated Na^+ channels at a higher density than anywhere else in the cell. If the graded potential that reaches it is still at threshold, about -55 mV , those channels open and an action potential begins. If the graded potential has faded below threshold, nothing is sent. The axon has voltage gated Na^+ and K^+ channels, which make the action potential again and again along its length. The axon terminal has voltage gated Ca^{2+} channels. Free Ca^{2+} is about 1 mM outside the cell and 0.0001 mM inside, and E_{Ca} is about $+123 \text{ mV}$, so when the action potential opens these channels, Ca^{2+} enters. The rise in Ca^{2+} inside the terminal makes vesicles fuse with the membrane and release neurotransmitter, and the electrical signal becomes a chemical one.

If you shade the regions where graded potentials happen, then the regions where voltage gated Na^+ channels are dense, and then the region where the signal becomes chemical, only one region is shaded more than once. Graded potentials spread as far as the trigger zone, and the trigger zone has the densest voltage gated Na^+ channels. That overlap is what makes it the decision point of the neuron. One detail is different in a sensory neuron: its trigger zone is not at the cell body. It is at the far end, where the receptor endings join the axon.

You can test your map of the neuron by blocking one kind of channel at a time. Tetrodotoxin, from pufferfish, blocks voltage gated Na^+ channels. The input zone still produces graded potentials, but no action potential can start at the trigger zone. Omega-conotoxin, from cone snail venom, blocks the voltage gated Ca^{2+} channels of axon terminals. Action potentials arrive at the terminal normally, but no Ca^{2+} enters and no neurotransmitter is released. Curare, a plant arrow poison, blocks the ligand gated channels on skeletal muscle, so the whole neuron works and the signal stops at the input zone of the next cell. The muscle fails to contract in all three cases, and to say which region failed you have to know which channel is found there.

One neuron, four working regions

SLIDES 6 TO 8 OF THE DECK

A neuron can be divided into four regions by what each one does. The input zone is the dendrites and the cell body. It receives signals. The trigger zone is the axon hillock, where the axon leaves the cell body, together with the first stretch of the axon, called the initial segment. It decides whether a signal is sent. The conducting zone is the axon, which carries the signal over a distance. The output zone is the axon terminal, which passes the signal to the next cell.

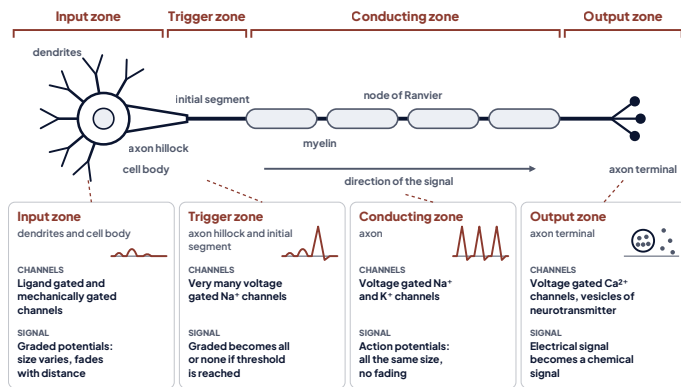


Figure 3. A motor neuron read from left to right, with the channels and the signal type for each region.

Input zone

Dendrites and cell body. Ligand-gated and mechanically-gated channels. Graded potentials.

Trigger zone

Axon hillock and initial segment. The highest density of voltage-gated Na^+ channels. The place where a graded potential becomes an action potential if threshold is reached.

Conducting zone

The axon. Voltage-gated Na^+ and K^+ channels. Action potentials.

Output zone

The axon terminal. Voltage-gated Ca^{2+} channels and vesicles of neurotransmitter. The electrical signal becomes a chemical signal.

Graded potential and action potential

A graded potential varies in size and fades with distance. An action potential is all or none: always the same size, with no fading.

The glia of the CNS

SLIDES 9 TO 10 OF THE DECK

The CNS has four kinds of glial cell. Astrocytes are star-shaped cells that control the fluid around neurons. Oligodendrocytes make the myelin of the CNS, and one oligodendrocyte wraps segments on several axons. Microglia are the immune cells of the CNS, and they engulf debris, dead cells and microbes.

Ependymal cells form the lining of the fluid-filled spaces of the brain and spinal cord. Their cilia move the cerebrospinal fluid, and specialized ependymal cells in the choroid plexus produce it.

Astrocytes do four things that matter for signaling. First, they take up K^+ . Every action potential lets a little K^+ out of the neuron, and the space between cells in the brain is very narrow, so that K^+ would build up quickly. Astrocytes take it up and pass it through a network of connected astrocytes to where the concentration is lower. Second, they take up neurotransmitter. Astrocytes wrap around synapses and remove glutamate and other neurotransmitters, which ends the signal, and they return the material to the neuron to be used again. Third, they pass fuel to neurons. They take glucose from the blood and supply neurons with lactate, which the neurons use to make ATP.

Fourth, astrocytes help form the blood-brain barrier. The barrier itself is the tight junctions between the cells of the capillary wall,

which seal brain capillaries far more tightly than capillaries elsewhere. Astrocyte end feet cover the capillaries and signal those cells to form the tight junctions and keep them. Small lipid-soluble molecules such as O_2 , CO_2 and alcohol cross freely. Glucose and amino acids cross on transport proteins. Many drugs cannot cross, which is why dopamine cannot be given for Parkinson disease and levodopa, which rides on an amino acid transporter, is given instead.

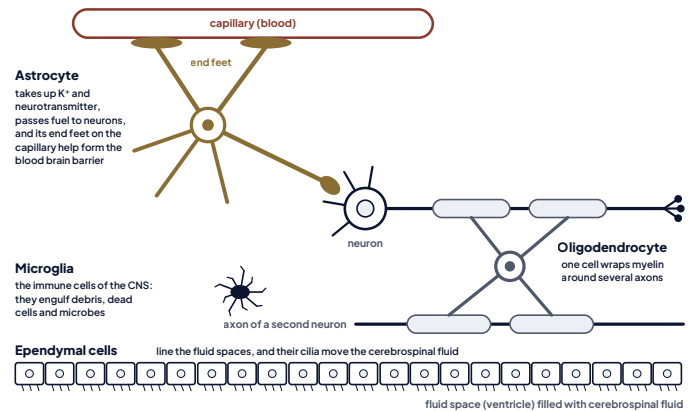


Figure 4. The four glial cells of the CNS, each drawn doing its job.

WORKED PROBLEM 2

The astrocytes around a busy group of neurons stop clearing K^+ , and the K^+ outside those neurons rises from 4 mM to 8 mM. Find the new E_K and the new resting potential.

Keep 140 mM K^+ inside, Na^+ at 145 out and 15 in, and $P_{\text{Na}} = 0.04$, as in the last lecture.

THE WORK

- Potassium equilibrium potential with 8 mM outside:
 $E_K = 61.5 \times \log_{10}(8/140) = 61.5 \times (-1.243) = -76 \text{ mV}$.
 With the normal 4 mM it was -95 mV .
- Goldman equation, with $P_K = 1$ and $P_{\text{Na}} = 0.04$ as in the last lecture. Top: $8 + (0.04 \times 145) = 8 + 5.8 = 13.8$. Bottom: $140 + (0.04 \times 15) = 140.6$.
- Ratio: $13.8/140.6 = 0.0981$. $\log_{10}(0.0981) = -1.008$.
 $V_m = 61.5 \times (-1.008) = -62 \text{ mV}$.
- The resting potential of every neuron in that region has moved from -71 mV to -62 mV . Threshold is about -55 mV , so these neurons are now only 7 mV away from firing, and they were never stimulated.
- This is what astrocytes prevent. Every action potential lets a little K^+ out of the neuron, and the space between cells in the brain is very narrow, so the K^+ concentration there can rise quickly. Astrocytes take the K^+ up and pass it through the network of astrocytes to places where the concentration is lower. When that clearing fails, activity in one group of neurons can spread to its neighbors, which is one of the ways a seizure spreads. The 8 mM here is an illustration and not a measured value for a particular condition.

Astrocyte

CNS glial cell that takes up K^+ and neurotransmitter, passes fuel to neurons and helps form the blood-brain barrier.

Oligodendrocyte

CNS glial cell that makes myelin. One cell wraps segments on several axons.

Microglia

The immune cells of the CNS. They engulf debris, dead cells and microbes.

Ependymal cell

Lines the fluid filled spaces of the brain and spinal cord, moves the cerebrospinal fluid with its cilia, and in the choroid plexus produces it.

Blood brain barrier

Tight junctions between the cells of brain capillary walls, formed and kept under the influence of astrocyte end feet, that limit what passes from blood into brain tissue.

The glia of the PNS, and what myelin is

SLIDE 11 OF THE DECK

The PNS has two kinds of glial cell. Satellite cells surround neuron cell bodies in ganglia. They support the cell body and control the fluid around it, much as astrocytes do in the CNS. Schwann cells make the myelin of the PNS. Each Schwann cell wraps one segment, about 1 mm long, of one axon, so a long axon needs hundreds of Schwann cells in a row. The gaps of bare axon membrane between segments are the nodes of Ranvier.

Myelin is the glial cell's own membrane wound around the axon many times, up to about 150 layers in a large axon. Membrane is mostly lipid, and ions cannot cross lipid, so myelin is electrical insulation. Two of the six glial cells make myelin, and they differ in one structural way. An oligodendrocyte sends out many arms, each arm wraps one segment, and one oligodendrocyte can supply segments to several dozen axons. A Schwann cell wraps its whole body around a single segment of a single axon. This matters in disease, because losing one oligodendrocyte strips a segment from many axons at once.

In a ganglion and a nerve of the PNS

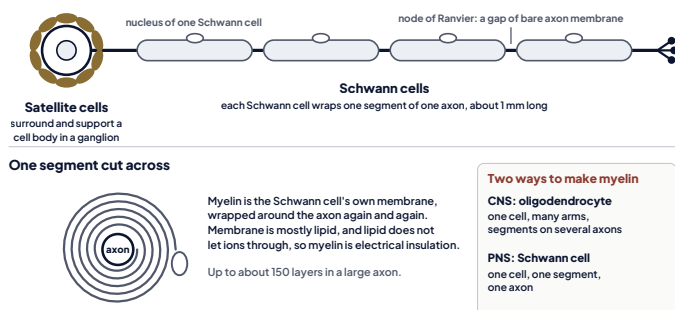


Figure 5. Satellite cells and Schwann cells, with one myelin segment cut across to show its layers.

Schwann cell

PNS glial cell that makes myelin. One cell wraps one segment of one axon.

Satellite cell

PNS glial cell that surrounds and supports a neuron cell body in a ganglion.

Myelin

Many layers of glial cell membrane wound around an axon. It is electrical insulation.

Node of Ranvier

A short gap of bare axon membrane between two myelin segments, packed with voltage gated Na^+ channels.

How myelin speeds conduction, and what happens when it is lost

SLIDES 12 TO 15 OF THE DECK

An action potential does not slide along the axon. Each patch of membrane has to be depolarized to threshold by current from the patch before it, and then it has to open its own Na^+ channels. Opening channels is the slow step. Current flowing along the inside of the axon is fast, but in a bare axon it leaks out through the membrane, so it reaches only a short distance ahead, and the action potential has to be made again at every patch. An unmyelinated axon conducts at about 0.5 to 2 meters per second.

Myelin carries no signal itself. It stops the leak. Under the myelin, current from one node flows along the inside of the axon and reaches the next node, about 1 mm away, still strong enough to bring that node to threshold. The action potential is made again only at the nodes, and this node to node pattern is called saltatory conduction. A large myelinated axon conducts at up to about 120 meters per second. A wider axon also conducts faster, because a wide axon has less resistance to current flowing along it, and myelin gives the same speed in a far thinner axon. The full electrical explanation comes with the action potential lectures.

Voltage gated Na^+ channels are not spread evenly along a myelinated axon. They are packed densely at the nodes, and there are very few under the myelin. When a segment of myelin is stripped away, two things go wrong together. Current leaks out through the bare membrane, and the bare membrane has too few Na^+ channels to make a new action potential. The depolarization fades before it reaches the next node, that node stays below threshold, and the signal stops. So conduction in a demyelinated axon does not drop to the speed of an unmyelinated axon. It fails, and this is called conduction block. If the myelin is only thinned, the signal gets through, but late. Over weeks the axon can add Na^+ channels to the bare stretch, and the glial cells can partly remyelinate it, so some function returns.

Two diseases show the difference between CNS and PNS myelin. In multiple sclerosis the immune system attacks the myelin of the CNS, which is made by oligodendrocytes, in the brain, the spinal cord and the optic nerves. What the person notices depends on which pathway is hit, such as blurred vision, numbness, weakness or poor balance, and episodes come and go over years. Recovery is partial, because oligodendrocytes remyelinate poorly. In Guillain-Barré syndrome the immune system attacks the myelin of the PNS, which is made by Schwann cells, often a few weeks after an infection. Weakness and tingling start in the feet and hands and move upward over days, and reflexes are lost. Most people recover over weeks to months, because Schwann cells remyelinate well.

Two axons, each 1 meter long. A dot marks every place the action potential has to be made again.

Unmyelinated axon



Myelinated axon

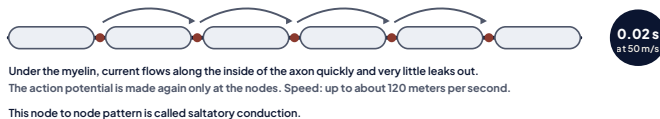


Figure 6. A dot marks every place the action potential has to be made again. Myelin reduces the number of those places.

WORKED PROBLEM 3

You stub your toe. A touch signal travels in a large myelinated axon at about 50 m/s, and the dull ache travels in a thin unmyelinated axon at about 1 m/s. The toe is about 1 m from the spinal cord. How long does each signal take to arrive?

time = distance ÷ speed.

THE WORK

- 1 Use time = distance ÷ speed, and keep the units matched: meters and meters per second.
- 2 Touch, in a large myelinated axon: $1 \text{ m} \div 50 \text{ m/s} = 0.02 \text{ s}$, which is 20 milliseconds.
- 3 Dull ache, in a thin unmyelinated axon: $1 \text{ m} \div 1 \text{ m/s} = 1 \text{ s}$.
- 4 The touch signal arrives 50 times sooner. You have felt this. When you stub your toe you know at once that you hit something, and the ache arrives about a second later. That delay is the difference between a myelinated and an unmyelinated axon.
- 5 The sharp first pain that comes before the ache travels in thinly myelinated axons at about 5 to 30 m/s, which puts it between the two. All of these speeds are approximate, and books give ranges.

Intact myelin



Middle segment stripped bare

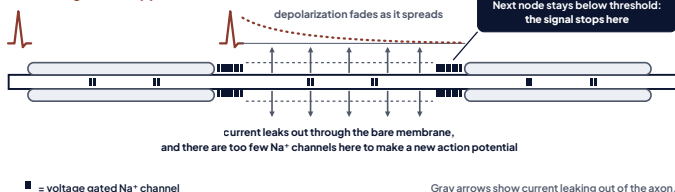


Figure 7. The same axon with its myelin intact, and with the middle segment stripped.

Saltatory conduction

Conduction in a myelinated axon, in which the action potential is regenerated only at the nodes of Ranvier.

Demyelination

Loss of myelin from an axon.

Conduction block

Failure of an action potential to pass a damaged stretch of axon.

Multiple sclerosis

A disease in which the immune system attacks CNS myelin, made by oligodendrocytes.

Guillain-Barré syndrome

A disease in which the immune system attacks PNS myelin, made by Schwann cells.

Axonal transport

SLIDES 16 TO 20 OF THE DECK

Almost every protein in a neuron is made in the cell body, because that is where the nucleus, the rough ER and the Golgi are. The axon and its terminal have almost no ribosomes. Yet the terminal needs a steady supply of channels and pumps for its membrane, the proteins of synaptic vesicles, enzymes that make neurotransmitter, and mitochondria, and proteins wear out in days to weeks. The distances are large. The cell body of a large motor neuron is about 0.1 mm across, and its axon to the foot is about 1 m long, which is 10,000 times the width of the cell body. If the cell body were the size of a tennis ball, the axon would be a tube about 670 m long, which is about 0.4 mile. Diffusion cannot cover that distance, because the time diffusion takes rises with the square of the distance. A protein can diffuse across a cell body in seconds, but to diffuse 1 m would take hundreds of years at the least. A neuron therefore needs a transport system that carries cargo in one direction, along a track, using energy.

The tracks are microtubules, which run the length of the axon. Every microtubule in an axon points the same way, with its plus end toward the terminal and its minus end toward the cell body. Two motor proteins walk along them, and each step splits ATP. Kinesin walks toward the plus end, so it carries cargo anterograde, from the cell body to the terminal. Dynein walks toward the minus end, so it carries cargo retrograde, from the terminal back to the cell body.

Fast anterograde transport carries vesicles, membrane proteins such as channels, and mitochondria at about 400 mm per day. Fast retrograde transport is somewhat slower, about 200 to 300 mm per day. It brings worn out membrane and old organelles back to the cell body, where lysosomes break them down and the parts are reused. It also carries information: growth factors released by the target cell are taken up at the terminal and carried back to the nucleus, and they tell the neuron that its connection is alive. Slow transport moves cytoskeleton proteins and dissolved enzymes from the cell body toward the terminal at about 0.2 to 8 mm per day. Current evidence is that this cargo rides on the same motors but spends most of its time paused, so its average speed is low. Some books describe slow transport as a flow of the cytoplasm, which was the earlier explanation. All of these speeds are approximate.

Some viruses and toxins reach the cell body by riding on dynein. Rabies virus enters the axon terminals of motor neurons near a bite, travels to the spinal cord, and spreads from neuron to neuron to the brain. A bite on the foot gives the longest path, which is one reason the illness usually takes longer to appear after a bite on the

leg than after a bite on the face. Herpes simplex virus enters sensory axon terminals in the skin of the lip and is carried to the cell body in a sensory ganglion, where it stays dormant. When it reactivates, new virus travels back down the same axon on kinesin, so the cold sore returns in the same place. Tetanus toxin enters motor axon terminals near a dirty wound and is carried to the spinal cord. There it crosses to inhibitory interneurons and blocks their release of neurotransmitter, so muscles cannot relax.

Now change the conditions. If ATP runs out, every motor stops, fast and slow, in both directions. If a drug takes the microtubules apart, the motors have nothing to walk on, and the terminals farthest from their cell bodies run short first. Those belong to the longest axons, so the person notices numbness and tingling that begin in the feet and the hands. This happens with vincristine, a chemotherapy drug that blocks the assembly of microtubules. It is given to stop cancer cells from dividing, and the effect on long axons is a side effect.

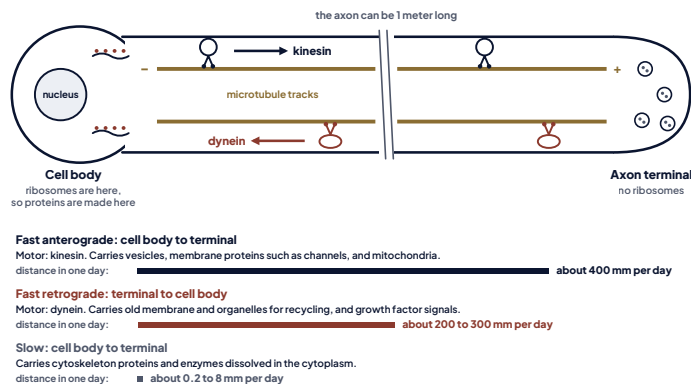


Figure 8. Three transport streams in one axon. The bars compare the distance each one covers in a day.

WORKED PROBLEM 4

A motor neuron has an axon 1 m long. How long does a protein take to reach the terminal by fast anterograde transport at 400 mm per day, and by slow transport at 1 mm per day? How long does the return take by fast retrograde transport at 200 mm per day?

time = distance ÷ speed, with 1 m = 1000 mm.

THE WORK

- 1** Use time = distance ÷ speed, with the distance in millimeters: 1 m = 1000 mm.
- 2** Fast anterograde transport: 1000 mm ÷ 400 mm per day = **2.5 days**. A sodium channel made in the cell body on Monday morning is in the membrane near the foot by Wednesday evening.
- 3** Slow transport at 1 mm per day: 1000 mm ÷ 1 mm per day = 1000 days, and 1000 ÷ 365 = **about 2.7 years**.
- 4** Slow transport covers a range. At 8 mm per day the same distance takes 125 days. At 0.2 mm per day it takes 5000 days, which is more than 13 years. The cytoskeleton proteins are the slowest cargo.
- 5** Fast retrograde transport at 200 mm per day: 1000 ÷ 200 = **5 days** from the terminal back to the cell body.
- 6** What the numbers say: fast transport can resupply the far end of the longest axon in the body within a few days, which is about how often membrane proteins need replacing. Slow transport is several hundred times slower, and that matters when an axon has to regrow, as you will see on slide 21.

Axonal transport

Movement of material along an axon by motor proteins walking on microtubules, using ATP.

Anterograde and retrograde

Anterograde is from the cell body toward the terminal. Retrograde is from the terminal toward the cell body.

Kinesin

The motor protein for anterograde transport. It walks toward the plus end of the microtubule.

Dynein

The motor protein for retrograde transport. It walks toward the minus end of the microtubule.

Fast and slow transport

Fast: about 400 mm per day anterograde and 200 to 300 retrograde, for vesicles, membrane proteins and organelles. Slow: about 0.2 to 8 mm per day, for cytoskeleton proteins and dissolved enzymes.

When an axon is cut

SLIDE 21 OF THE DECK

When an axon is cut, the part that is separated from the cell body degenerates, because it no longer receives anything by axonal

transport. What happens next depends on the glial cells. In the PNS, Schwann cells survive. They help clear the debris, line up as a tube and release growth factors, and the stump of the axon can regrow along that tube at about 1 to 2 mm per day. That is about the speed of slow axonal transport, and it is not a coincidence, because a growing axon is built from cytoskeleton proteins that arrive by slow transport. In the CNS, oligodendrocytes and astrocytes release molecules that inhibit growth, and astrocytes form a scar. There is very little regrowth, so damage to the brain or spinal cord is largely permanent.

WORKED PROBLEM 5

A nerve is cut at the wrist and the two ends are stitched together. The fingertip is 200 mm from the cut. If the axons regrow at 1 to 2 mm per day, how long before feeling could begin to return in the fingertip? How long does an action potential at 50 m/s take to cover the same 200 mm?

time = distance ÷ speed.

THE WORK

- 1 The axon has to regrow from the cut to the fingertip, 200 mm.
- 2 At 2 mm per day: $200 \div 2 = 100$ days. At 1 mm per day: $200 \div 1 = 200$ days.
- 3 So feeling could begin to return in the fingertip after roughly 3 to 7 months, and there is also a delay of days to weeks before regrowth starts. These numbers are approximate.
- 4 Compare that with signaling. Once the axon is rebuilt, an action potential covers the same 200 mm at 50 m/s in $0.2 \div 50 = 0.004$ s, which is 4 milliseconds.
- 5 The regrowth speed is about the same as slow axonal transport, and that is not a coincidence. A growing axon is built from cytoskeleton proteins, and those arrive from the cell body by slow transport.

The patient, explained

SLIDE 22 OF THE DECK

The optic nerve is part of the CNS, so its axons are myelinated by oligodendrocytes. In multiple sclerosis the immune system attacks CNS myelin. Her first episode was a patch of inflammation in the right optic nerve that stripped myelin from many axons. Those axons went into conduction block, and her vision blurred. Then the inflammation settled, oligodendrocytes partly remyelinated the patch, and the axons added Na^+ channels to the bare membrane. Signals get through again, but slowly, which is the 142 ms against 102 ms in her timing test.

A repaired axon has very little to spare. In a healthy myelinated axon, each node receives several times more current than it needs to reach threshold. In her repaired axons the current that arrives is only just enough. Ion channels open and close faster when they are warmer, so in a warm axon the Na^+ channels inactivate sooner and each action potential delivers less current to the next node. A healthy axon has plenty to spare, so nothing changes. In her right optic nerve, a rise in body temperature of about 0.5°C (0.9°F) is

enough to drop the current below what the next node needs, and those axons stop conducting until she cools down. No new damage has been done. This is called the Uhthoff phenomenon.

Look at it as an experiment. The variable that fails is the current reaching the next node, compared with the current that node needs to reach threshold. Anything that makes each action potential last longer delivers more current. Lowering the temperature does that, because cooler channels inactivate later, and this is why her vision clears when she cools down. Blocking voltage gated K^+ channels does it too. Losing myelin uncovers K^+ channels that are normally hidden beneath it, and when they open, positive charge leaves the axon and the action potential is cut short. A drug that blocks these channels lengthens the action potential and improves conduction in some people with multiple sclerosis. Neither change repairs the myelin. They act on the electrical problem, and the immune attack is a separate target.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture.

CHEMISTRY

What an ion is, and the charges on Na^+ , K^+ and Ca^{2+}
Lipids do not let ions through, which is why a membrane is an electrical insulator
ATP is the molecule cells split to pay for work

Chemistry review page

FROM EARLIER LECTURES

The resting potential, -70 mV, and threshold, about -55 mV (last lecture)
The Nernst and Goldman equations, and the typical ion concentrations (last lecture)
The four channel types by what opens them, and the words depolarize and hyperpolarize (last lecture)
Rough ER, Golgi, vesicles, lysosomes and the cytoskeleton (Week 2)
Diffusion is fast over short distances and slow over long ones (Week 3)

The only anatomy you need is this: the CNS is the brain and spinal cord, the PNS is everything else, and a ganglion is a cluster of neuron cell bodies outside the CNS.

MATH

time = distance ÷ speed
Unit conversions: $1\text{ m} = 1000\text{ mm}$, and $1\text{ s} = 1000\text{ ms}$
Converting days to years by dividing by 365

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one. Pick one prompt per competency. Blank paper, nothing open. Most of these prompts ask for a drawing, so practice the drawings from the figures above.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

07

Neuron structural and functional classes

Map the organization of the nervous system into the central nervous system, the sensory and efferent divisions of the peripheral nervous system, the somatic motor and autonomic branches, and the enteric nervous system, then classify a neuron as multipolar, bipolar, or pseudounipolar by structure and as sensory, motor, or interneuron by function, and match each class to a location in the nervous system.

Do one of the two. Head your paper with the number.

7-1

Draw a multipolar, a bipolar, and a pseudounipolar neuron side by side, each with its dendrites, cell body, and axon in the right arrangement and the cell body in the right position relative to the signal path. Under each write one real location in the body where you would find it. One line: one of the three has its cell body off to the side of the signal path, so say what advantage that gives a sensory neuron.

7-2

Draw a reflex arc across your box, from a receptor in the skin to a skeletal muscle, with three neurons in it. Label every neuron twice, once by structure and once by function, and mark whether each cell body sits inside or outside the CNS. One line: name the functional class that outnumbers the other two enormously in a human, and say what that tells you about what the CNS mostly does.

08

Functional regions of a neuron

Label the dendrites, cell body, axon hillock, trigger zone, axon, and axon terminal and state which signal type each region carries.

Do one of the two. Head your paper with the number.

8-1

Draw one motor neuron big, across the whole width of your box. Label the dendrites, cell body, axon hillock, trigger zone, axon, myelin, nodes, and axon terminal. Above each region write the signal type it carries, graded or all or none, and sketch the shape of that signal there. One line: mark the spot where graded becomes all or none and say what has to be true at that spot for the change to happen.

8-2

Draw the same neuron three times as three narrow strips stacked up. In the first, shade every region where graded potentials happen. In the second, shade every region with a high density of voltage gated sodium channels. In the third, shade where chemical signaling takes over from electrical. One line: name the region that appears in more than one strip and say why that makes it the decision point.

09

Glial cell functions

Match astrocytes, oligodendrocytes, microglia, ependymal cells, satellite cells, and Schwann cells to their physiological roles.

Do one of the two. Head your paper with the number.

9-1

Draw a CNS scene and a PNS scene side by side. In the CNS scene put an astrocyte touching both a capillary and a neuron, an oligodendrocyte myelinating more than one axon, a microglial cell, and ependymal cells lining a fluid space. In the PNS scene put a Schwann cell on one axon segment and satellite cells wrapped around a cell body. One line: two of these six make myelin, so say the one structural difference in how they do it.

9-2

Draw one neuron in the middle of your box with six arrows pointing at it, one from each glial type. On each arrow write the single job that glia does for the neuron or its surroundings. Then cross out one arrow of your own choosing and draw what goes wrong. One line: name the glial cell you removed and the first symptom you would expect, not the last.

10

Myelin and its loss

Explain how myelin and the nodes of Ranvier speed conduction and predict the functional consequence of demyelination.

Do one of the two. Head your paper with the number.

10-1

Draw two axons the same length side by side, one bare and one myelinated with nodes of Ranvier. On each, mark with dots every place the action potential has to be regenerated. Put a clock reading beside each showing which signal arrives first. One line: myelin is an insulator and does nothing electrical itself, so say in one sentence how adding insulation makes a signal faster.

10-2

Draw a myelinated axon with three intact internodes, then draw the same axon with the middle internode stripped bare. Show the action potential traveling along it and mark where it slows or fails. On both, draw the sodium channel distribution, dense at the nodes and sparse under the myelin. One line: the demyelinated stretch has few sodium channels, so say why conduction fails there rather than simply slowing down.

11

Axonal transport

Distinguish fast anterograde and retrograde axonal transport from slow transport and state what each carries.

Do one of the two. Head your paper with the number.

11-1

Draw a long neuron with the cell body on the left and the terminal on the right, and run a microtubule track down the axon. On the track draw fast anterograde transport with kinesin and fast retrograde transport with dynein, arrows in opposite directions, with the cargo written beside each. One line: the cell body makes everything, so say what the retrograde direction is even for.

11-2

Draw the same axon and add slow transport as a third, much slower stream with its cargo written beside it. Draw a scale bar showing the real speed difference, hundreds of mm per day for fast transport against a few mm per day for slow. Then draw a virus entering at the terminal and mark the route it takes to the cell body. One line: say which transport system it hijacked and why that is bad news for the neuron.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · Neurons and glia. Conduction speeds, transport speeds and regrowth rates are typical textbook values and are approximate. All figures are original drawings for this course.

The resting membrane potential

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The four worked problems from the slides are written out in full here. Silverthorn chapter 5 (the resting membrane potential) and chapter 9.

COMPETENCIES 1 TO 5 OF WEEK 4

26 SLIDES · 4 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each worked problem yourself before you read the work. Every one asks you to predict a direction and explain it, so a pencil is all you need.

The problem for today

SLIDES 1 TO 2 OF THE DECK

Every cell in your body has a voltage across its membrane. It is small, about 70 thousandths of a volt, and the inside of the cell is the negative side. In this lecture we build that voltage starting from zero. We start with a cell that has no voltage at all, add one membrane protein at a time, and read a voltmeter after each step, so that you can see what each protein contributes. Then we read two equations in plain words and use them to predict which way ions move, and last we change the blood chemistry and predict what happens.

Here is the problem we solve by the end. A construction worker is pinned under a collapsed wall for five hours with both legs crushed. He is awake and talking the whole time. The rescue team lifts the wall, and within minutes his heart rhythm turns chaotic and his heart stops. Nothing new injured him. The only thing that changed is that blood began flowing through his legs again.

Blood drawn in the ambulance shows a plasma potassium of 7.0 mM, where the normal range is 3.5 to 5.0 mM. His sodium (139 mM, range 135 to 145), chloride (102 mM, range 98 to 106), glucose (94 mg/dL, range 70 to 100) and temperature, 36.8 °C (98.2 °F), are all in range. So potassium rose by about 3 mM and everything else held. That does not sound like much, and sodium can move by 3 mM without anyone noticing. Write down your best guess for why 3 mM of potassium can stop a heart. By the end of these notes you will be able to explain the answer.

See it move: build the resting potential yourself

SLIDE 20 OF THE DECK

This is a working model of the cell in this lecture. Press the steps in order, 1 to 6, and watch the ions and the voltmeter. Each step matches a section of the notes below, so keep it open and come back to it as you read. Step 2 goes with the pump, step 3 with the potassium leak channels, step 4 with the sodium cell, step 5 with the real resting cell, and step 6 is a preview of the action potential. You can pause it at any time, turn the pump on and off, and change the potassium outside the cell with the slider.

Moving model: run it on the course site, [drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-membrane-potential-notes.html](https://stack.github.io/human-physiology-Fa26/biol005-w04-membrane-potential-notes.html)

THREE THINGS TO TRY

- 1 On step 3, wait for the voltmeter to settle near -95 mV and then watch the potassium channels. Ions are still crossing in both directions. That is what no net movement looks like.
- 2 On step 5, slide the potassium outside the cell from 4 up to 7 mM, and then down to 2.5 mM. Write down the voltmeter reading each time and compare it with worked problem 4.
- 3 On step 5, turn the pump off and watch for a full minute. The voltage barely moves at first and then drifts toward zero as the gradients run down. A real cell takes minutes to hours to do this.

Two terms, and what a voltmeter reads

SLIDES 3 TO 4 OF THE DECK

Two terms need to be kept apart from the start. The resting membrane potential is a steady voltage across the membrane of a cell that is not being stimulated. Almost every cell in the body has one: liver cells, fat cells and skin cells as well as neurons and muscle cells. An action potential is a brief, rapid change in that voltage that travels along the membrane. Only excitable cells can produce one, which means neurons and the three kinds of muscle. A liver cell has a resting potential because it has the sodium potassium pump and potassium leak channels, which are covered below. It cannot fire an action potential because it does not have voltage gated sodium channels. This lecture covers the resting potential. The action potential is the next lecture, and you need this one to understand it.

A membrane potential is always a comparison. One electrode goes inside the cell, one stays outside, and the meter reports the inside compared with the outside. By agreement the outside is called zero, so -70 mV means the inside is 70 millivolts more negative than the outside. A millivolt is one thousandth of a volt. A AA battery is 1,500 mV, so a cell holds about one twentieth of that across a membrane only 8 nanometers thick. Typical resting values are about -70 mV in a neuron and about -90 mV in skeletal and heart muscle.

One point is easy to get wrong. The cell is not filled with negative charge. The fluid inside and the fluid outside are each electrically neutral, with equal numbers of positive and negative charges. The only place the charges are unequal is a very thin layer on the two

faces of the membrane, slightly more negative charge on the inner face and slightly more positive charge on the outer face.

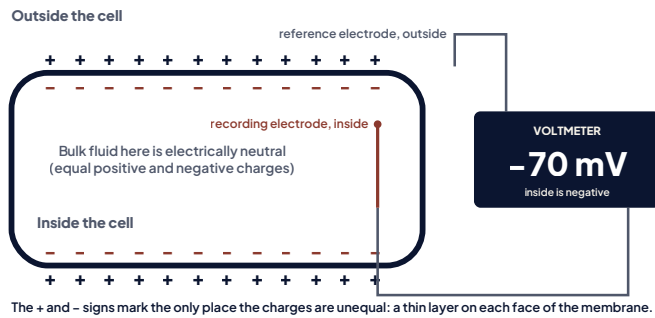


Figure 1. One electrode inside, one outside. The meter reports the inside compared with the outside.

Resting membrane potential

The steady voltage across the membrane of an unstimulated cell, inside compared with outside. About -70 mV in a neuron.

Action potential

A brief, rapid change in membrane potential that travels along the membrane. Only neurons and muscle cells produce one.

Millivolt (mV)

One thousandth of a volt.

Excitable cell

A cell that can fire action potentials: neurons and skeletal, cardiac and smooth muscle cells.

Where the membrane proteins come from

SLIDE 5 OF THE DECK

Every pump and channel in this lecture is a protein, and every protein is built from a gene. All of your cells contain the same genes. What differs from one cell type to another is which genes are switched on. All cells switch on the genes for the sodium potassium pump and for potassium leak channels, which is why almost every cell has a resting potential. Neurons and muscle cells also switch on the genes for voltage gated sodium and potassium channels, which is why only they can fire action potentials.

A membrane protein follows a fixed route from its gene to the membrane. In the nucleus the gene is copied into mRNA. Ribosomes on the rough endoplasmic reticulum read the mRNA and build the protein directly into the ER membrane. The Golgi finishes and sorts the protein. A vesicle carries it to the surface, and when the vesicle fuses with the cell membrane, the protein becomes part of that membrane. Transcription and translation are on the review page, so those steps are not repeated here.

A cell can change how many copies of a channel are in its membrane by adding more or taking some out. More open channels for an ion means a higher permeability to that ion, and permeability decides what the membrane potential will be. This is also why a mutation in one channel gene causes disease mainly in the tissues that use that gene. One inherited form of long QT syndrome comes from a faulty potassium channel in heart muscle: heart cells take longer to repolarize, and the person is at risk of dangerous rhythms. Diseases caused by a faulty ion channel gene are called channelopathies. Cystic fibrosis is another one, caused by a faulty chloride channel.

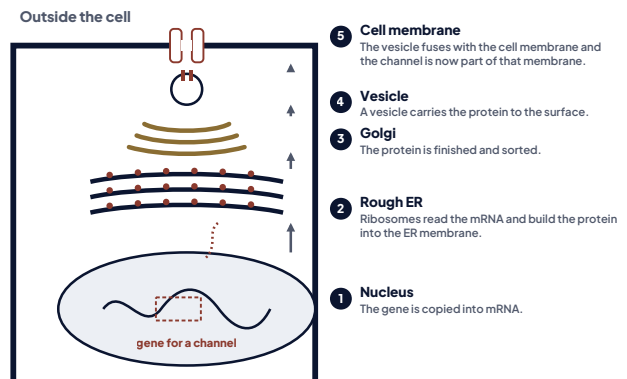


Figure 2. The route a channel or pump takes from its gene to the cell membrane.

Gene expression

Using a gene to build its protein. Cells differ in which genes they express, not in which genes they have.

Channelopathy

A disease caused by a faulty ion channel gene, such as some forms of long QT syndrome or cystic fibrosis.

Building the voltage: the pump

SLIDES 6 TO 8 OF THE DECK

Start with an imaginary cell that has a lipid membrane and nothing else. Potassium, sodium and everything else are mixed the same way inside and outside, and ions cannot cross the oily middle of a bilayer, so nothing moves. Inside there are large negative molecules that can never leave, proteins and the phosphate groups on ATP and nucleic acids, but each one is paired with a positive ion, so the inside is neutral. The voltmeter reads zero. For a voltage to appear, positive and negative charges have to end up unequally shared across the membrane. That takes two things: something to make the two sides different, and a way for some ions, and not others, to cross.

The first protein we add is the sodium potassium pump, the Na^+/K^+ ATPase. It is in the membrane of every cell you have. Each cycle moves 3 sodium ions out and 2 potassium ions in, and splits one ATP to pay for it, because both ions are being moved toward the side where they are already more concentrated. The pump uses a lot of energy: roughly a fifth to a third of the ATP a resting body uses, and more than that in neurons.

Because three positive charges leave for every two that enter, the pump does make the inside slightly negative. A pump that moves net charge like this is called electrogenic. But look at the voltmeter. Even with the pump running at full speed, the cell is only a few millivolts negative. A real neuron is at -70 mV , so the pump alone does not explain the resting potential.

The main thing the pump does is create two steep concentration gradients. Potassium ends up at about 140 mM inside and 4 mM outside, 35 times more concentrated inside. Sodium ends up at about 145 mM outside and 15 mM inside, roughly 10 times more concentrated outside. Chloride is about 110 mM outside and 10 mM inside, and free calcium is about 1 mM outside and only 0.0001 mM inside. The inside of a cell is high in potassium and the fluid around it is high in sodium. A concentration difference is a form of stored energy, like water held behind a dam, and the

pump uses ATP to build it. No ions cross yet, because there is still no channel for them to cross through.

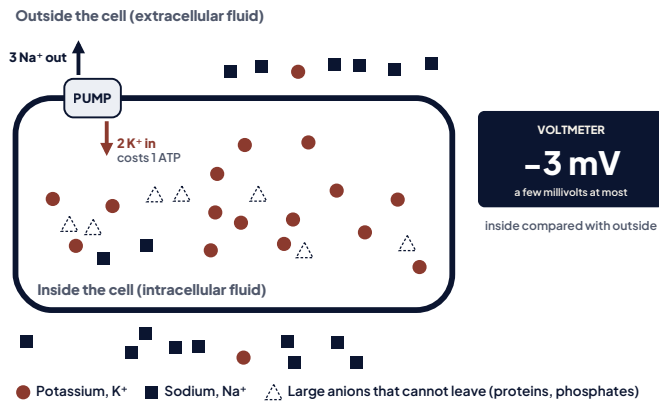


Figure 3. The pump sorts the ions and produces very little voltage on its own.

Sodium potassium pump (Na^+/K^+ ATPase)

A membrane protein that moves 3 Na^+ out and 2 K^+ in for each ATP it splits. It builds and maintains the ion gradients.

Electrogenic

Describes a pump that moves net charge across the membrane and so contributes directly to the membrane potential.

Concentration gradient

A difference in the concentration of a substance between two places. Substances diffuse from the higher concentration toward the lower.

Typical concentrations

K^+ 4 mM outside, 140 inside. Na^+ 145 outside, 15 inside. Cl^- 110 outside, 10 inside. Free Ca^{2+} about 1 outside, 0.0001 inside.

Potassium leak channels and the equilibrium potential

SLIDES 9 TO 11 OF THE DECK

The second protein we add is a potassium leak channel. A leak channel is a protein pore with no gate, so it is open all the time. This one is lined so that potassium fits through and sodium does not. The channel does not choose a direction. It only makes movement possible. Potassium is 35 times more concentrated inside, so it diffuses out.

Each potassium ion that leaves carries one positive charge out and leaves behind the negative partner it was paired with. Those large anions are too big for any channel. So every potassium ion that leaves makes the inner face of the membrane a little more negative. A voltage produced this way, by an ion diffusing down its concentration gradient, is called a diffusion potential.

Now two forces act on every potassium ion. The chemical force comes from the concentration gradient and pushes potassium out. The electrical force comes from the voltage: a negative interior attracts a positive ion and pulls potassium back in. Together they are the electrochemical gradient. At the start the electrical pull is very small, so potassium leaves quickly. Each ion that leaves makes the inside more negative, and that makes it harder for the next one to leave. At one particular voltage the two forces are exactly equal and opposite. Potassium ions still cross in both directions, but as many enter as leave, so there is no net movement, even though the channels are still open. That voltage

is the equilibrium potential for potassium, written E_K , and it is also called the Nernst potential. With 140 mM inside and 4 mM outside it is about -95 mV .

Many students picture potassium leaving until the cell is nearly empty. That is not what happens. Setting up -95 mV across the membrane of a typical cell takes far fewer than one potassium ion out of every thousand in the cell, and the concentrations do not measurably change. The reason so few ions can do so much is where they sit. The potassium ions that left stay pressed against the outer face of the membrane, held by the negative charges they left on the inner face only 8 nanometers away. Opposite charges held that close together across a thin insulator produce a large voltage from a small number of charges, which is how a capacitor works. This is why a membrane potential can change in a thousandth of a second, and why a cell can do it again and again without running down.

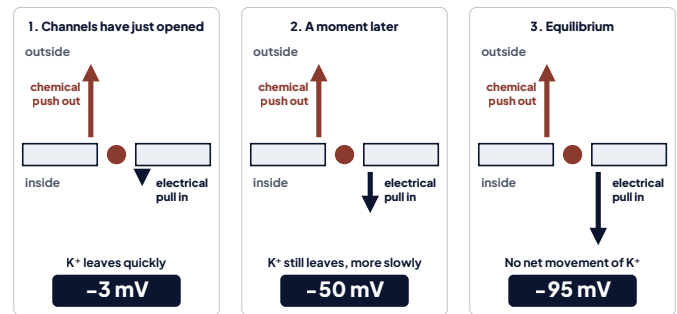


Figure 4. The push from concentration stays the same. The pull from voltage grows with every ion that leaves, until the two are equal.

Leak channel

An ion channel with no gate. It is always open. Potassium leak channels are the most numerous and set the resting potential.

Diffusion potential

A voltage produced across a membrane when an ion diffuses down its concentration gradient and leaves its oppositely charged partner behind.

Electrochemical gradient

The combined effect of the chemical force (concentration) and the electrical force (voltage) on an ion.

Equilibrium potential (Nernst potential)

The membrane potential at which the electrical force on an ion exactly balances its concentration gradient, so there is no net movement of that ion. E_K is about -95 mV .

A sodium cell, and one voltage scale

SLIDES 12 TO 13 OF THE DECK

Take a second imaginary cell. Give it the same pump, so it has the same gradients, but close the potassium channels and open channels that only sodium fits through. Sodium is concentrated outside, so the chemical force pushes it in. At the start the inside is slightly negative, so the electrical force pulls it in as well. Both forces point the same way, so sodium enters quickly. Each sodium ion brings a positive charge with it, so the inside goes from slightly negative, through zero, to positive. Once the inside is positive, the electrical force reverses and begins pushing sodium back out. The two forces balance at about $+61 \text{ mV}$, the equilibrium

potential for sodium, E_{Na} . The same rules gave the opposite sign, because the gradient points the opposite way.

Put the two cells on one voltage scale. Whenever an ion can cross the membrane, its movement shifts the membrane potential toward that ion's equilibrium potential: -95 mV for potassium and $+61$ mV for sodium. The more open channels an ion has, the larger its effect, so the ion with the highest permeability has the most influence on the membrane potential. A resting neuron sits at -70 mV, which is 25 mV from E_K and 131 mV from E_{Na} . That tells you the resting membrane is far more permeable to potassium than to sodium, roughly 25 to 40 times more. It is not exactly at E_K , which means a small amount of another ion is also crossing. That ion is sodium.

Where each ion would hold the voltage if it were the only ion that could cross

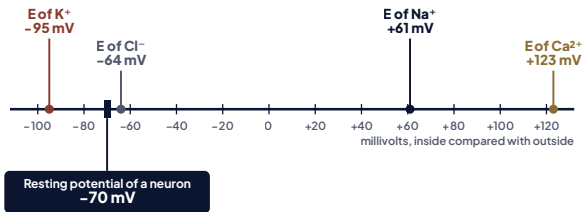


Figure 5. Equilibrium potentials for the typical concentrations, with a neuron's resting potential marked below the line.

The Nernst equation

SLIDES 14 TO 16 OF THE DECK

The Nernst equation gives the equilibrium potential. You will learn to read it, not calculate with it. It turns a concentration ratio into the voltage that would exactly cancel it. At body temperature, 37°C (98.6°F), it can be written $E_{ion} = (61.5 \text{ mV} \div z) \times \log_{10} ([ion] \text{ outside} \div [ion] \text{ inside})$, where z is the charge on the ion: $+1$ for potassium and sodium, -1 for chloride, $+2$ for calcium. The 61.5 mV bundles the gas constant, the Faraday constant, the conversion to log base 10 and body temperature, and many books round it to 61 or 60 . The full form is $E = (RT \div zF) \times \ln(\text{out} \div \text{in})$, which is the same equation before the constants are multiplied out.

Three things are worth reading out of it. Only the ratio of outside to inside matters, so the steeper the gradient, the larger the voltage needed to balance it. Every tenfold difference is worth about 61 mV for an ion with a single charge. The sign comes from two things together, the direction of the gradient and the charge on the ion. A positive ion that is more concentrated inside needs a negative interior to hold it in, so its equilibrium potential is negative, and a positive ion that is more concentrated outside needs a positive interior to hold it out, so its equilibrium potential is positive. A negative ion flips the sign. And an ion with a double charge feels the voltage twice as strongly, so it needs only half the voltage to balance the same ratio, which is why z is in the denominator.

For the concentrations in these notes, the values to know are E_K about -95 mV (many books say -90), E_{Na} about $+61$ mV, E_{Ca} about $+120$ mV and E_{Cl} about -64 mV, close to the resting potential. The most useful thing an equilibrium potential tells you is direction. When an ion's channels open, the ion moves so that it pulls the membrane toward its own equilibrium potential, and the farther the membrane is from that value, the larger the driving force.

$$E_{ion} = (61.5 \text{ mV} \div z) \times \log_{10} ([ion] \text{ outside} \div [ion] \text{ inside})$$

WORKED PROBLEM 1

Potassium is 4 mM outside the cell and 140 mM inside. Without calculating anything, predict whether E_K is positive or negative and explain why. Then say which way potassium moves when its channels are open in a neuron resting at -70 mV, and which way that moves the membrane potential.

Use the two forces, not arithmetic. Potassium carries one positive charge, so $z = +1$.

THE WORK

- Start with the chemical force. Potassium is 35 times more concentrated inside, so its concentration gradient pushes it **out**.
- Ask what electrical force would balance that push. Potassium is a positive ion, so the force that holds it back has to pull it in, and a positive ion is pulled in by a **negative** interior. That is why E_K has to be negative.
- The equation says the same thing in its own way. Outside over inside is 4 over 140 , a ratio smaller than 1 , and for an ion with a positive charge that always gives a negative answer. For these concentrations E_K is about **-95 mV**. Books that use slightly different concentrations say -90 mV, and either value is fine.
- Read that number as a sentence: if potassium were the only ion that could cross, the inside would settle about 95 mV negative, and at that voltage potassium would stop moving on net.
- Now the direction. The neuron is at -70 mV, which is not negative enough to hold potassium in. The outward push from concentration is still larger than the inward pull from voltage, so potassium **leaks out**, slowly and continuously. Each potassium ion that leaves carries a positive charge out, so the membrane moves **down, toward -95 mV**.
- The rule to keep: an ion whose channels are open always moves the membrane toward its own equilibrium potential. The gap between the membrane potential and E_K , here about 25 mV, is potassium's driving force, and it is small.

WORKED PROBLEM 2

Sodium is 145 mM outside and 15 mM inside. Chloride is 110 mM outside and 10 mM inside. Predict the sign of each equilibrium potential from the direction of the gradient and the charge on the ion. Then, using E_{Na} of about +61 mV and E_{Cl} of about -64 mV, say which way each ion moves when its channels open in a neuron resting at -70 mV, and rank potassium, sodium and chloride by the size of their driving force.

Both gradients point inward. Watch the charge on chloride.

THE WORK

- 1** Sodium. It is about 10 times more concentrated outside, so the chemical force pushes it **in**. To hold back a positive ion that is being pushed in, the inside has to repel it, which means the inside has to be positive. E_{Na} is **positive**, about +61 mV.
- 2** Chloride. It is also more concentrated outside, so the chemical force pushes it in as well. But chloride carries a negative charge, and a negative ion is pushed back out by a **negative** interior. So E_{Cl} is **negative**, about -64 mV, even though its gradient points the same way as sodium's. The charge flips the sign, which is what $z = -1$ does in the equation.
- 3** Direction for sodium. The cell is at -70 mV and sodium's equilibrium potential is +61 mV. Concentration and voltage both point inward, so when sodium channels open, sodium rushes **in** and the membrane moves **up** toward +61 mV. Very little gets in at rest only because very few sodium channels are open.
- 4** Direction for chloride. A cell at -70 mV is only about 6 mV more negative than E_{Cl} , so chloride is nearly balanced. Opening chloride channels moves very little of it, and the small net movement, outward at this voltage, nudges the membrane toward -64 mV.
- 5** Rank the driving forces by the gap between -70 mV and each equilibrium potential: sodium about 131 mV, potassium about 25 mV, chloride about 6 mV. Sodium has by far the largest driving force, and at rest it is held back only by closed channels.
- 6** What the three say together: at rest, potassium is slightly off balance, chloride is close to balanced, and sodium is far off balance. That is why opening sodium channels, the subject of the next lecture, changes the voltage so quickly.

and opposite charge, and the pump returns the ions that leaked. This is a steady state and not an equilibrium. An equilibrium needs no energy to stay where it is. A steady state stays constant only while energy keeps being supplied, and here that energy is ATP.

The Goldman equation, also called the Goldman-Hodgkin-Katz or GHK equation, handles several ions at once. As with the Nernst equation, you will learn to read it, not calculate with it. In words, it says that the membrane potential is a blend of the ions' equilibrium potentials, and each ion's share is weighted by the membrane's permeability to that ion, P , which in practice means how many of that ion's channels are open. Written out for potassium and sodium it reads $V_m = 61.5 \times \log_{10} \left(\frac{P_K [K^+]_{out} + P_{Na} [Na^+]_{out}}{P_K [K^+]_{in} + P_{Na} [Na^+]_{in}} \right)$. Each ion's concentrations are multiplied by its permeability, so an ion with few open channels has little say. Chloride can be added as a third term. It is left out here because chloride is already close to balanced at rest.

The pump therefore contributes in two ways. Its direct contribution, from moving 3 charges out for every 2 in, is only a few millivolts. Its indirect contribution is everything else, because without the gradients the leak channels have nothing to work with. The resting potential is produced mainly by potassium leaking out through leak channels, and the pump makes that possible by keeping the cell full of potassium. If a poison blocked every pump, the voltmeter would lose only a few millivolts in the first second, because the gradients are still there. Over the following hours sodium would leak in and potassium out with nothing to return them, both equilibrium potentials would drift toward zero, the membrane potential would follow, and the cell would swell as water followed the sodium in.

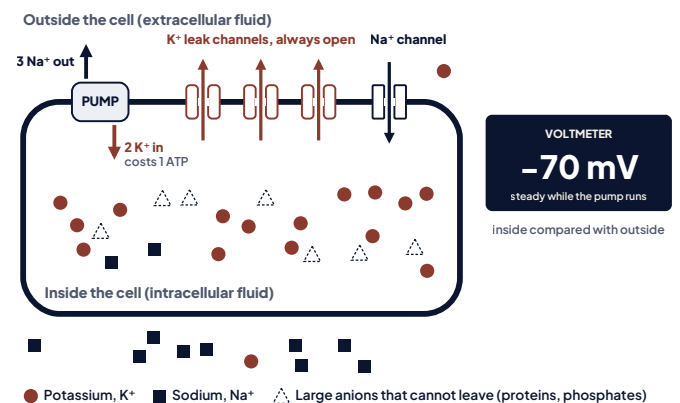


Figure 6. Many potassium leak channels, a few sodium leak channels, and the pump putting both ions back.

The real resting cell, and the Goldman equation

SLIDES 17 TO 19 OF THE DECK

A real membrane has many potassium leak channels and a few sodium leak channels. The small inward sodium leak brings positive charge in and raises the potential from -95 mV to about -70 mV. At -70 mV neither ion is at its own equilibrium, so potassium keeps leaking out and sodium keeps leaking in, all the time. The voltage holds steady because the two leaks carry equal

WORKED PROBLEM 3

At rest the membrane is about 25 times more permeable to potassium than to sodium. Without calculating, predict whether V_m sits nearer E_K (about -95 mV) or E_{Na} (about $+61$ mV), and roughly where. Then predict V_m if P_{Na} fell to zero, and if P_{Na} became very large.

E_K about -95 mV and E_{Na} about $+61$ mV. P_K is about 25 times P_{Na} .

THE WORK

- 1 Read the equation as a tug of war. Each ion pulls the membrane toward its own equilibrium potential, and its pull is weighted by its permeability, P , which in practice means how many of its channels are open.
- 2 Potassium's weight is about 25 times sodium's, so the membrane sits **close to E_K** and nowhere near halfway between the two. Halfway would be about -17 mV, which is nothing like a resting cell.
- 3 Sodium's small weight still counts. Sodium is far from its own equilibrium potential and there is a lot of it outside, so the few open sodium channels let in enough positive charge to lift the potential about 24 mV above E_K . That puts rest at about **-70 mV**. The equation gives about -71 mV for these values, which matches what an electrode measures in a real neuron.
- 4 Check the first extreme. If P_{Na} fell to zero, potassium would be the only ion with any weight, the equation would become the Nernst equation for potassium, and V_m would sit at **-95 mV**. That was cell 1.
- 5 Check the other extreme. If P_{Na} became very large, sodium's weight would swamp potassium's and V_m would head for **$+61$ mV**. That was cell 2, and the next lecture shows a neuron doing something close to it for a thousandth of a second.
- 6 The sentence to keep: the resting potential is a blend of the equilibrium potentials, weighted by permeability, and it sits near E_K because the resting membrane is far more permeable to potassium than to anything else.

Steady state

A condition that stays constant only because energy is continually spent to hold it there. The resting potential is a steady state.

Equilibrium

A condition that stays constant with no energy input. A single ion at its Nernst potential is at equilibrium.

Permeability (P)

How easily an ion crosses the membrane, set mainly by how many of its channels are open.

Goldman (GHK) equation

Predicts the membrane potential from the concentrations of several ions, each weighted by its permeability.

Channels, and the words for a potential that moves

SLIDES 21 TO 22 OF THE DECK

Every ion channel is a protein pore, and channels are sorted by what makes the pore open. Leak channels have no gate and are always open. They are in every cell, and potassium leak channels set the resting potential. Voltage gated channels open when the membrane potential changes. Sodium and potassium versions run along axons, and calcium versions sit at axon terminals and in heart muscle. Ligand gated channels, also called chemically gated, open when a specific molecule binds to them. The acetylcholine receptor on skeletal muscle at the neuromuscular junction is one, and so are the receptors on a neuron's dendrites. Mechanically gated channels open when the membrane is stretched, pressed or bent, as in touch receptors in the skin and hair cells in the inner ear.

Only leak channels are open in a resting cell, which is why this lecture needed only them. A cell uses the other three to change its membrane potential. The Goldman equation tells you what opening any of them does: it raises the permeability for whichever ion fits through, which gives that ion a bigger weight, so the membrane potential moves toward that ion's equilibrium potential. The concentrations do not need to change at all.

Four words describe a membrane potential that moves. A resting membrane is polarized, meaning it has a negative side and a positive side. Depolarization is any change that makes the inside less negative than rest. Overshoot is the part of a depolarization that goes past 0 mV, where the inside is briefly positive, so define it by the zero line and not by the height of the peak. Repolarization is the return toward the resting potential after a depolarization. Hyperpolarization is any change that makes the inside more negative than rest. A hyperpolarized cell is farther from threshold and harder to excite. A depolarized cell is closer to threshold and, at least at first, easier to excite.

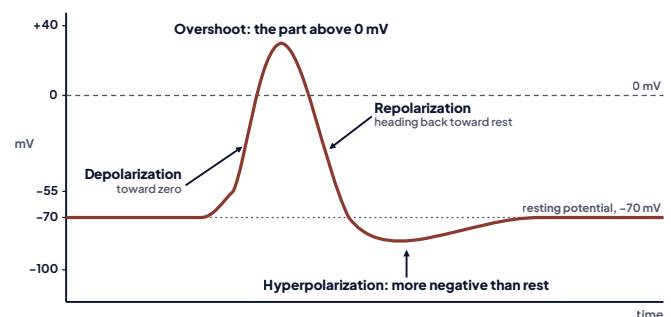


Figure 7. Each word describes a direction of change compared with the resting potential.

Voltage gated channel

Opens in response to a change in membrane potential. Found along axons, at axon terminals and in muscle.

Ligand gated channel

Opens when a specific molecule, such as a neurotransmitter, binds to it.

Mechanically gated channel

Opens when the membrane is stretched, pressed or bent.

Threshold

The membrane potential, about -55 mV in a neuron, at which voltage gated sodium channels open and the cell fires. Covered in the next lecture.

Changing the blood

SLIDES 23 TO 24 OF THE DECK

Plasma potassium is the outside potassium for every cell in the body, so a change in the blood is a change in the Nernst ratio for every cell at once. More potassium outside means a weaker gradient pushing potassium out, less negative charge left behind, and every cell depolarizes. Less potassium outside means a steeper gradient, and every cell hyperpolarizes.

With low potassium, rest moves away from threshold. Nerve and skeletal muscle cells become harder to excite, and the patient has muscle weakness and cramps. Heart muscle responds to low potassium in a more complicated way, which we take up in the cardiac weeks. With high potassium, rest moves toward threshold, and for a short time cells fire more easily. Then a second effect becomes more important. Voltage gated sodium channels that stay at a depolarized potential for too long switch to an inactivated state and cannot reopen until the membrane repolarizes. Cells become unable to fire. In skeletal muscle that is weakness. In the heart it is slowed, disorganized conduction.

Now compare sodium. Dropping plasma sodium by 20 mM, from 145 to 125 , moves E_{Na} from $+61$ to $+57$ mV and the resting potential from -71 to -73 mV, a shift of 2 mV. Raising potassium by only 3 mM moved the resting potential 7 mV. There are two reasons, and both are in the equations. The resting membrane is about 25 times more permeable to potassium, so the resting potential follows E_K and mostly ignores E_{Na} . And the Nernst equation works on ratios: 4 to 7 mM nearly doubles the outside potassium, while 145 to 125 mM changes sodium by only 14 percent. Low sodium is still dangerous, but for a different reason. It lowers the osmolarity of the extracellular fluid, water moves into brain cells, and they swell.

Same cell, three different blood potassium values

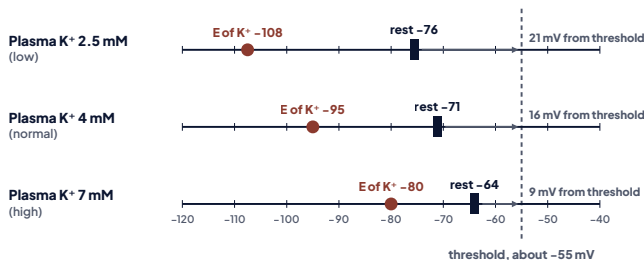


Figure 8. The same cell at three blood potassium values. Threshold is covered fully in the next lecture.

WORKED PROBLEM 4

Predict which way E_K and the resting potential move when plasma potassium rises from 4 to 7 mM, and when it falls to 2.5 mM. For each, say whether the cell moves toward threshold or away from it, and why.

Inside potassium stays at 140 mM, and nothing else changes.

THE WORK

- 1 High, 7 mM.** More potassium outside makes the gradient less steep, so less negative charge inside is needed to balance it. E_K becomes **less negative**, moving from about -95 toward about -80 mV. The resting potential follows E_K because potassium has most of the permeability, so the cell **depolarizes**, from about -71 to about -64 mV, and moves closer to threshold.
- 2 Low, 2.5 mM.** Less potassium outside makes the gradient steeper, so a more negative interior is needed to hold potassium in. E_K becomes **more negative**, about -108 mV, and the resting potential follows it down to about -76 mV. The cell **hyperpolarizes** and moves farther from threshold.
- 3** Why the direction surprises people: adding a positive ion outside makes the inside less negative, not more. The reason is not the charge on the added potassium. A weaker push out means less potassium leaves, so less negative charge is left behind inside.
- 4** With low potassium, rest moves away from threshold. Nerve and skeletal muscle cells become harder to excite, and the patient has muscle weakness and cramps. Heart muscle responds to low potassium in a more complicated way, which we take up in the cardiac weeks.
- 5** With high potassium, rest moves toward threshold, and for a short time cells fire more easily. Then a second effect becomes more important. Voltage gated sodium channels that stay at a depolarized potential for too long switch to an inactivated state and cannot reopen until the membrane repolarizes. Cells become unable to fire. In skeletal muscle that is weakness. In the heart it is slowed, disorganized conduction.

The patient, explained

SLIDE 25 OF THE DECK

Most of the potassium in the body is inside skeletal muscle cells, at 140 mM. Five hours of crushing killed a large mass of muscle, and dead cells release their contents. While the worker was pinned, the wall acted like a tourniquet. No blood flowed through his legs, so the released potassium stayed there. When the wall was lifted, blood flow returned and carried that potassium into his circulation all at once, and plasma potassium rose from 4 to 7 mM.

From there it is worked problem 4. E_K moved from -95 to -80 mV in every cell of his body, and resting potentials followed it upward by about 7 mV. Heart cells held at a depolarized potential have many of their sodium channels held in the inactivated state. The electrical signal crosses the heart slowly and unevenly, the rhythm breaks down, and the heart stops pumping. That is why a 3 mM change in potassium affects the heart when a 3 mM change in sodium has almost no effect.

Look at the treatment as an experiment. The variable that failed is plasma potassium, and through it the gap between rest and threshold. Calcium given into a vein does not change potassium at all. Raising the calcium outside the cell moves threshold to a less negative value, which restores the gap, and it works within minutes. Insulin with glucose, and albuterol, both speed up the sodium potassium pump, so potassium moves out of the plasma and into cells within the hour, although it is still in the body. Dialysis, or working kidneys, is the only step that removes potassium from the body. The same physiology shows up whenever many cells break open at once, as in large burns, and whenever the kidneys cannot excrete potassium, as in kidney failure.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture. The review pages are linked from each card.

CHEMISTRY

What an ion is, and the charges on Na^+ , K^+ , Cl^- and Ca^{2+}
Opposite charges attract and like charges repel
Diffusion: particles move from higher concentration toward lower
Concentration units: moles per liter and millimolar (mM)

[Chemistry review page](#)

FROM EARLIER WEEKS

The lipid bilayer, and why an ion cannot cross its oily middle unaided (Week 3)
Intracellular and extracellular fluid, and that a blood test reads the extracellular side (Week 3)
Channels and pumps as membrane proteins, and that ATP pays for uphill transport

No new anatomy is needed for this lecture, so there is no anatomy list this time.

MATH

Reading a voltage scale with negative values: -90 mV is more negative, and lower on the scale, than -70 mV
The gap between two points on that scale, for example -70 mV to $+60$ mV is 130 mV
What it means when a ratio such as outside over inside is bigger than 1 or smaller than 1

Competency 6, the membrane potential simulation, is done in lab. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

01

Ion distribution and electrochemical gradients

State the typical intracellular and extracellular concentrations of sodium, potassium, chloride, and calcium and separate the chemical from the electrical component of the driving force on each ion.

Do one of the two. Head your paper with the number.

1-1

Draw a cell and give each ion a pair of bars, inside and outside, height for concentration: sodium 145 mM out against 15 mM in, potassium 4 out against 140 in, chloride 110 out against 10 in, calcium about 1 out against 0.0001 free in. Mark the membrane inside negative. On each ion draw two separate arrows, chemical force and electrical force. One line: two ions have arrows pointing opposite ways, so name both and say which force wins.

1-2

Draw four vertical voltage scales side by side, one per ion, from minus 100 mV up to plus 150 mV. Mark each ion's equilibrium potential: potassium near minus 90, sodium near plus 60, calcium near plus 120, chloride near minus 65. Draw one horizontal line at minus 70 mV across all four. One line: each gap from that line is a driving force, so name the ion with the largest gap and say what stops it moving at rest.

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one.

02

Nernst equation

Explain what an ion's equilibrium potential means, predict its sign from the direction of the ion's concentration gradient and its charge, and use it to predict which way that ion will move when its channels open.

Do one of the two. Head your paper with the number.

2-1

Draw a cell with potassium high inside and sodium high outside. For each ion, draw the chemical arrow (down its concentration gradient) and the electrical arrow that would balance it. Beside each ion, write its equilibrium potential and whether the sign is positive or negative, and one line on why the sign comes out that way.

2-2

Draw a vertical voltage scale from minus 100 to plus 70 mV. Mark E_K , E_{Na} and a resting potential of minus 70 mV. Now open potassium channels, then sodium channels: for each, draw an arrow showing which way the ion moves and which way the membrane potential moves, and label which one has the bigger driving force and why.

03

Resting membrane potential

Explain why the resting membrane potential sits near the potassium equilibrium potential, explain in words how the Goldman equation combines each ion's concentration gradient and permeability, and predict how the potential shifts when membrane permeability to potassium or sodium changes.

Do one of the two. Head your paper with the number.

3-1

Draw a cell at rest with minus 70 mV written inside. On its membrane draw potassium leak channels as many, sodium leak channels as few, and the sodium potassium pump. Beside the cell draw a voltage scale with the potassium equilibrium potential near minus 90 and the sodium equilibrium potential near plus 60, and mark where minus 70 sits between them. One line: the resting potential sits close to one of those two and nowhere near halfway, so say what makes it lean.

3-2

Draw a voltage scale from minus 90 to plus 60 mV with the resting potential marked, then draw four labeled arrows off it showing where the potential moves when you open more potassium channels, open sodium channels, block the potassium leak, and stop the sodium potassium pump. One line: one of those four shifts the potential only a few mV even though it eventually ruins the cell, so name it and say why the immediate effect is so small.

04

Ion channel gating

Compare leak, voltage gated, ligand gated, and mechanically gated channels by what opens each one and give a physiological location for each.

Do one of the two. Head your paper with the number.

4-1

Draw four channels in a row in a strip of membrane: leak, voltage gated, ligand gated, and mechanically gated. Draw each one twice, closed and open, and draw whatever opens it right beside it, a voltage change, a bound molecule, a stretch or a bend. Under each write one real place in the body it is found. One line: only one of the four is open at rest, so say what job that gives it.

4-2

Draw one neuron stretched across the whole width of your box. Along it place each channel type where it actually works: ligand gated on the dendrites, voltage gated sodium and potassium along the axon, leak channels everywhere, voltage gated calcium at the terminal. Add a mechanically gated channel on a sensory ending drawn off to the side. One line: number the order these open during one real signal and say which one starts it.

05

Depolarization and hyperpolarization

Define depolarization, repolarization, hyperpolarization, and overshoot and label each on a membrane potential tracing.

Do one of the two. Head your paper with the number.

5-1

Draw a membrane potential tracing with time on the x and mV on the y, starting at minus 70. Take it up past zero to about plus 30 and back down below minus 70 before it returns. Bracket and label depolarization, overshoot, repolarization, hyperpolarization, and the return to rest. Draw zero mV as a dashed line. One line: define overshoot using that dashed line, not using the peak.

5-2

Draw one set of axes with a dashed line at minus 70 mV and another at zero. Plot four traces from rest: a small depolarization that fades away, a hyperpolarization, a depolarization that crosses threshold, and a repolarization from plus 30 back down. Label each with its correct term. One line: say which of your traces make the cell easier to fire and which make it harder, and how you can tell from direction alone.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · The resting membrane potential. Concentrations are typical teaching values, not laboratory reference ranges. All figures are original drawings for this course.

Graded potentials and the action potential

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The three worked problems from the slides are written out in full here. Silverthorn chapter 9.

COMPETENCIES 12 TO 14 OF WEEK 4

23 SLIDES · 3 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each problem yourself before you read the answer.

The problem for today

SLIDES 1 TO 2 OF THE DECK

A neuron uses two kinds of electrical signal. Graded potentials are small, local and variable in size. Action potentials are large, always the same size, and able to travel the full length of an axon. In this lecture we build both from the channels that produce them. We start with graded potentials, then build the voltage gated sodium channel one gate at a time, add the voltage gated potassium channel, and use the two channels to explain every phase of the action potential. Last, we block one channel at a time and predict what happens.

Here is the problem we solve by the end. A man eats pufferfish that was not prepared correctly. Within an hour his lips and tongue are numb. Then his arms and legs grow weak, and a few hours later he cannot breathe without help. He is fully conscious the entire time. The toxin in the fish, tetrodotoxin, does one thing: it plugs voltage gated sodium channels from the outside.

Measurements from a single axon show what the toxin changes and what it leaves alone. The resting membrane potential is -70 mV before and after. Potassium inside is still 140 mM and sodium outside is still 145 mM. A weak stimulus still produces a small depolarization of about 5 mV that fades. Only one thing is different. A strong stimulus, which normally makes the membrane rise to $+30$ mV in under a millisecond, now produces a depolarization of about 20 mV that fades. Everything from the last lecture is normal, so nothing in that lecture explains this patient. Write down your best guess for why losing that one large response would stop a person from breathing.

See it move: fire an action potential yourself

SLIDE 20 OF THE DECK

This is a working model of the trigger zone in this lecture. On the left it draws the membrane potential against time. On the right is a patch of membrane with two voltage gated sodium channels and two voltage gated potassium channels, and their gates open and close in step with the tracing. Set the stimulus strength and press Stimulate, or press Step to stop at each phase and read what is happening. Keep it open and come back to it as you read the sections below. Rest in the model is the Goldman value, -71 mV, and threshold is -55 mV, so it takes a stimulus of 16 mV or more to reach threshold.

Moving model: run it on the course site, [drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-action-potential-notes.html](https://stack.github.io/human-physiology-Fa26/biol005-w04-action-potential-notes.html)

THREE THINGS TO TRY

- 1 Stimulate at 8 mV, 15 mV, 16 mV and 25 mV, and write down the highest and lowest values each time. Below threshold the response grows with the stimulus. At 16 mV and at 25 mV the highest value is $+30$ mV and the lowest is -84 mV both times.
- 2 Set the stimulus to 20 mV and use the Step button. At the peak and at the bottom of the after-hyperpolarization, read the two permeabilities and the Goldman value, and compare them with worked problem 2.
- 3 Turn on the tetrodotoxin button and stimulate at 25 mV. Then turn it off, block the voltage gated potassium channels, and stimulate again. Compare both tracings with your predictions in the section on changing the conditions.

Two kinds of electrical signal

SLIDE 3 OF THE DECK

Both signals are changes in the membrane potential away from rest, and both are produced by ion channels opening. They differ in which channels open and in what happens next. A graded potential is a small change in membrane potential whose size depends on the size of the stimulus. It happens in the input zone of the neuron, the dendrites and cell body, and it fades within a millimeter or two. An action potential is a large, brief depolarization that is the same size every time. It starts at the trigger zone, which in most neurons is the first part of the axon, and it reaches the axon terminals at full size.

The two work together. Graded potentials carry information into the trigger zone, and the trigger zone decides whether to send an action potential down the axon.

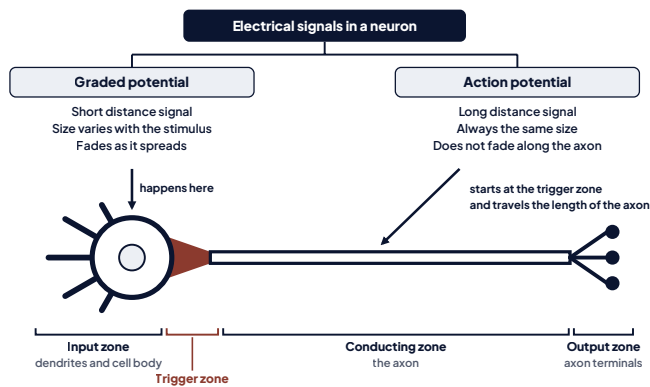


Figure 1. Graded potentials happen in the input zone. Action potentials start at the trigger zone and travel along the axon.

Graded potential

A change in membrane potential whose size varies with the strength of the stimulus and which fades with distance.

Action potential

A large, brief, all or none depolarization produced by voltage gated channels, which travels along an axon without getting smaller.

Trigger zone

The region of a neuron with a very high density of voltage gated Na^+ channels, where action potentials start. In most neurons it is the axon hillock and the first part of the axon.

Graded potentials

SLIDES 4 TO 5 OF THE DECK

A graded potential begins when a stimulus opens gated channels. At a synapse, a neurotransmitter opens ligand gated channels. In a touch receptor, stretch of the membrane opens mechanically gated channels. If the channels that open let sodium in, the membrane depolarizes. If they let potassium out or chloride in, the membrane hyperpolarizes. The rule from the last lecture still holds: the membrane potential moves toward the equilibrium potential of the ion whose permeability went up.

A stronger stimulus opens more channels, or holds them open longer, so more ions cross and the change in voltage is larger. That is what graded means. A graded potential has no threshold and no fixed size. It can be 1 mV or 20 mV, and any stimulus that opens channels produces one.

A graded potential gets smaller as it spreads. The sodium that enters is positive charge, and it spreads along the inside of the dendrite in both directions. This spread of charge is called local current flow. Two things reduce it. The first is current leak. The membrane is not a perfect insulator. It has open leak channels along its whole length, and some of the positive charge leaves the cell through them before it gets very far. The second is cytoplasmic resistance. The cytoplasm itself resists the flow of charge, in the way a thin wire resists current more than a thick one. With charge being lost through the membrane and slowed by the cytoplasm, the depolarization is smaller at every point farther from the channels that opened.

A wider dendrite has a lower cytoplasmic resistance, because there is more cross section for charge to move through, so a graded potential spreads farther in it before it fades. Widening

the tube does nothing to current leak, which depends on the leak channels in the membrane.

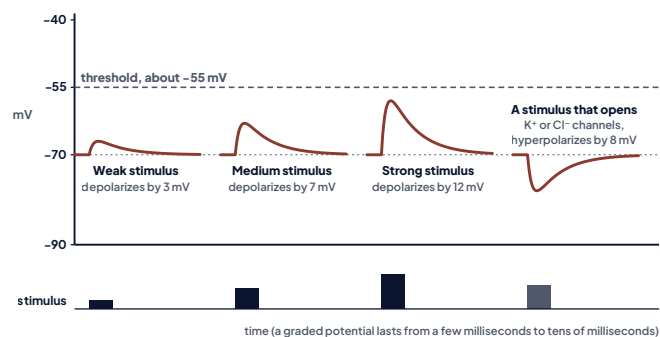
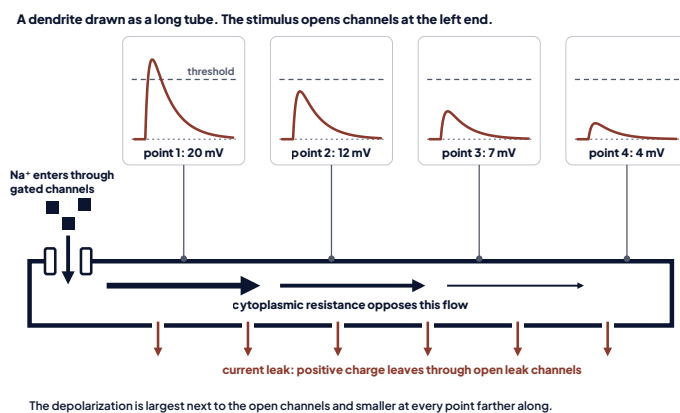


Figure 2. Three depolarizing graded potentials and one hyperpolarizing graded potential, recorded at the same spot.



The depolarization is largest next to the open channels and smaller at every point farther along.

Figure 3. The same graded potential recorded at four points along a dendrite.

Local current flow

The spread of charge through the cytoplasm away from the place where ions entered.

Current leak

Loss of charge across the membrane through open leak channels as a graded potential spreads.

Cytoplasmic resistance

The resistance of the cytoplasm to the flow of charge along the inside of a dendrite or axon. It is lower in a wider process.

What reaches the trigger zone

SLIDE 6 OF THE DECK

Whether a neuron fires depends on how large the graded potential still is when it arrives at the trigger zone. A graded potential that is still large enough to reach threshold there is called suprathreshold. One that is too small is called subthreshold, and it fades without producing an action potential.

Graded potentials that overlap in time add together, which is called summation. Depolarizing graded potentials add, and hyperpolarizing graded potentials subtract. The trigger zone responds to the total.

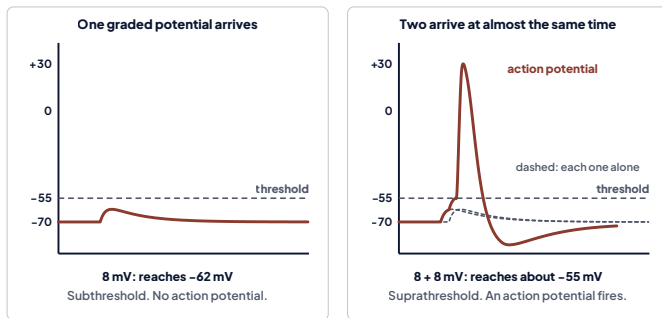


Figure 4. A graded potential that is below threshold at the trigger zone fades. Two that arrive together can add up to threshold.

WORKED PROBLEM 1

A neuron rests at -70 mV with threshold at -55 mV. One excitatory synapse produces 20 mV at the dendrite, and 8 mV of it reaches the trigger zone. Does the neuron fire with one such synapse active? With two at once? With two, plus an inhibitory synapse worth -5 mV at the trigger zone?

Work out the membrane potential at the trigger zone in each case and compare it with threshold.

THE WORK

- 1** Start from rest. The trigger zone is at -70 mV and threshold is about -55 mV, so the membrane has to depolarize by **15 mV** to reach threshold.
- 2** One synapse alone. The graded potential was 20 mV at the dendrite, but only 8 mV of it is left at the trigger zone. $-70 + 8 = -62 \text{ mV}$. That is 7 mV short of threshold, so this graded potential is **subthreshold** and no action potential fires. The membrane returns to -70 mV.
- 3** Two synapses together. Graded potentials add, so the trigger zone receives $8 + 8 = 16 \text{ mV}$. $-70 + 16 = -54 \text{ mV}$. That is past threshold, so the sum is **suprathreshold** and the neuron fires one action potential.
- 4** Now add the inhibitory synapse, which by itself hyperpolarizes the trigger zone by 5 mV. $8 + 8 - 5 = 11 \text{ mV}$. $-70 + 11 = -59 \text{ mV}$. The neuron is below threshold again and does not fire.
- 5** What the three answers say together: the neuron adds up every graded potential that reaches the trigger zone, depolarizing and hyperpolarizing, and the only question that matters is whether the total reaches threshold.
- 6** Adding graded potentials as plain numbers is a simplification. Real graded potentials that overlap add to slightly less than their arithmetic sum. The reasoning is the same.

Subthreshold

A graded potential too small to bring the trigger zone to threshold.

Suprathreshold

A graded potential large enough to bring the trigger zone to threshold.

Summation

The adding together of graded potentials that overlap in time.

The voltage gated sodium channel and its two gates

SLIDES 7 TO 10 OF THE DECK

The action potential depends on a channel that was not needed for the resting potential. A voltage gated sodium channel is a pore that only sodium fits through, and part of the protein senses the membrane potential. It has two gates in series, so sodium can pass only when both gates are open. It helps to picture a hallway with a door at each end. You can walk through only when both doors are open at the same time.

Step 1, the resting state. At the resting potential the activation gate is closed and the inactivation gate is open. One closed gate is enough to stop all sodium, so the channel is closed. This closed channel is able to open. It is waiting for a depolarization.

Step 2, the activated state. When the membrane depolarizes, the voltage sensor moves and the activation gate opens within a fraction of a millisecond. The inactivation gate was already open, so now both gates are open and sodium enters the cell. Sodium has a large driving force. At -55 mV the membrane is 116 mV away from the sodium equilibrium potential, and the concentration gradient and the voltage both point inward. The more the membrane depolarizes, the more of these channels open.

Step 3, the inactivated state. Depolarization sends the same signal to both gates. The activation gate responds quickly and opens. The inactivation gate responds slowly and closes, about half a millisecond later. So the channel is open only for the time between the fast gate opening and the slow gate closing. Once the inactivation gate has closed, the channel is inactivated. It is closed, but it is not the same as the resting state: the activation gate is open and the inactivation gate is blocking the pore. An inactivated channel cannot reopen, even if the membrane is still depolarized or is depolarized again. The channel shuts itself off, and that is what ends sodium entry.

Step 4, recovery. When the membrane repolarizes toward -70 mV, both gates return to their resting positions. The activation gate closes and the inactivation gate reopens, and the channel is closed and able to open again. So the channel has three states, closed, open and inactivated, and it moves through them in that order. It matters which gate is closed. In the resting state a depolarization will open the channel. In the inactivated state it will not, and only time spent near the resting potential resets it. A membrane whose sodium channels are inactivated cannot fire, however strongly it is stimulated. You saw this in the crush injury patient of the last lecture, and in the next lecture it explains the refractory period.

Many drugs act on this channel. Local anesthetics such as lidocaine block it, so the axons near the injection cannot carry action potentials and no pain signal reaches the brain. Some antiarrhythmic drugs and some antiseizure drugs, such as phenytoin and carbamazepine, also act on voltage gated sodium channels.

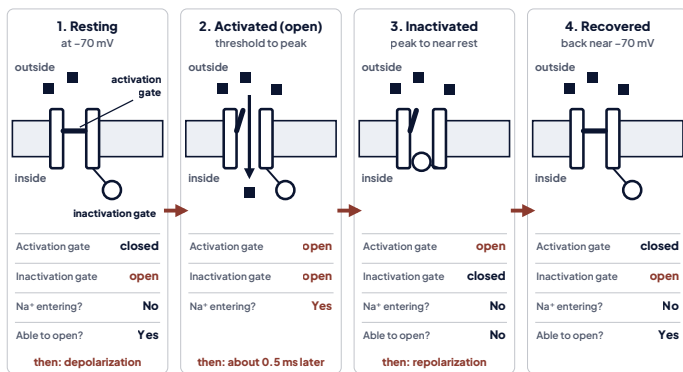


Figure 5. One channel, four drawings, three states. The bar is the activation gate and the ball on a chain is the inactivation gate. The recovered channel is back in the resting state.

Voltage gated Na⁺ channel

A sodium channel that opens when the membrane depolarizes. It has an activation gate and an inactivation gate.

Activation gate

Closed at rest. Opens within a fraction of a millisecond when the membrane depolarizes. Closes when the membrane repolarizes.

Inactivation gate

Open at rest. Closes about half a millisecond after depolarization. Reopens only after the membrane has repolarized.

Inactivated

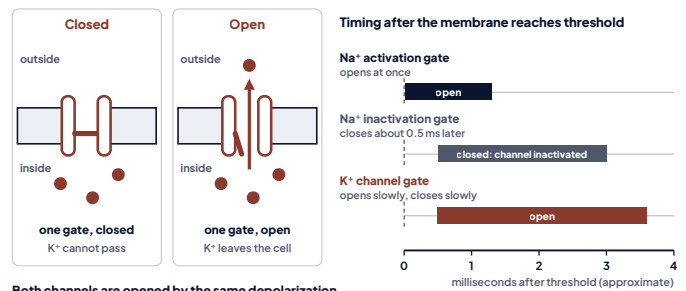
The state in which the inactivation gate blocks the pore. The channel is closed and cannot be opened by depolarization.

The voltage gated potassium channel

SLIDE 11 OF THE DECK

The second channel of the action potential is a voltage gated potassium channel. It is a different protein from the potassium leak channel that sets the resting potential. It has a single gate, which is closed at rest. The same depolarization that opens the sodium channels also opens these, but this gate is slow. It opens about when the sodium channels are inactivating. When it opens, potassium leaves the cell, because at any potential above -95 mV the net force on potassium is outward. The gate is also slow to close. It stays open until the membrane is back near rest, and for a little while after that.

The delay is essential. If the potassium channel were as fast as the sodium channel, sodium entering and potassium leaving at the same moment would cancel each other, and the membrane would hardly depolarize at all. The delay gives sodium about half a millisecond with no opposition, which is what allows the membrane to rise all the way to +30 mV. Then potassium leaves with no opposition, because the sodium channels have inactivated.



Both channels are opened by the same depolarization. The sodium channel responds in a fraction of a millisecond. The potassium channel responds about half a millisecond later, at about the time the sodium channels are inactivating.

Figure 6. The voltage gated K⁺ channel, and the timing of all three gates after the membrane reaches threshold. The times are approximate.

Threshold and the positive feedback loop

SLIDES 12 TO 13 OF THE DECK

The trigger zone has a very high density of voltage gated sodium channels. A small depolarization opens a few of them and a larger one opens more. Two currents then compete. Sodium entering depolarizes the membrane. Potassium leaving through the leak channels repolarizes it, and that outward current also grows as the membrane moves away from the potassium equilibrium potential. Below threshold the potassium current is larger, so the membrane returns to rest, and the response is only a graded potential.

Threshold is the membrane potential at which the inward sodium current just exceeds the outward potassium current. In many neurons it is near -55 mV. Books give values from about -50 to -55 mV, and it varies from cell to cell. Because threshold is a balance between two currents, it is not a fixed property of the cell. Anything that reduces the number of available sodium channels, or increases potassium permeability, moves threshold farther from rest and makes the cell harder to fire.

Past threshold there is a loop. Depolarization opens sodium channels, sodium entry depolarizes the membrane further, and that opens more sodium channels. This is positive feedback: the response increases the stimulus that caused it. It is why the rising phase is so steep, and why it cannot stop partway. Positive feedback needs something outside the loop to stop it. Here there are two things, and both were started by the same depolarization about half a millisecond earlier. Inactivation gates close, which stops sodium entry. Voltage gated potassium channels open, which lets potassium leave.

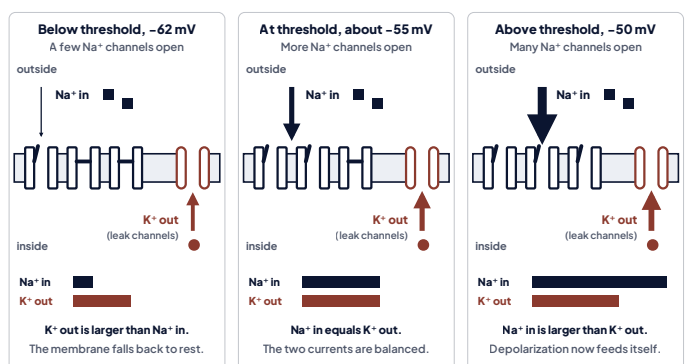


Figure 7. The trigger zone at three potentials. The bars compare the two currents.

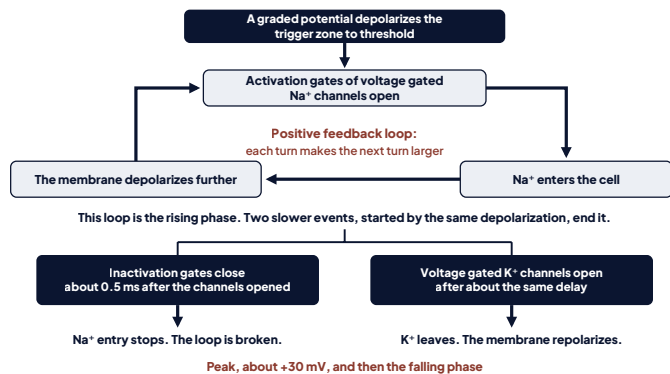


Figure 8. The positive feedback loop of the rising phase, and the two slower events that end it.

Threshold

The membrane potential at which Na⁺ entry through voltage gated channels exceeds K⁺ exit, so that depolarization sustains itself. About -55 mV in many neurons.

Positive feedback

A loop in which the response increases the stimulus that caused it. It continues until something outside the loop stops it.

The phases of the action potential

SLIDES 14 TO 16 OF THE DECK

1. Rest. Only leak channels are open. Potassium permeability is about 25 times sodium permeability, and the membrane sits near -70 mV. The voltage gated sodium channels are closed and able to open, and the voltage gated potassium channels are closed.
2. Depolarization to threshold. A graded potential arrives at the trigger zone and brings it to about -55 mV.
3. Rising phase. Activation gates open, sodium permeability rises several hundred times, and sodium enters. The membrane moves toward the sodium equilibrium potential, passes 0 mV and becomes positive inside. This takes well under a millisecond.
4. Peak, about +30 mV. Inactivation gates are closing and potassium channels are opening, so the rise stops. The membrane never reaches +61 mV. It would need the sodium channels to stay open, and they do not. Sodium entry also slows as the membrane approaches the sodium equilibrium potential, because the driving force on sodium shrinks.
5. Falling phase. Sodium channels are inactivated and voltage gated potassium channels are open. Potassium leaves, carrying positive charge out, and the membrane repolarizes.
6. After-hyperpolarization, also called the undershoot. The potassium channels are slow to close. For a few milliseconds potassium permeability is higher than at rest, so the membrane goes below -70 mV and approaches the potassium equilibrium potential. During this time the sodium channels recover: their activation gates close and their inactivation gates reopen.
7. Return to rest. The voltage gated potassium channels finish closing. Only the leak channels are open again, and they bring the membrane back to -70 mV. The sodium potassium pump does not do this. The pump is far too slow to shape a signal that lasts two milliseconds, and so few ions crossed that there is almost nothing for it to correct.

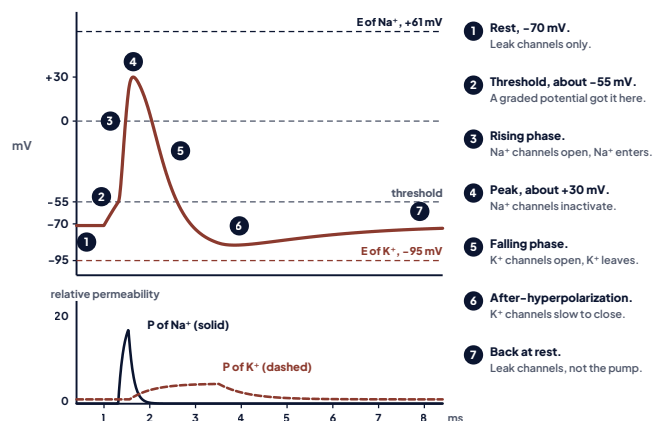


Figure 9. One action potential with the permeability to Na⁺ and K⁺ lined up under it.

WORKED PROBLEM 2

Use the Goldman equation, read in words, to predict where the membrane potential sits at four moments: at rest ($P_{Na} : P_K = 0.04 : 1$), with every voltage gated Na^+ channel open ($20 : 1$), at the real peak (about $4.3 : 1$), and during the after-hyperpolarization ($0.04 : 3$). For each, say whether it lies nearer E_K (about -95 mV) or E_{Na} (about $+61$ mV), and why.

$V_m = 61.5 \times \log_{10} \left(\frac{P_K [K^+]_{out} + P_{Na} [Na^+]_{out}}{P_K [K^+]_{in} + P_{Na} [Na^+]_{in}} \right)$. K^+ 4 out and 140 in. Na^+ 145 out and 15 in. The equation describes a membrane that has had time to settle. At the peak and at the bottom of the after-hyperpolarization the potential has stopped changing for an instant, so it applies exactly at those two moments.

THE WORK

- 1** At rest, $P_{Na} : P_K = 0.04 : 1$. Potassium has about 25 times the weight of sodium, so the membrane sits close to E_K , at about -71 mV. You met this value last lecture.
- 2** Every voltage gated Na^+ channel open, $P_{Na} : P_K = 20 : 1$. Now sodium has 20 times the weight of potassium, so the membrane swings most of the way to E_{Na} , to about $+50$ mV. This is where the membrane would settle if the sodium channels stayed open.
- 3** At the real peak, $P_{Na} : P_K$ is only about $4.3 : 1$, because many sodium channels have already inactivated and potassium channels have started to open. Sodium still has the larger weight, but by less, so the membrane sits lower, at about $+30$ mV.
- 4** After-hyperpolarization, $P_{Na} : P_K = 0.04 : 3$. Sodium permeability is back to its resting value, and potassium permeability is about three times its resting value because the voltage gated K^+ channels have not closed yet. Potassium's weight is now even larger than at rest, so the membrane sits closer to E_K than rest does, at about -84 mV.
- 5** Read the four answers in order: -71 , toward $+50$, $+30$, -84 , and then back to -71 when the potassium channels close. That is the whole shape of the action potential, and the concentrations never changed. Only the permeabilities did.
- 6** The ratios 20 to 1, 4.3 to 1 and 3 times resting are round teaching values. Measured values differ from one kind of neuron to another.

After-hyperpolarization (undershoot)

The few milliseconds after an action potential when the membrane is more negative than rest, because voltage gated K^+ channels are still open.

All or none

SLIDE 17 OF THE DECK

A stimulus that does not reach threshold gives a graded potential and no action potential. A stimulus that reaches threshold gives a full action potential. There is nothing in between. This is the all or none property. In Figure 10 the fourth stimulus is much stronger than the third, and the two action potentials have the same peak and the same shape.

The reason is that the stimulus only has to bring the trigger zone to threshold. After that the positive feedback loop runs on the energy stored in the sodium gradient, and the stimulus no longer contributes. How high the membrane goes depends on the sodium equilibrium potential, which comes from the concentrations, and on how quickly the sodium channels inactivate and the potassium channels open, which are properties of the channel proteins. None of those depends on the stimulus. Striking a match is a fair comparison. Once the match head reaches its ignition temperature it burns with the energy stored in the match, and striking it harder does not make a bigger flame.

If amplitude cannot carry information about stimulus strength, something else must. That is the first topic of the next lecture.

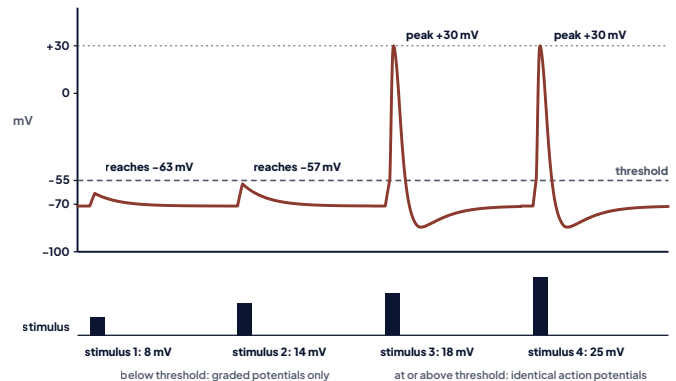


Figure 10. Four stimuli and four responses. The third and fourth responses are identical.

All or none

An action potential either happens at full size or does not happen. Its amplitude does not depend on the strength of the stimulus.

How few ions move

SLIDE 18 OF THE DECK

In the last lecture you learned that almost no ions have to move to set up the resting potential. The same is true of an action potential. The charge that has to cross the membrane to change its potential depends on the area of the membrane, and for all cell membranes it is about 6,000 single charges per square micrometer for a change of 100 mV. That figure comes from the electrical capacitance of cell membranes, about 1 microfarad per square centimeter.

Rising phase

The rapid depolarization from threshold to the peak, produced by Na^+ entering through voltage gated Na^+ channels.

Falling phase

The repolarization from the peak back toward rest, produced by Na^+ channel inactivation and K^+ leaving through voltage gated K^+ channels.

Because so few ions cross, the concentrations inside and outside the cell are the same after an action potential as before it, and the equilibrium potentials have not moved. That is why the pump is not needed to end an action potential, and why a neuron whose pumps were stopped could still fire a thousand or more action potentials before its gradients ran down.

WORKED PROBLEM 3

Take a neuron cell body as a sphere 20 μm across. How many Na⁺ ions must enter to take the membrane from -70 to +30 mV? What fraction of the sodium already inside is that?

Radius 10 μm. About 6,000 charges per μm² for 100 mV. Sodium inside is 15 mM. One mole is 6.02 × 10²³ ions.

THE WORK

- 1

Surface area of a sphere is 4πr². With r = 10 μm, the area is 4 × 3.14 × 100 = **1,257 μm²**.
- 2

Charges needed: 1,257 μm² × 6,000 per μm² = about **7.5 million Na⁺ ions** to carry the membrane from -70 to +30 mV.
- 3

Now count the sodium ions already inside. The volume of the sphere is (4 ÷ 3)πr³ = 4,189 μm³, which is 4.19 × 10⁻¹² liters. At 15 mM that is 6.3 × 10⁻¹⁴ moles, or about **38 billion Na⁺ ions**.
- 4

Compare them. 7.5 million ÷ 38 billion is about 1 in 5,000, or **0.02 percent**. Sodium inside the cell goes from 15 mM to about 15.003 mM. E_{Na} does not measurably change.
- 5

The same number of K⁺ ions leaves during the falling phase. The cell holds about 350 billion K⁺ ions, so it loses roughly 1 in every 47,000.
- 6

A real neuron moves somewhat more than this minimum, because for part of each action potential sodium is entering while potassium is leaving. The conclusion is the same: one action potential does not change the concentrations, so the pump is not needed to end it. The pump matters over thousands of action potentials, when it returns the small amounts that crossed.

	GRADED POTENTIAL	ACTION POTENTIAL
Where it happens	Input zone: dendrites and cell body. Also sensory receptor endings.	Starts at the trigger zone and travels along the axon
Channels that produce it	Ligand gated or mechanically gated channels	Voltage gated Na ⁺ and K ⁺ channels
Direction of the change	Depolarizing or hyperpolarizing	Always a depolarization, followed by repolarization and a brief after-hyperpolarization
Size	Varies with the strength of the stimulus	All or none. Always about 100 mV.
Over distance	Gets smaller, because of current leak and cytoplasmic resistance	Arrives at the axon terminal at full size
Threshold	None	Requires a suprathreshold graded potential at the trigger zone
Can two add together?	Yes	No
Refractory period	None	Yes. Covered in the next lecture.
Its job	To carry the strength of the input to the trigger zone	To carry a signal over a long distance without loss

Changing the conditions

SLIDE 21 OF THE DECK

Block the voltage gated potassium channels and leave everything else normal. The rising phase and the peak are normal, because they depend on sodium channels. The falling phase is much slower. Sodium channels still inactivate, so sodium entry stops, but the only way for potassium to leave is the leak channels. The action potential lasts several times longer and there is no after-hyperpolarization. Tetraethylammonium, a laboratory potassium channel blocker, does this.

Stop the inactivation gate from closing. The membrane rises normally and then stays depolarized, because sodium keeps entering while potassium leaves. Repolarization is delayed and the action potential is greatly prolonged. Some scorpion and sea anemone toxins work this way. The experiment shows what inactivation is for: it is the step that ends sodium entry.

Block the voltage gated sodium channels with tetrodotoxin. The stimulus still depolarizes the membrane, because the channels that produce graded potentials are not voltage gated sodium channels. The depolarization passes -55 mV and nothing happens, because the channels that would start the positive feedback loop cannot conduct. There is no rising phase, so there is no action potential at any stimulus strength.

The patient, explained

SLIDE 22 OF THE DECK

Tetrodotoxin plugs the outer mouth of the voltage gated sodium channel. It does not touch leak channels, the pump, or ligand gated channels, so the diner's resting potentials are normal and his graded potentials are normal. What he has lost is the rising phase. With the sodium channels blocked, reaching threshold

The two signals side by side

SLIDE 19 OF THE DECK

The graded potential is how a neuron measures its input. Its size reports how strong the stimulus was, and summation lets the neuron combine many inputs. But it fades, so it cannot carry a message farther than a millimeter or two. The action potential is how the neuron sends its decision over a distance. It does not fade, because it is regenerated by voltage gated channels all along the axon, and the cost of that is a fixed size.

GRADED POTENTIAL AND ACTION POTENTIAL COMPARED

opens nothing. There is no positive feedback loop, so no action potential forms in the axon.

Sensation fails first. Numb lips and tongue come first, because sensory axons can no longer carry action potentials. Then motor neurons can no longer carry commands to skeletal muscle, and skeletal muscle fibers, which use the same kind of sodium channel, cannot fire either. When the diaphragm stops receiving action potentials from the phrenic nerve, breathing stops. His heart keeps beating because heart muscle uses a different voltage gated sodium channel that is far less sensitive to tetrodotoxin. The toxin also crosses poorly from the blood into the brain, so the neurons of his brain keep working, and he is conscious while unable to move.

Look at it as an experiment. The problem is that no action potentials reach the diaphragm or the other breathing muscles, so air does not move. The target variable is ventilation of the lungs, and through it blood oxygen and carbon dioxide. The intervention is a ventilator, which moves the air that the muscles cannot. It does nothing to the toxin or to the channels. Recovery comes from the fact that tetrodotoxin binding is reversible. As the body clears the toxin over hours to days, it leaves the channels, and they work normally again, because they were never damaged. With ventilation maintained, strength and sensation return as the toxin is cleared.

Local anesthetics block the same channel from the inside of the membrane. The numb lip after dental work and the numb lips of this patient are the same physiology at very different scales.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture.

CHEMISTRY

What an ion is, and the charges on Na^+ and K^+
Opposite charges attract and like charges repel
Diffusion: particles move from higher concentration toward lower

Chemistry review page

FROM EARLIER LECTURES

The resting membrane potential: E_K is -95 mV , E_{Na} is $+61 \text{ mV}$, and rest is -70 mV
The Goldman equation, and that raising the permeability to an ion moves the membrane potential toward that ion's equilibrium potential
Leak, voltage gated, ligand gated and mechanically gated channels
Depolarization, repolarization, hyperpolarization and overshoot
The four zones of a neuron: input zone, trigger zone, conducting zone and output zone

No new anatomy is needed for this lecture, so there is no anatomy list this time.

MATH

Scientific notation, such as 6.02×10^{23}
Area and volume of a sphere (the formulas are given in the problem)

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

12

Graded potentials

Describe how a graded potential is produced and explain why it varies with stimulus strength and decays with distance, using the terms current leak and cytoplasmic resistance, and relate current, voltage, and resistance with Ohm's law, including how opening channels raises conductance.

Do one of the two. Head your paper with the number.

12-1

Draw a dendrite as a long tube with a stimulus at one end. At four points along it, draw the membrane potential as a small trace, showing the signal shrinking as you move away. Draw current leaking out through channels along the way and label the cytoplasmic resistance inside the tube. One line: name the two reasons the signal shrank, and say which one you could reduce by making the tube wider.

12-2

Draw one set of axes, membrane potential against time, and plot three graded potentials from the same spot produced by a weak, a medium, and a strong stimulus. Then plot two of them arriving close together and summing. Mark threshold as a dashed line. One line: a graded potential changes size with stimulus strength and an action potential does not, so say what the graded potential's actual job is.

13

Action potential phases

Diagram an action potential and state the channel state and ion movement responsible for depolarization to peak and for repolarization and afterhyperpolarization.

Do one of the two. Head your paper with the number.

13-1

Draw an action potential tracing, mV on the y and ms on the x, from minus 70 up to about plus 30 and back down through an afterhyperpolarization near minus 80. Label threshold, depolarization, peak, repolarization, and afterhyperpolarization on the curve. Underneath, draw two permeability traces lined up in time with it, one for sodium and one for potassium. One line: say what closes at the peak and what is still open during the afterhyperpolarization.

13-2

Draw a voltage gated sodium channel four times across the top of your box, in its resting, activated, inactivated, and recovered states, with both of its gates drawn separately every time. Under each, write the membrane voltage at which that state occurs. Beneath that row, draw a voltage gated potassium channel in its two states with its timing. One line: both open on depolarization, so say what makes potassium arrive late and why the cell needs that delay.

14

Threshold and all or none

Explain what threshold represents at the trigger zone and why action potential amplitude does not change with a stronger stimulus.

Do one of the two. Head your paper with the number.

14-1

Draw the trigger zone as a close up patch of membrane with voltage gated sodium channels in it, three times over: below threshold with a few channels open and the leak winning, exactly at threshold with the two currents balanced, and above threshold with the positive feedback running away. Draw the current arrows in and out in every panel. One line: define threshold from your three drawings in terms of two currents, not a number.

14-2

Draw one set of axes with four stimuli of increasing strength as bars on a bottom trace and the resulting membrane potential plotted above them. Show the first two failing to reach threshold and the last two producing identical full sized action potentials. Mark threshold. One line: the fourth stimulus was far stronger than the third and the spike came out the same size, so say what actually fixes the amplitude.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · Graded potentials and the action potential. Concentrations and permeability ratios are typical teaching values. All figures and the moving model are original drawings made for this course.

How action potentials carry information

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The four worked problems from the slides are written out in full here. Silverthorn chapter 9.

COMPETENCIES 15 TO 18 OF WEEK 4

22 SLIDES · 4 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each problem yourself before you read the answer.

The problem for today

SLIDES 1 TO 2 OF THE DECK

An action potential is the same size every time a neuron fires one. That raises three questions, and they are the plan for this lecture. How can a signal that never changes size tell the brain that one touch is light and another is hard? How does it travel a meter along an axon without fading? And what decides how fast it gets there? We answer the first with the refractory periods and frequency coding, the second with local current, and the third with axon diameter and myelin. Then we change the conditions and predict what happens.

Here is the problem we solve by the end. A dentist injects lidocaine beside the nerve that serves the lower teeth. Within a few minutes the lip and the teeth on that side are numb, and the drill causes no pain. The patient can still feel the dentist pushing on the tooth. A few hours later all feeling is back. The drug never reached the pain receptors in the tooth, and they still respond to the drill.

You need three facts to make a first prediction. Lidocaine blocks voltage gated Na^+ channels. It stays close to where it was injected, partway between the tooth and the brainstem. And the endings of pain receptors respond to the drill with channels that are not voltage gated Na^+ channels. So graded potentials still form in the tooth, and action potentials still start in the axon near the tooth. They stop in the stretch of nerve beside the injection, and nothing reaches the brainstem. Pain is felt only when action potentials reach the brain, and one blocked stretch is enough to stop them. The harder question is why pressure still gets through, when the axons that carry it lie in the same nerve and the same drug. Write down a guess before you read on.

See it move: race two axons

SLIDE 15 OF THE DECK

This is a working model of two axons that are stimulated at the same moment. The clock shows real milliseconds. You can choose which two axons to compare, change the diameter of axon A and take its myelin off, change the middle third of both axons, and change how often stimuli arrive at the trigger zone. The strip at the bottom shows which stimuli fire an action potential and which ones land inside the absolute refractory period. Come back to it as you read the sections below.

Moving model: run it on the course site, [drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-ap-conduction-notes.html](https://stack.github.io/human-physiology-Fa26/biol005-w04-ap-conduction-notes.html)

THREE THINGS TO TRY

- 1 Choose the motor fiber and the slow pain fiber and watch the clock. Axon A arrives in 11 ms and axon B in 1 s. Then take the myelin off axon A. It is still 15 times wider than axon B, and it now conducts at only 3.9 m/s. Compare that with worked problem 2.
- 2 Put the myelin back on axon A and press Remove the myelin, which strips only the middle third. Read the new arrival time. Then press Remove the myelin and warm the body, and last press Apply local anesthetic. Say in your own words why the action potential stops in each case.
- 3 Slide the stimuli up from 100 to 500 per second, and then to 800. Read how many action potentials are fired each time, and compare the ceiling with worked problem 1. Then shorten the absolute refractory period to 1 ms and read the ceiling again.

The refractory periods

SLIDES 3 TO 6 OF THE DECK

Start with what you already have from the last lecture. The voltage gated Na^+ channel has two gates, an activation gate and an inactivation gate, and so it has three states. When it is closed and ready, the activation gate is closed and the inactivation gate is open, and a depolarization to threshold will open it. When it is open, both gates are open, for less than a millisecond. Then the inactivation gate closes and the channel is inactivated. An inactivated channel cannot open, however strongly the membrane is depolarized. It resets to closed and ready only after the membrane has repolarized. Everything in this lecture follows from that one fact.

A refractory period is a stretch of time after an action potential during which that patch of membrane is harder, or impossible, to fire again. The absolute refractory period starts when the membrane reaches threshold and lasts about 1 to 2 ms, which is most of the action potential. During it the Na^+ channels are open at first and then inactivated. No stimulus of any size can start a

second action potential, because a stimulus works by opening closed channels, and there are none.

The relative refractory period follows and lasts a few more milliseconds. Some Na^+ channels have reset and some have not. The voltage gated K^+ channels are still open, so K^+ is still leaving, and the membrane is hyperpolarized, farther from threshold than usual. Two things now work against a stimulus: open K^+ channels let positive charge leave as fast as the stimulus brings it in, and the membrane has farther to go. A second action potential is possible, but it takes a larger than normal stimulus, and it may be smaller than normal because only the channels that have reset can take part. An everyday comparison is a toilet that has just been flushed. While the tank is empty, pressing the handle does nothing, however hard you press. While it is refilling you can flush again, but the flush is weaker.

Refractory periods do three jobs. They keep action potentials separate, so they never overlap or add together the way graded potentials do. They set a maximum firing frequency. And they make conduction one way, because the membrane just behind a traveling action potential is refractory.

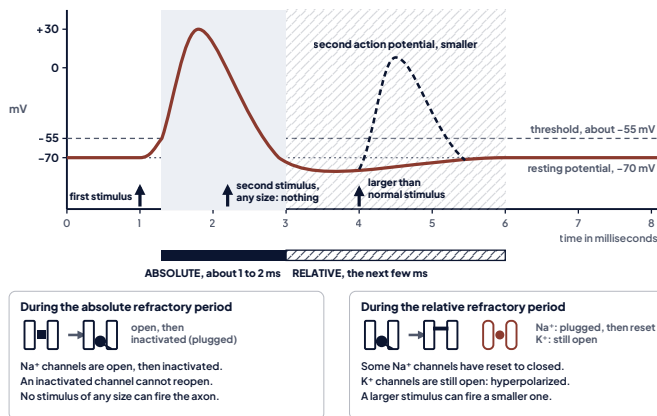


Figure 1. The two refractory periods, with the state of the channels during each one and the result of a second stimulus.

WORKED PROBLEM 1

An axon has an absolute refractory period of 2 ms. What is the largest number of action potentials it can send in one second? What if the refractory period were 1 ms?

maximum frequency = $1 \div \text{absolute refractory period}$, with the period in seconds.

THE WORK

- 1** Put the refractory period into seconds. 2 ms = **0.002 s**.
- 2** One action potential needs 0.002 s before the next one can start, so the most that fit into one second is $1 \div 0.002 = \mathbf{500 \text{ action potentials per second}}$.
- 3** Repeat it for an absolute refractory period of 1 ms. $1 \div 0.001 = \mathbf{1,000 \text{ per second}}$. A shorter refractory period gives a higher ceiling.
- 4** Read the answer as a sentence: however strong the stimulus is, this axon cannot send more than 500 action potentials in one second, because for 2 ms after each one its sodium channels cannot open.
- 5** This is a ceiling and not a typical rate. Most neurons fire far below it, commonly well under 100 per second, because the relative refractory period comes next and a stimulus has to be unusually strong to fire the axon during it.

Absolute refractory period

About 1 to 2 ms from threshold. Na^+ channels are open and then inactivated, so no stimulus of any size can fire a second action potential.

Relative refractory period

The next few milliseconds. Some Na^+ channels have reset, K^+ channels are still open and the membrane is hyperpolarized. A larger than normal stimulus can fire a smaller than normal action potential.

Inactivated

The state of the voltage gated Na^+ channel after it has opened. It cannot reopen until the membrane repolarizes and the channel resets.

Maximum firing frequency

$1 \div \text{the absolute refractory period}$. With 2 ms it is 500 per second. Most neurons fire far below their ceiling.

How a neuron codes the strength of a stimulus

SLIDES 7 TO 8 OF THE DECK

Because every action potential is the same size, strength cannot be coded in the size of the signal. It is coded in the number of action potentials per second. A stronger stimulus opens more channels in the receptor ending, so it produces a larger graded potential, and one that lasts longer. At the trigger zone, a graded potential that is still above threshold when the absolute refractory period ends fires another action potential, and the larger it is, the sooner in the relative refractory period it can do that. So a larger, longer graded potential produces more action potentials per second. This is frequency coding. The ceiling is the one set by the

absolute refractory period, and the length of the burst codes how long the stimulus lasted.

At the axon terminal, each action potential releases about the same amount of neurotransmitter, so more action potentials per second release more neurotransmitter. That is how the next cell receives the information about strength. The next lecture covers how this release works.

A single axon has only that one way to report a stronger stimulus. The nervous system as a whole has a second way. A stronger stimulus reaches more receptors, including ones with a higher threshold, so more neurons fire. This is called recruitment, or population coding. The brain reads both: how fast each axon is firing, and how many axons are firing. Recruitment matters most at high intensities. An axon with a 2 ms absolute refractory period that is already firing at 400 per second cannot double its rate, because its ceiling is 500, but a stimulus twice as strong still recruits receptors that were silent before.

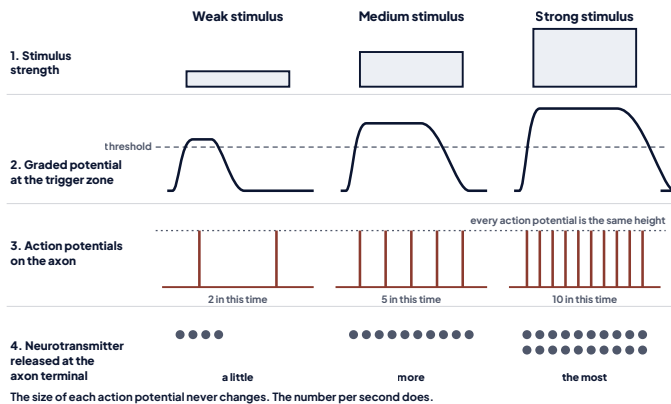


Figure 2. One sensory neuron and three strengths of stimulus. Read each column from top to bottom.

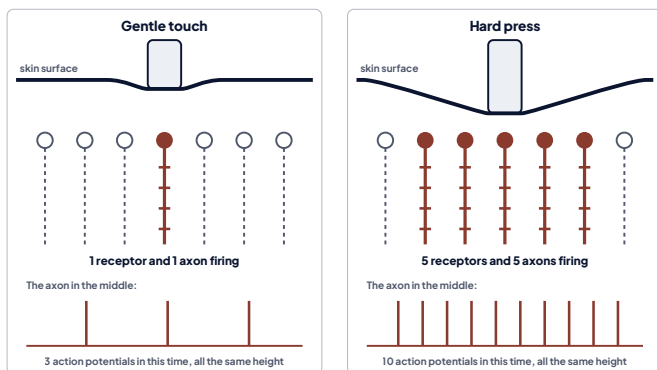


Figure 3. The same patch of skin touched gently and pressed hard.

Frequency coding

Signaling the strength of a stimulus by the number of action potentials per second, because their size is fixed.

Recruitment (population coding)

Signaling a stronger stimulus by the number of neurons that fire.

Trigger zone

The part of the neuron with the highest density of voltage gated Na^+ channels, where graded potentials are turned into action potentials.

How the action potential moves along the axon

SLIDES 9 TO 10 OF THE DECK

Stop a traveling action potential at one moment and look at the membrane. In the active zone, Na^+ has entered and the inside of the membrane is positive, while the neighboring stretches are still negative inside. Positive charge flows along the inside of the axon toward the negative regions on both sides. This flow of charge is called local current. Ahead of the action potential, the local current depolarizes resting membrane. When that patch reaches threshold, its own Na^+ channels open and it fires a full action potential, and then it does the same for the patch after it.

Behind the action potential, the local current arrives at membrane that fired a moment ago. Its Na^+ channels are inactivated and its K^+ channels are open, so it is refractory and cannot fire. By the time it has recovered, the active zone has moved too far away to reach it. That is why an action potential that starts at the trigger zone travels only toward the axon terminal. The axon itself has no preferred direction. If an axon is stimulated in the middle in the laboratory, two action potentials travel away from the electrode in opposite directions, because the membrane on both sides is at rest.

The action potential that reaches the axon terminal is therefore not the one that left the trigger zone. It is a copy, made fresh by each patch of membrane along the way, using the Na^+ and K^+ gradients at that patch. That is what regenerated means, and it is why the signal is the same size a meter away as it was at the start. A graded potential is different. It spreads by local current only, nothing renews it, charge leaks out through the membrane as it goes, and it is gone within a few millimeters. A row of falling dominoes is a fair comparison on two points: each domino falls with its own stored energy, so the last falls as hard as the first, and a domino that has fallen cannot fall again until it is set back up.

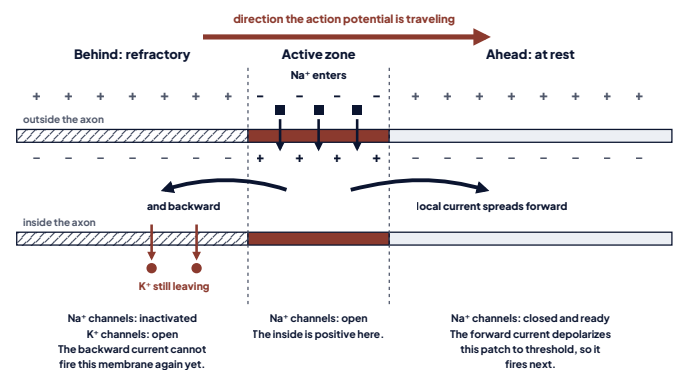


Figure 4. Local current spreads forward and backward. Only the forward direction finds membrane that can fire.

Local current

The flow of positive charge along the inside of the axon from the active zone toward neighboring membrane that is still negative inside.

Regenerated

Made again at each patch of membrane. The action potential does not decrease with distance because each patch fires its own.

Conduction

The movement of an action potential along an axon at full size.

What sets conduction velocity

SLIDES 11 TO 14 OF THE DECK

Two things decide how fast an axon conducts. The first is diameter. Local current flows through the cytoplasm, and a wider axon has more room for charge to flow, so its internal resistance is lower, in the same way that a wide pipe carries water more easily than a narrow one. The local current reaches farther ahead before it fades, so membrane farther down the axon is brought to threshold sooner. The second is how much current leaks out through the membrane on the way, because current that leaks out is lost to the job of depolarizing the next patch. Myelin, which is many layers of glial cell membrane wrapped around the axon, raises the resistance of the membrane so very little leaks out. It also means less charge is needed to change the voltage across that stretch. Both effects let local current spread farther and faster.

Myelin gives far more speed for the space it takes up. The squid giant axon has no myelin. It is about 500 μm across and conducts at about 25 m/s. Your largest myelinated axons are about 20 μm across and conduct at up to about 120 m/s. Area goes with the square of the diameter, and $(500 \div 20)^2$ is 625, so about 625 of your axons fit in the space of one squid axon, and each is nearly five times faster. That is what lets a nerve as thick as a finger carry many thousands of fast axons down a limb.

In an unmyelinated axon, voltage gated channels are spread along the whole membrane, and every patch has to depolarize and fire in turn. This is continuous conduction, and it is slow because opening channels takes time at every patch. In a myelinated axon the voltage gated Na^+ channels are concentrated at the nodes of Ranvier. Local current from one node flows inside the axon, under the myelin, and reaches the next node still strong enough to bring it to threshold, so the action potential is regenerated only at the nodes. This is saltatory conduction, from the Latin word for jumping. Nothing actually jumps. What travels between the nodes is local current, a very fast spread of charge along the inside of the axon. The time is spent at the nodes, and there are far fewer of those stops. It also costs less ATP, because Na^+ enters and K^+ leaves only at the nodes, so the Na^+/K^+ pump has far fewer ions to move back.

Real human nerve fibers conduct at anywhere from about 0.5 to about 120 m/s. Large myelinated A alpha fibers, 12 to 20 μm , conduct at 70 to 120 m/s and carry commands to skeletal muscle. A beta fibers, 5 to 12 μm , carry touch and pressure at 30 to 70 m/s. Thin myelinated A delta fibers, 2 to 5 μm , carry fast, sharp pain and cold at 12 to 30 m/s. Unmyelinated C fibers, 0.4 to 1.2 μm , carry slow, aching pain and warmth at 0.5 to 2 m/s. These are approximate ranges, and books differ a little on the boundaries. Two rough rules reproduce them: for a myelinated axon, velocity in m/s is about 6 times the diameter in μm , and for an unmyelinated axon it is about the square root of the diameter in μm . To rank any set of axons, put the myelinated ones first, and within each group put the larger diameter first.

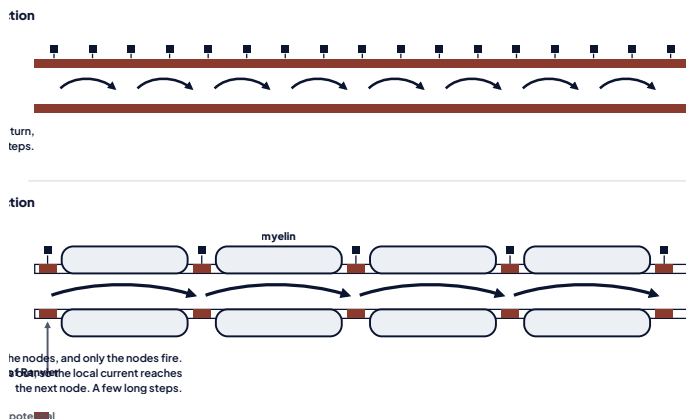


Figure 5. Two axons of the same diameter. The shaded membrane is the membrane that depolarizes and fires.

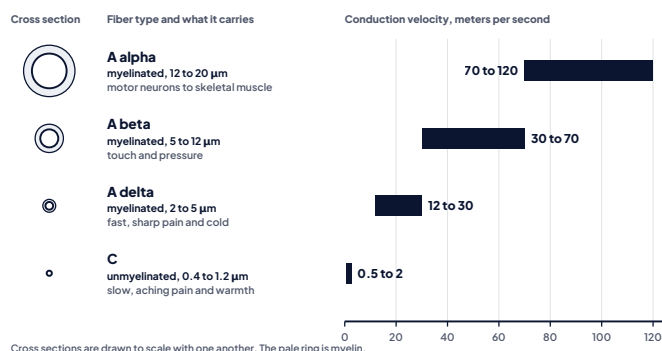


Figure 6. Approximate ranges for human nerve fibers.

WORKED PROBLEM 2

Sensory axons from your toe run about 1 m to the spinal cord. How long does an action potential take in a large myelinated fiber at 100 m/s, and in a small unmyelinated fiber at 1 m/s? Then explain why a stubbed toe hurts twice.

time = distance $\div$ velocity.

THE WORK

- The relationship is time = distance $\div$ velocity.
- Large myelinated fiber: $1 \text{ m} \div 100 \text{ m/s} = 0.010 \text{ s} = \mathbf{10 \text{ ms}}$.
- Small unmyelinated fiber: $1 \text{ m} \div 1 \text{ m/s} = \mathbf{1 \text{ s}}$. That is 100 times longer, for the same stretch of leg.
- Now the two kinds of pain fiber. A delta fibers are thin and myelinated. At about 20 m/s, 1 m takes $1 \div 20 = 0.05 \text{ s} = \mathbf{50 \text{ ms}}$. C fibers are unmyelinated. At about 1 m/s, 1 m takes $\mathbf{1 \text{ s}}$.
- So when you stub your toe, the action potentials in the A delta fibers reach the spinal cord about a second before the ones in the C fibers. You feel a sharp, well located pain first, and a dull, aching, spreading pain about a second later. These are called first pain and second pain. Both sets of action potentials started at the same moment.
- The withdrawal reflex uses the faster fibers, which is why your foot has already moved before the ache begins.

Continuous conduction

Conduction in an unmyelinated axon, where every patch of membrane fires in turn.

Saltatory conduction

Conduction in a myelinated axon, where the action potential is regenerated only at the nodes of Ranvier and local current carries the depolarization between them.

Node of Ranvier

A short bare gap between segments of myelin, where voltage gated Na^+ channels are concentrated.

First pain and second pain

The sharp pain carried by A delta fibers, followed about a second later by the aching pain carried by slower C fibers.

Changing the conditions: myelin, potassium and calcium

SLIDES 16 TO 18 OF THE DECK

Remove the myelin from part of an axon, as happens in the central nervous system in multiple sclerosis. Where myelin is lost, local current leaks out through the bare membrane, so less of it reaches the next node. That node takes longer to reach threshold, and conduction slows. If too little arrives, the node never reaches threshold and conduction fails. The bare stretch cannot make up for it, because it was under myelin and has very few voltage gated Na^+ channels of its own. Warming makes this worse. If body temperature rises from 37.0°C (98.6°F) to 37.5°C (99.5°F), ion channels gate faster, the Na^+ channels at each node inactivate sooner, and each action potential is briefer and sends less current on. A healthy node receives several times more current than it needs, so it is unaffected. A node beyond a demyelinated stretch is receiving barely enough, so it fails. Cooling brings it back, and no new damage has occurred.

Change the potassium in the blood. Plasma K^+ is the outside K^+ for every neuron and muscle cell. In the resting potential lecture you calculated that 7 mM moves the resting potential to about -64 mV and 2.5 mM moves it to about -76 mV . Threshold is set by the Na^+ channels and stays at about -55 mV , so what changes is the gap between rest and threshold. Worked problem 3 follows both cases through.

Change the calcium in the blood. Ca^{2+} ions in the extracellular fluid bind to the outside of the voltage gated Na^+ channel, and while they are bound the channel needs a larger depolarization to open. With low extracellular Ca^{2+} the channels open more easily, so threshold moves closer to the resting potential. Nerves begin to fire without being stimulated, and the muscles they supply contract. The result is tingling, cramps and sustained spasm, called tetany. In Chvostek's sign, tapping the facial nerve in front of the ear makes the face twitch. In Trousseau's sign, a blood pressure cuff kept inflated on the arm for about three minutes brings on a spasm of the hand and wrist. With high extracellular Ca^{2+} the reverse happens, and nerve and muscle are less excitable. Students often expect low calcium to weaken muscle. The Ca^{2+} that makes skeletal muscle contract is stored inside the muscle cell, and a low plasma Ca^{2+} hardly changes that store. It also explains why calcium given into a vein helps in hyperkalemia: it moves threshold away from the depolarized resting potential and restores the gap.

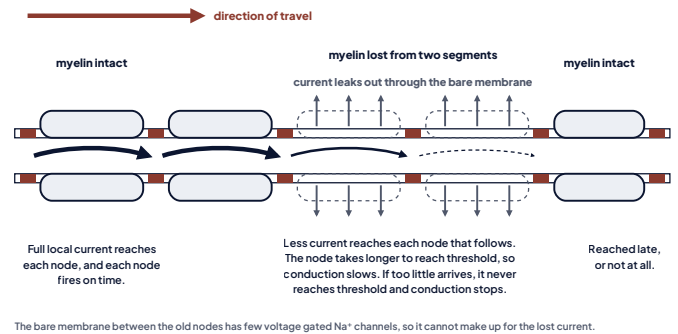


Figure 7. Demyelination. Less current reaches each node beyond the bare stretch.

Plasma potassium moves the resting potential. Threshold stays at about -55 mV .

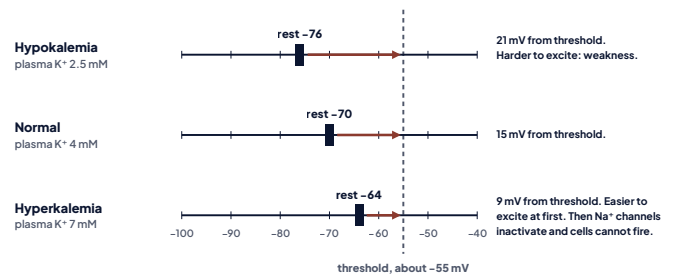


Figure 8. The gap between the resting potential and threshold at three plasma potassium values.

WORKED PROBLEM 3

For a plasma K^+ of 2.5 mM, 4 mM and 7 mM, work out how far the resting potential is from threshold and predict what happens to excitability. Explain both stages of hyperkalemia.

Resting potentials from the resting potential lecture: about -76, -70 and -64 mV. Threshold about -55 mV.

THE WORK

- 1 Normal, 4 mM: the resting potential is about -70 mV and threshold is about -55 mV, so the gap is **15 mV**. A graded potential has to depolarize the trigger zone by 15 mV to fire the neuron.
- 2 Hypokalemia, 2.5 mM: from the resting potential lecture, the resting potential is about -76 mV. The gap to threshold is $-55 - (-76) = 21 \text{ mV}$. The same graded potential that used to reach threshold now falls short, so neurons and skeletal muscle are **less excitable**. The patient is weak.
- 3 Hyperkalemia, 7 mM: the resting potential is about -64 mV. The gap is $-55 - (-64) = 9 \text{ mV}$. At first a smaller graded potential is enough, so cells are **more excitable**.
- 4 Now add what you know about the sodium channel. A membrane that stays depolarized holds some of its voltage gated Na^+ channels in the inactivated state, and an inactivated channel cannot open. The longer and the further the membrane is depolarized, the fewer channels are closed and ready.
- 5 With too few channels ready, the action potential rises slowly and is small, and if enough are inactivated the cell cannot fire at all. So hyperkalemia makes cells more excitable first and then less excitable, and what changes in between is the number of Na^+ channels that are available.
- 6 The Goldman equation gives -71 mV for the normal resting potential with the values from that lecture. It is rounded to -70 mV here. Books also vary by a few millivolts on threshold.

Hyperkalemia

High plasma K^+ . The resting potential depolarizes toward threshold. Cells are more excitable at first, and then Na^+ channels inactivate and cells cannot fire.

Hypokalemia

Low plasma K^+ . The resting potential hyperpolarizes, farther from threshold, and cells are less excitable.

Tetany

Sustained muscle spasm caused by nerves firing on their own, as in low extracellular Ca^{2+} .

Blocking the sodium channel, and the dentist's chair explained

SLIDES 19 TO 21 OF THE DECK

There are two different ways to interfere with Na^+ entry, and they give different results. Lower the Na^+ outside the axon, and E_{Na} falls, so the peak of the action potential is lower, but the channels still open and the action potential is still conducted. Block the channel, and there is no action potential at all. Local anesthetics such as lidocaine cross the membrane, enter the voltage gated Na^+ channel from the inside of the cell and block the pore. Tetrodotoxin blocks the same channel from the outside.

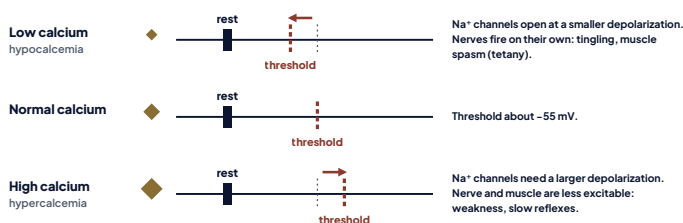
Now the dentist's chair. The pain receptors and the first stretch of axon work normally, so graded potentials still form and action potentials still start and travel toward the brainstem. In the injected stretch the Na^+ channels cannot open. Lidocaine binds most readily to channels that are open or inactivated, so axons that are firing are blocked sooner. Local current from the last working patch still flows into the blocked stretch, but no channels open, so it is only a graded depolarization and it fades within a few millimeters. Nothing reaches the brainstem, so there is no pain.

Small fibers are blocked before large ones. To stop a myelinated axon, the drug has to block about three nodes in a row. In thin A delta fibers the nodes are close together, so a short stretch of drug covers enough of them. In thick A beta fibers, which carry pressure, the nodes are up to a millimeter or two apart, and the same stretch of drug may not cover three. C fibers are thin and have no myelin to keep the drug out. That is why the patient feels pushing and no pain. This order is the usual teaching rule and it is a simplification, because where a fiber lies within the nerve and how often it is firing also matter. Everything returns when the drug diffuses away, because nothing was damaged.

Look at it as an experiment. The problem is that action potentials from the tooth reach the brain. The target variable is the number of Na^+ channels that can open in one stretch of axon. The intervention is a drug that blocks them. The mechanism is that the action potential cannot be regenerated in that stretch. The same mechanism predicts what happens if a large amount of the drug reaches the blood. Every neuron and every heart muscle cell has these channels, so the first effects are in the brain, with numbness around the mouth, ringing in the ears and then seizures, and in the heart conduction slows. This is why a local anesthetic is kept local.

Every perturbation in this lecture acted on the same thing: how many voltage gated Na^+ channels are closed and ready to open. Hyperkalemia inactivates them. Calcium changes how easily they open. A local anesthetic blocks them. Demyelination reduces the current that reaches them.

Extracellular calcium changes how easily voltage gated Na^+ channels open



Drawn to show direction only. The resting potential barely changes, because the resting membrane lets almost no calcium through.

Figure 9. Extracellular calcium changes how easily the Na^+ channels open. Direction only.

Demyelination

Loss of myelin from part of an axon. Current leaks out between nodes, and conduction slows or fails.

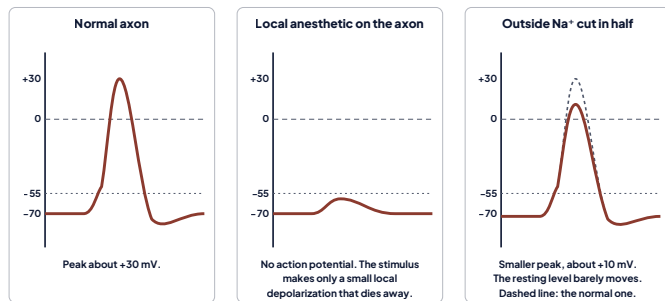


Figure 10. The same axon and the same stimulus under three conditions.

WORKED PROBLEM 4

An axon is bathed in fluid with 72.5 mM Na^+ , half the normal 145 mM. Inside stays at 15 mM. Calculate the new E_{Na} , predict what happens to the peak of the action potential, and compare that with an axon whose Na^+ channels are blocked.

$$E_{\text{ion}} = (61.5 \text{ mV} \div z) \times \log_{10}(\text{outside} \div \text{inside}), \text{ with } z = +1.$$

THE WORK

- Normal sodium: $E_{\text{Na}} = 61.5 \times \log_{10}(145 \div 15) = 61.5 \times 0.985 = +61 \text{ mV}$.
- Half the outside sodium: $E_{\text{Na}} = 61.5 \times \log_{10}(72.5 \div 15) = 61.5 \times 0.684 = +42 \text{ mV}$. Halving the outside concentration lowers E_{Na} by $61.5 \times \log_{10}(2) = 18.5 \text{ mV}$.
- At the peak of an action potential the membrane is far more permeable to Na^+ than to anything else, so the peak follows E_{Na} . When E_{Na} drops by about 19 mV, the peak drops by about the same amount, from about +30 mV to about +10 mV. Hodgkin and Katz did this experiment on the squid giant axon in 1949, and the peak fell by about as much as the Nernst equation predicts.
- The resting potential changes only a little, because the resting membrane is barely permeable to Na^+ . You saw the same thing in the resting potential lecture.
- The action potential is smaller, but it is still there. The channels still open, Na^+ still enters, and each patch of membrane still fires the next. With a local anesthetic or tetrodotoxin the channels cannot open at all, so there is no action potential to conduct.

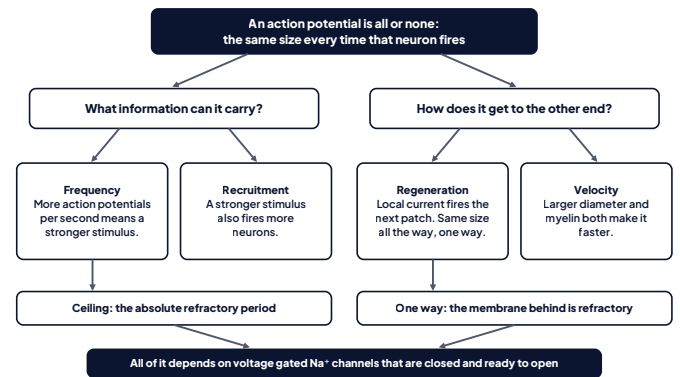


Figure 11. The whole lecture on one map. Read it from the top down.

Local anesthetic

A drug such as lidocaine that blocks voltage gated Na^+ channels from the inside of the cell, so no action potential is conducted past the blocked stretch.

Tetrodotoxin

A toxin that blocks the voltage gated Na^+ channel from the outside.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture.

CHEMISTRY

What an ion is, and the charges on Na^+ , K^+ and Ca^{2+}
Opposite charges attract, and current is a flow of charge
Resistance: the harder it is for charge to flow, the higher the resistance

Chemistry review page

FROM EARLIER LECTURES

The resting membrane potential, the Nernst equation, and how plasma K^+ moves the resting potential (this week)
Neurons and glia: axon, trigger zone, axon terminal, myelin and nodes of Ranvier (this week)
Graded potentials and the action potential: threshold, all or none, the two gates of the Na^+ channel, tetrodotoxin (this week)

The anatomy you need is from the neurons and glia lecture: axon, trigger zone, axon terminal, myelin and node of Ranvier.

MATH

Dividing by a decimal: $1 \div 0.002 = 500$
Changing milliseconds to seconds: $2 \text{ ms} = 0.002 \text{ s}$
 $\text{time} = \text{distance} \div \text{velocity}$
The log button, for worked problem 4

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one. Competencies 19 and 20, the action potential simulation and the

nerve conduction measurement, are done in lab. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

15

Coding of stimulus intensity

Explain how the nervous system encodes stimulus strength when action potential amplitude is fixed.

Do one of the two. Head your paper with the number.

15-1

Draw three stimulus bars of increasing strength across the bottom of your box, weak, medium, and strong. Above each, draw the spike train one sensory neuron sends, as vertical marks on a time line. Keep every spike exactly the same height. One line: say precisely what changed between your three trains, and what could not have changed.

15-2

Draw a patch of skin with a stimulus pressed into it, gentle on the left and hard on the right. In each, draw how many receptors and how many axons get recruited, and draw the spike train on one of those axons underneath. One line: the nervous system has two ways to say louder, so name both from your drawing and say which one a single axon can do on its own.

16

Refractory periods

Distinguish the absolute from the relative refractory period by channel state and explain how the refractory period sets maximum firing frequency and prevents backward conduction.

Do one of the two. Head your paper with the number.

16-1

Draw an action potential tracing and bracket the absolute and relative refractory periods underneath it. Along each bracket, draw the state of the voltage gated sodium channels, inactivated and then recovering, and show whether the potassium channels are still open. Then draw a second stimulus arriving inside each period and show the result. One line: say why no stimulus at all, however large, works during the absolute period.

16-2

Draw an axon with an action potential halfway along it and an arrow for the direction it is traveling. Shade the membrane behind it and mark what state its sodium channels are in. Draw the local current spreading both forward and backward, and show why only one direction succeeds. One line: an absolute refractory period of about 1 to 2 ms sets a ceiling on firing rate, so work out that ceiling in spikes per second.

17

Conduction velocity

Rank axons by conduction velocity using diameter and myelination and contrast continuous with saltatory conduction.

Do one of the two. Head your paper with the number.

17-1

Draw four axons stacked vertically: large myelinated, small myelinated, large unmyelinated, and small unmyelinated. Rank them fastest to slowest and write a rough velocity beside each, using the real range of about 0.5 up to about 120 m per second. One line: say which buys more speed per unit of space taken up, diameter or myelin, and why that matters for a nerve running down a limb.

17-2

Draw one myelinated and one unmyelinated axon of similar diameter side by side, the same length. On each, shade where the membrane actually depolarizes, continuously along one and node to node on the other. Draw the local current loops on both. One line: saltatory means jumping, but the action potential does not really jump, so say what is actually moving between the nodes.

18

Effects of altered extracellular ions

Predict the effect of hyperkalemia and hypokalemia on resting potential and excitability and explain why local anesthetics that block sodium channels abolish the action potential.

Do one of the two. Head your paper with the number.

18-1

Draw a voltage scale from minus 100 to plus 30 mV with the resting potential at minus 70 and threshold at about minus 55 marked on it. Using the Nernst logic for potassium, draw where the resting potential moves in hyperkalemia and in hypokalemia, and show the remaining gap to threshold in each. One line: hyperkalemia first makes the cell more excitable and then less, so say what happens to the sodium channels in between.

18-2

Draw a normal action potential tracing, then draw the same axon after a local anesthetic such as lidocaine has blocked its voltage gated sodium channels, and show what is left of the trace. Beside it, draw the axon in low extracellular sodium and show what changes about the peak, not about the resting level. One line: say why blocking the channel abolishes the signal while low extracellular sodium only shrinks it.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · How action potentials carry information. Velocities, diameters and times are approximate teaching values. All figures are original drawings for this course.

The chemical synapse

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The four worked problems from the slides are written out in full here. Silverthorn chapter 9.

COMPETENCIES 21 TO 23 OF WEEK 4

20 SLIDES · 4 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each problem yourself before you read the answer.

The problem for today

SLIDES 1 TO 2 OF THE DECK

An action potential reaches the end of an axon and stops there. The next cell is across a gap of extracellular fluid, and an electrical signal cannot cross that gap. In this lecture we follow how the signal is turned into a chemical, carried across, and turned back into an electrical signal on the other side. The place where this happens is a chemical synapse. We take the events in strict order, because each step causes the next one, and once you know the order you can predict what happens when any single step fails.

Here is the problem we solve by the end. A five month old girl has been constipated for several days. Now she feeds poorly, her cry is weak, her eyelids droop, and she cannot hold up her head. Within two days she needs a ventilator to breathe. She is alert and she feels every touch. A week earlier her pacifier had been dipped in honey.

Testing narrows it down. She is awake, follows faces and grimaces when her foot is pinched, so her sensory neurons and her brain are working. Action potentials travel down her motor axons at normal speed, so conduction is working. But when a motor nerve is stimulated, the electrical response of the muscle is far smaller than normal. In laboratory studies of muscle poisoned in this way, acetylcholine applied straight onto the muscle still makes it contract normally, so the muscle and its receptors work. Her gut, which is also driven by neurons that release acetylcholine, is hardly moving.

Put those results together. The axon conducts, and the muscle responds when acetylcholine reaches it. So the failure lies between the action potential arriving at the end of the axon and acetylcholine arriving at the muscle's receptors. That stretch is the synapse, and this lecture divides it into nine steps. As you read, decide which step has failed in this baby.

See it move: run one synapse yourself

SLIDE 15 OF THE DECK

This is a working model of the synapse in this lecture. Press Send one action potential and follow the step that is named under the picture. The table on the right reports what happened in numbers, and the graph shows the response of the postsynaptic cell. Then block one step at a time. Each block matches a toxin or drug described later in these notes, so keep the model open and come back to it as you read. The note under the model says which parts are real and which are illustrative.

Moving model: run it on the course site, [drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-synapse-notes.html](https://stack.github.io/human-physiology-Fa26/biol005-w04-synapse-notes.html)

THREE THINGS TO TRY

- 1 Send one action potential with nothing blocked and read the table. Then turn on Botulinum toxin and send another. Calcium still enters, but vesicles released falls from 3 to 0 and the response falls from 1.00 to 0. Now turn that off and block the Ca^{2+} channels. The postsynaptic cell is just as silent, but this time calcium entry reads no. The readouts upstream of the block are what tell you where the sequence stopped.
- 2 Clear the blocks and block the removal enzyme. The peak of the response rises from 1.00 to about 1.80 and it lasts about 39 ms where it lasted about 28 ms before. Block reuptake as well, so that only diffusion is left, and it rises to about 3.69 and lasts about 75 ms. Compare the solid line on the graph with the dashed line each time.
- 3 Clear the blocks, send a steady train, and move the slider from 10 to 100 action potentials per second. The average response over the train rises from about 0.14 to about 1.35, because more action potentials per second release more transmitter per second. Then block the removal enzyme at 100 per second and watch the response stop returning to baseline.

The parts of a chemical synapse

SLIDE 3 OF THE DECK

A synapse is the place where a neuron passes its signal to another cell. The cell sending the signal is the presynaptic cell, and the cell receiving it is the postsynaptic cell, which may be another neuron, a muscle cell or a gland cell. A chemical synapse has three parts.

The first part is the axon terminal of the presynaptic neuron. This is the output zone from the lecture on neurons. It holds synaptic vesicles, which are small membrane sacs filled with neurotransmitter. Some vesicles are docked against the

membrane that faces the next cell, ready to release, and others wait in reserve farther back. The terminal has mitochondria to supply ATP, and its membrane contains voltage gated Ca^{2+} channels, which are closed at rest.

The second part is the synaptic cleft, a gap of extracellular fluid about 20 to 40 nm wide between neurons, and a little wider at the neuromuscular junction. The two membranes do not touch, which is why the action potential cannot continue across.

The third part is the postsynaptic membrane, which holds receptors. A receptor is a protein that binds one particular transmitter. Astrocytes wrap many synapses in the central nervous system, and they take up some transmitters from the cleft, as you saw in the lecture on glia.

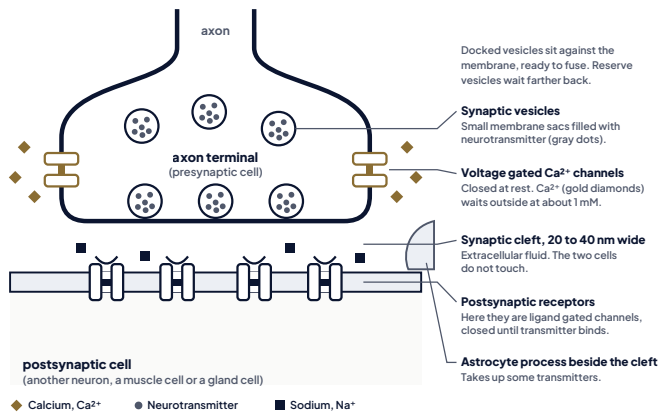


Figure 1. The presynaptic terminal, the synaptic cleft and the postsynaptic membrane. The cleft is drawn much wider than it really is.

Synapse

The place where a neuron passes its signal to another cell.

Presynaptic and postsynaptic

The presynaptic cell sends the signal. The postsynaptic cell receives it.

Synaptic vesicle

A small membrane sac in the axon terminal, filled with neurotransmitter.

Synaptic cleft

The gap of extracellular fluid between the two cells, about 20 to 40 nm wide.

Receptor

A protein in the postsynaptic membrane that binds one particular transmitter and starts the response.

Steps 1 to 5: inside the axon terminal

SLIDES 4 TO 5 OF THE DECK

Step 1. An action potential travels down the axon and depolarizes the membrane of the terminal. Step 2. That depolarization opens voltage-gated Ca^{2+} channels. These channels respond to voltage in the same way as the voltage-gated sodium channels of the axon, but the ion they pass is calcium.

Step 3. Ca^{2+} enters the terminal. Free calcium is about 1 mM outside the cell and about 0.0001 mM inside, which is a 10,000 fold difference, and the inside of the cell is negative. Both the concentration gradient and the voltage drive calcium in. Worked problem 1 puts numbers on this.

Step 4. The Ca^{2+} that enters binds a sensor protein called synaptotagmin on vesicles that are already docked at the membrane. Step 5. That triggers the SNARE proteins. One SNARE protein sits on the vesicle, and its partners sit on the terminal membrane. They wind around each other and pull the two membranes together until they fuse. The inside of the vesicle is now open to the cleft, and the transmitter empties out. Release by fusion of a vesicle with the cell membrane is exocytosis, which was covered in Week 2. Afterward the vesicle membrane is taken back in by endocytosis and refilled.

Calcium is the link between the electrical signal and the chemical one. Without Ca^{2+} entry there is no release, however many action potentials arrive. It also explains how the frequency code from the last lecture is passed on: more action potentials per second let in more Ca^{2+} , more vesicles fuse, and more transmitter is released.

One classic experiment shows that calcium entry is required. Put the terminal in a bath with no calcium and send an action potential. Steps 1 and 2 happen normally, because the channels respond to voltage and not to calcium. Step 3 fails, because there is no Ca^{2+} outside to enter, and nothing after it happens.

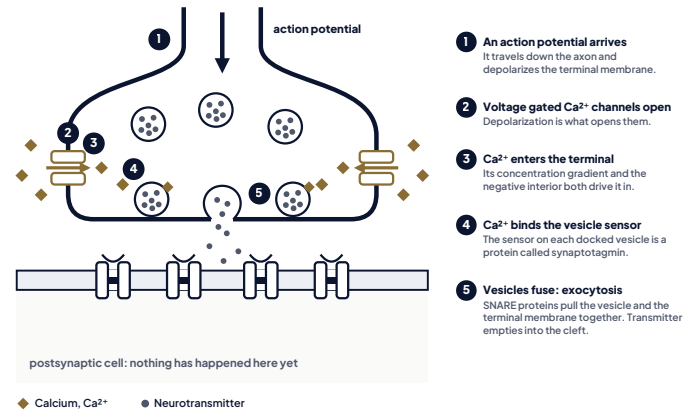


Figure 2. The first five steps. Each one causes the next.

WORKED PROBLEM 1

Free Ca^{2+} is about 1 mM outside the terminal and about 0.0001 mM inside. Find E_{Ca} . Then find the driving force on Ca^{2+} at rest, -70 mV, and at the peak of an action potential, +30 mV, and say which way Ca^{2+} is driven each time.

Use the Nernst equation from the resting potential lecture.
Driving force = membrane potential – equilibrium potential of the ion. For a positive ion, a negative answer means the ion is driven into the cell.

THE WORK

- 1 Write the ratio, outside over inside. $1 \div 0.0001 = 10,000$. The log of 10,000 is 4.
- 2 Calcium carries two positive charges, so $z = +2$. $E_{\text{Ca}} = (61.5 \div 2) \times 4 = 30.75 \times 4 = +123 \text{ mV}$. This is the same value as on the voltage scale in the resting potential lecture.
- 3 The driving force on an ion is the membrane potential minus that ion's equilibrium potential. At rest: $-70 - (+123) = -193 \text{ mV}$. For a positive ion, a negative driving force means the ion is driven **into** the cell.
- 4 At the peak of the action potential: $+30 - (+123) = -93 \text{ mV}$. That is smaller, but it still points inward. The membrane never gets anywhere near +123 mV, so at every moment of an action potential calcium is driven inward.
- 5 So why does calcium not enter at rest, when the driving force is largest? Because its channels are closed. The gradient is always there. The action potential supplies the one thing that was missing, which is an open channel.
- 6 Because there is so little free calcium inside to begin with, a small amount entering raises the concentration right beside the channels a hundredfold or more within a fraction of a millisecond. A signal that starts from almost nothing and rises that fast is easy for the cell to detect, and the vesicle sensor sits right beside those channels.

Voltage gated Ca^{2+} channel

A channel in the terminal membrane that opens when the membrane depolarizes and lets calcium in.

Synaptotagmin

The calcium sensor on a docked vesicle. When calcium binds it, fusion is triggered.

SNARE proteins

Proteins on the vesicle and on the terminal membrane that pull the two membranes together so that they fuse.

Exocytosis

Release of a vesicle's contents by fusion of the vesicle with the cell membrane.

Driving force

Membrane potential minus an ion's equilibrium potential. It tells you how hard, and which way, the ion is driven when a channel opens.

Steps 6 to 9, the delay, and the direction

SLIDES 6 TO 7 OF THE DECK

Step 6. The transmitter diffuses across the cleft. Nothing carries it. Because the gap is only 20 to 40 nm, diffusion takes about a microsecond. Step 7. The transmitter binds receptors on the postsynaptic membrane. Each receptor fits one transmitter, the way a lock fits one key, and the binding is reversible: a transmitter molecule binds, stays for a moment, and comes off again.

Step 8. The postsynaptic cell responds. In the simplest case the receptor is a ligand gated channel, which was covered in the resting potential lecture. With transmitter bound, it opens, Na^+ enters, and the membrane depolarizes a little. That response is a graded potential. Its size depends on how much transmitter was released, it fades with distance, and it has no threshold of its own. Whether it leads to an action potential in the postsynaptic cell is the subject of the next lecture.

Step 9. The transmitter is removed from the cleft. As the concentration in the cleft falls, the receptors empty, the channels close, and the signal ends. The three routes of removal have their own section below.

All of this takes time. The postsynaptic response begins about 0.5 to 1 ms after the action potential reaches the terminal. This is the synaptic delay. Most people guess that the delay is the time taken to diffuse across the cleft, but that is only about a microsecond, less than one percent of it. Most of the delay is spent in the terminal: the voltage gated Ca^{2+} channels take time to open, and then Ca^{2+} has to trigger fusion.

A chemical synapse also works in one direction only. Only the presynaptic side has vesicles ready to release, and only the postsynaptic side has the receptors facing them.

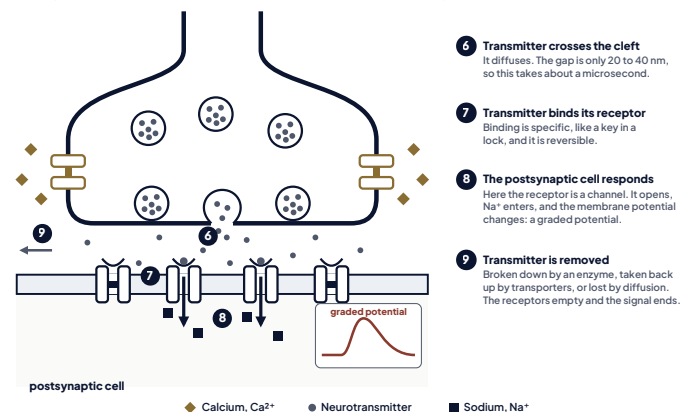


Figure 3. The last four steps. The response in the postsynaptic cell is a graded potential.

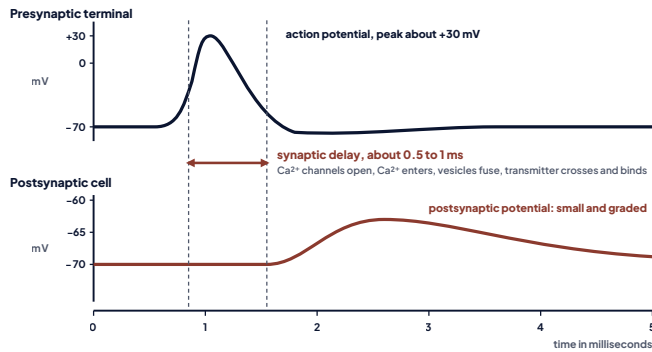


Figure 4. The action potential in the terminal and the response in the postsynaptic cell, on one time axis. Voltages are typical teaching values.

WORKED PROBLEM 2

A pathway has 1 m of axon in total, conducting at 50 m/s. How long does the signal take if the pathway contains 1 synapse? How long if it contains 5?

Use 1 ms for each synaptic delay.

THE WORK

- 1 Conduction time is distance divided by speed. $1 \text{ m} \div 50 \text{ m/s} = 0.02 \text{ s} = \mathbf{20 \text{ ms}}$.
- 2 One synapse adds about 1 ms. Total: $20 + 1 = \mathbf{21 \text{ ms}}$.
- 3 Five synapses add about 5 ms. Total: $20 + 5 = \mathbf{25 \text{ ms}}$. The same length of axon now takes about one fifth longer.
- 4 Put the delay in terms of distance. In 1 ms this axon conducts $50 \text{ m/s} \times 0.001 \text{ s} = 0.05 \text{ m}$. Each synapse costs as much time as **5 cm** of axon, although the cleft itself is only 20 to 40 nm across.
- 5 This is why the fastest reflexes use the fewest synapses. The knee jerk reflex has a single synapse in the spinal cord between the sensory neuron and the motor neuron.
- 6 The delay is a cost of using a chemical synapse. What the cell gets in return is a place where the signal can be made larger or smaller, which is the subject of the next lecture.

Graded potential

A change in membrane potential whose size depends on the strength of the stimulus, here the amount of transmitter. It fades with distance.

Synaptic delay

The time between the action potential reaching the terminal and the start of the postsynaptic response, about 0.5 to 1 ms.

and is over in milliseconds. The nicotinic acetylcholine receptor, the AMPA receptor for glutamate and the GABA-A receptor are examples.

A metabotropic receptor is not a channel. With transmitter bound, it activates a G protein on the inner face of the membrane, which is why it is also called a G protein coupled receptor. The G protein may act on a nearby ion channel directly, or it may switch on an enzyme that makes a second messenger such as cAMP. Second messengers can open channels, close channels, or change other proteins in the cell. More steps make the response slower to start, and it can last from hundreds of milliseconds to minutes. One receptor activates many G proteins, so the signal is also amplified. The muscarinic acetylcholine receptor, every receptor for norepinephrine and dopamine, and most receptors for serotonin are metabotropic.

Now the most important idea in this section. The receptor, and not the transmitter, decides the response. Acetylcholine shows this clearly. At skeletal muscle it binds the nicotinic receptor, whose channel passes both Na^+ and K^+ . A muscle fiber rests near -90 mV , so the driving force on Na^+ is $-90 - 61 = -151 \text{ mV}$, inward, and the driving force on K^+ is $-90 - (-95) = +5 \text{ mV}$, outward. Sodium is driven about 30 times harder, so much more Na^+ enters than K^+ leaves, and the muscle fiber depolarizes and contracts.

At the pacemaker cells of the heart, the same acetylcholine binds the muscarinic receptor. Its G protein opens K^+ channels, K^+ leaves, the cell moves away from threshold, and the pacemaker fires less often, so the heart rate slows. That is one of the actions of the muscarinic receptor in the heart and the easiest one to draw. We return to the others in the cardiac weeks.

So the transmitter is only the message that something has arrived. What the postsynaptic cell does about it depends on which receptor it has put in its membrane and on which ions move as a result. If the result brings the cell toward threshold, the synapse is excitatory. If it holds the cell away from threshold, the synapse is inhibitory. It is like a whistle at a sports field: the same whistle tells one team to start and the other to stop, and the meaning is in the listener.

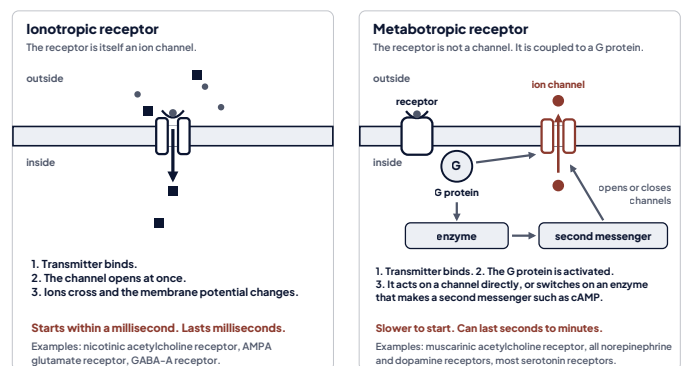


Figure 5. An ionotropic receptor is a channel. A metabotropic receptor works through a G protein.

Two families of receptor, and why the receptor decides

SLIDES 8 TO 9 OF THE DECK

Receptors for neurotransmitters come in two families. An ionotropic receptor is a ligand gated ion channel: the receptor and the channel are the same protein. Transmitter binds, the pore opens, and ions cross. The response begins within a millisecond

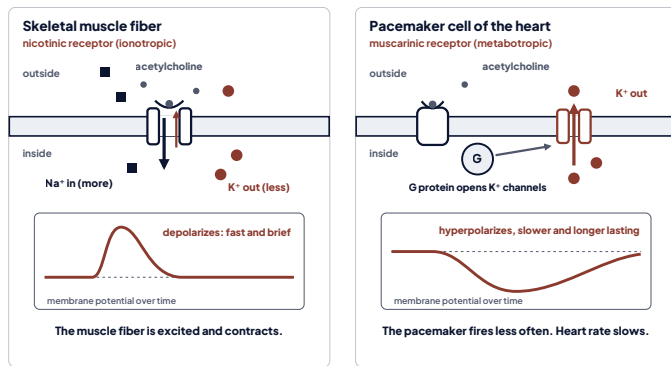


Figure 6. Acetylcholine arriving at two different cells.

Ionotropic receptor

A receptor that is itself an ion channel. Fast and brief.

Metabotropic receptor

A receptor coupled to a G protein. It may act through a second messenger. Slower to start and longer lasting.

Second messenger

A small molecule made inside the cell, such as cAMP, that carries the signal from the receptor to its targets.

Nicotinic and muscarinic receptors

The two families of acetylcholine receptor. Nicotinic receptors are ionotropic. Muscarinic receptors are metabotropic.

Seven classes of neurotransmitter

SLIDES 10 TO 12 OF THE DECK

Silverthorn groups neurotransmitters into seven classes by chemical structure. Acetylcholine is a class of its own. The amines are each made from a single amino acid: dopamine, norepinephrine and epinephrine from tyrosine (these three are the catecholamines), serotonin from tryptophan, and histamine from histidine. The amino acids do most of the fast signaling in the central nervous system: glutamate is the main excitatory transmitter, GABA is the main inhibitory transmitter of the brain, and glycine is the main inhibitory transmitter of the spinal cord. The purines are ATP and adenosine.

Peptides are short chains of amino acids. Substance P excites pain pathways in the spinal cord, and the opioid peptides, the endorphins and enkephalins, inhibit them. Morphine acts on the same receptors. A neuron often releases a peptide together with a small transmitter. The last two classes are unusual. Gases such as nitric oxide, and lipids such as the endocannabinoids, are made at the moment they are needed. They are not stored in vesicles, and they leave the cell by diffusing through its membrane.

The table gives the usual effect of each transmitter and one real place where it acts. Usual effect means the effect at its most common receptors, and the receptor has the final say. The GABA-A receptor and the glycine receptor are Cl^- channels. Opening them holds the membrane away from threshold, and the next lecture explains how. A few more facts are worth knowing. Epinephrine is mainly a hormone from the adrenal medulla and a transmitter in only a few brain pathways. Histamine neurons in the hypothalamus promote wakefulness, which is why older antihistamines cause drowsiness. Adenosine is inhibitory in the brain, and caffeine blocks its receptors.

Where a transmitter is made matters. Acetylcholine, the amines, the amino acids and the purines are small molecules. They are made in the axon terminal and loaded into vesicles there. The enzymes that make them are proteins, so those are built in the cell body and shipped down the axon. Peptides are built on ribosomes like any other protein, so they are made in the cell body, packaged into vesicles there, and carried to the terminal by fast axonal transport along microtubules, which was covered in the lecture on neurons. Nitric oxide is made from the amino acid arginine by an enzyme. It lasts only a few seconds, and it acts on whichever cells it diffuses into.

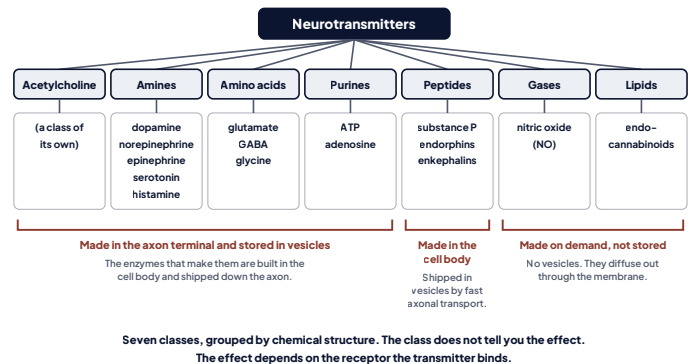


Figure 7. The seven classes, grouped by chemical structure, with where each one is made.

EACH TRANSMITTER, ITS USUAL EFFECT, AND ONE PLACE IT ACTS

TRANSMITTER	USUAL EFFECT	A REAL SITE OF ACTION
Acetylcholine	Excitatory at nicotinic receptors. Excitatory or inhibitory at muscarinic receptors.	Neuromuscular junction (excites skeletal muscle). Heart pacemaker (slows it). Autonomic ganglia and the brain.
Norepinephrine (a catecholamine)	Either one. Every receptor for it is metabotropic.	Released by sympathetic neurons. Speeds the heart through beta-1 receptors and relaxes airway smooth muscle through beta-2 receptors. Arousal pathways in the brain.
Dopamine (a catecholamine)	Either one. Every receptor for it is metabotropic.	Brain: pathways for movement in the basal nuclei, and for reward.
Serotonin	Either one. Most of its receptors are metabotropic.	Brainstem neurons that project widely through the brain: mood, sleep and appetite.
Glutamate	Excitatory. The main excitatory transmitter of the central nervous system.	Brain and spinal cord. Its AMPA and NMDA receptors are cation channels.
GABA	Inhibitory. The main inhibitory transmitter of the brain.	Brain. Its GABA-A receptor is a Cl^- channel, which holds the membrane away from threshold.
Glycine	Inhibitory. The main inhibitory transmitter of the spinal cord.	Spinal cord and brainstem. Its receptor is also a Cl^- channel.
Neuropeptides	Depends on the peptide. All of their receptors are metabotropic.	Substance P excites pain pathways in the spinal cord. Endorphins and enkephalins inhibit them.

WORKED PROBLEM 3

A motor neuron has its cell body in the spinal cord and its terminal in the foot, 1 m away. Fast axonal transport moves vesicles at up to about 400 mm per day. How long does a new supply of a peptide take to arrive? Compare that with the supply of acetylcholine.

Time is distance divided by speed. Keep the units the same.

THE WORK

- Convert the axon length to the units of the transport speed. 1 m = **1,000 mm**.
- Time = distance ÷ speed = 1,000 mm ÷ 400 mm per day = **2.5 days**. That is the fastest case. Any vesicle that moves more slowly takes longer.
- Now acetylcholine. The terminal makes it on the spot from choline and acetyl CoA, and about half of the choline is recovered from the cleft after each release. Emptied vesicle membrane is taken back in by endocytosis and refilled within about a minute.
- So the two supplies are refilled on very different time scales: about a minute for acetylcholine, and days for a peptide.
- Prediction: a neuron that fires rapidly for a long time can run low on its peptide long before it runs low on its small transmitter. A peptide used up at the terminal can only be replaced by the cell body, 2.5 days away.

Catecholamines

Dopamine, norepinephrine and epinephrine. All three are made from the amino acid tyrosine.

Glutamate

The main excitatory transmitter of the central nervous system.

GABA and glycine

The main inhibitory transmitters: GABA in the brain, glycine in the spinal cord.

Neuropeptide

A transmitter that is a short chain of amino acids, made in the cell body and shipped to the terminal.

Ending the signal: three routes of removal

SLIDES 13 TO 14 OF THE DECK

A signal carries information only if it ends. If transmitter stayed in the cleft, the receptors would stay bound, and the postsynaptic cell could not tell one action potential from the next. A doorbell whose button sticks tells you nothing about how many visitors there are. Three routes clear transmitter from the cleft, and each synapse relies mainly on one of them.

The first route is breakdown by an enzyme. Acetylcholinesterase is anchored in the cleft, and it splits acetylcholine into choline and acetate within milliseconds. Neither piece activates the receptor. A transporter carries the choline back into the terminal, where it is joined to acetyl CoA to make new acetylcholine.

The second route is reuptake. Transporter proteins in the presynaptic membrane carry whole transmitter molecules back into the terminal, where they are reloaded into vesicles or broken down. Serotonin, norepinephrine and dopamine are removed mainly this way. Glutamate and GABA are taken up by astrocytes as well as by neurons. These transporters are cotransporters that use the Na^+ gradient, the same kind of secondary active transport covered in Week 2.

The third route is diffusion. Some transmitter always drifts out of the open edges of the cleft. For peptides this is the main route, and peptidases outside the cleft then break them down.

Figure 9 shows what happens to the response to one action potential when removal is blocked, using the model from this lecture. Release is unchanged, so the same amount of transmitter enters the cleft. It then stays longer. Receptors that would have emptied are bound again and again, so the response keeps growing for longer, which makes it taller, and it takes longer to end, which makes it wider. In the model the peak goes from 1.00 to about 1.36 with reuptake blocked and to about 1.80 with the enzyme blocked. Those sizes are illustrative. Which block matters more at a real synapse depends on which route that synapse mainly uses.

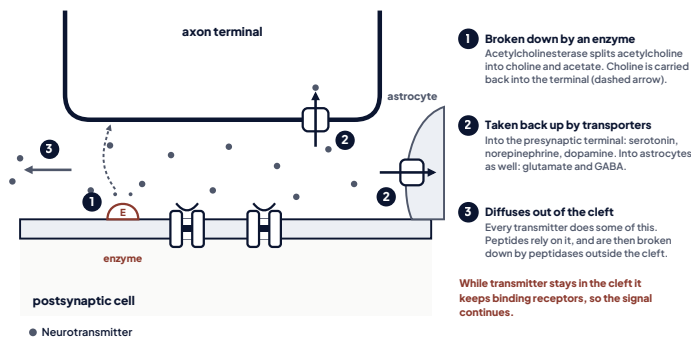


Figure 8. Breakdown by an enzyme, reuptake by transporters, and diffusion out of the cleft.

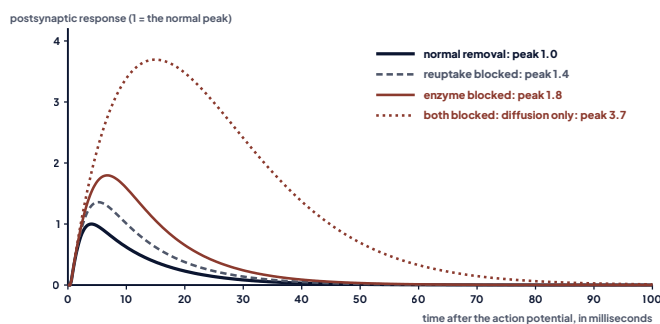


Figure 9. The response to one action potential with each removal route blocked. Relative units from the model. The shapes are illustrative.

Acetylcholinesterase

The enzyme in the cleft that splits acetylcholine into choline and acetate.

Reuptake

Transport of whole transmitter molecules back into the presynaptic terminal, or into glia.

Diffusion

Movement of transmitter out of the cleft down its concentration gradient.

Toxins and drugs as experiments on single steps SLIDES 16 TO 18 OF THE DECK

Much of what is known about synapses came from toxins and drugs that block one step at a time. For each one, find the step it blocks and then predict what the postsynaptic cell receives. Anything that blocks steps 1 to 8 gives the postsynaptic cell less signal. Anything that blocks step 9 gives it more, because removal is the only step whose job is to end the signal.

Curare and botulinum toxin both paralyze skeletal muscle, and they can be told apart by measuring acetylcholine release. With curare, acetylcholine is released in the normal amount, and it is the receptors that are occupied. With botulinum toxin no acetylcholine is released. Blocking acetylcholinesterase helps against curare, because curare competes with acetylcholine for the receptor and more acetylcholine displaces it. It cannot help against botulinum toxin, because there is no acetylcholine to preserve.

One honest note about SSRIs. Blocking the transporter raises serotonin in the cleft within hours, yet the improvement in mood usually takes several weeks. So the longer term changes that neurons make in response to the extra serotonin must matter too, and those are still being worked out. For this course, know the step that the drug blocks.

Worked problem 4 compares botulinum toxin with an organophosphate. They block opposite steps at the same synapses, and most of their signs are opposite. The exception is skeletal muscle, where both end in weakness.

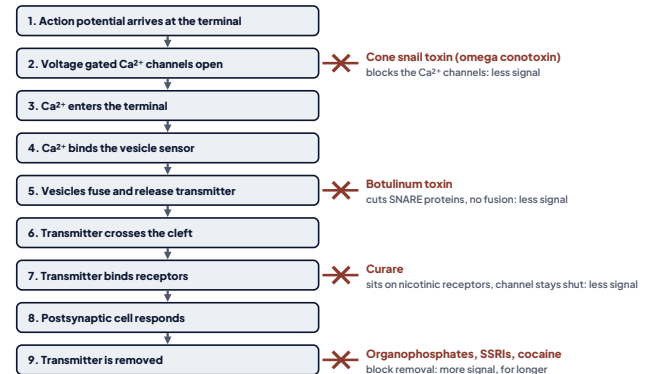


Figure 10. The nine steps from top to bottom, with the step each agent blocks.

SIX AGENTS, THE STEP EACH ONE BLOCKS, AND THE SIGNAL THE POSTSYNAPTIC CELL RECEIVES

AGENT	WHAT IT DOES	STEP	SIGNAL
Omega conotoxin, from cone snail venom	Blocks the voltage gated Ca^{2+} channels of the terminal	2	Less
Botulinum toxin	Cuts SNARE proteins, so vesicles cannot fuse	5	Less
Curare	Occupies nicotinic receptors without opening them	7	Less
Organophosphate insecticides and nerve agents	Inhibit acetylcholinesterase	9	More, for longer
SSRIs, such as fluoxetine and sertraline	Block the serotonin reuptake transporter	9	More, for longer
Cocaine	Blocks the dopamine reuptake transporter	9	More, for longer

WORKED PROBLEM 4

Botulinum toxin blocks release. An organophosphate blocks removal. Both act at every synapse that uses acetylcholine. Explain the signs of each at skeletal muscle, where the receptors are nicotinic, and at glands, gut, pupils and heart, where they are muscarinic, and explain why both end in muscle weakness.

Work out the acetylcholine in the cleft first. Then ask what each kind of target does with no signal, and with a signal that never ends.

THE WORK

- 1** **Botulinum toxin** acts at step 5. No vesicles fuse, so the acetylcholine in the cleft falls toward zero at every synapse that releases acetylcholine.
- 2** Skeletal muscle (nicotinic receptors) receives no signal: the muscles are weak and floppy. This is called flaccid paralysis. Glands and gut (muscarinic receptors) receive no signal either: dry mouth, dry eyes, constipation, and pupils that are wide and slow to react.
- 3** **An organophosphate** acts at step 9. Acetylcholinesterase is inhibited, so acetylcholine stays in the cleft and keeps binding at every one of the same synapses.
- 4** Glands, gut, airways and heart (muscarinic receptors) are stimulated continuously: saliva, tears, sweat and airway secretions increase, the gut is overactive, the airways narrow, the pupils become very small, and the heart rate slows.
- 5** Skeletal muscle (nicotinic receptors) is stimulated continuously too. At first the muscles twitch. Then they become weak. The reason is in the action potential lecture: a membrane that is held depolarized keeps its voltage gated sodium channels inactivated, so the muscle fiber can no longer fire.
- 6** So the two poisons give opposite signs at glands, gut, pupils and heart, and yet both can stop breathing, because a skeletal muscle needs a signal that turns on and then turns off.

The patient, explained

SLIDE 19 OF THE DECK

The baby has infant botulism. Honey can carry spores of the bacterium *Clostridium botulinum*. An adult's gut bacteria keep the spores from growing. An infant's gut bacteria are not yet established, so the spores grew in her large intestine and the bacteria made botulinum toxin. Honey is one known source of the spores. Soil and dust are another, and in many cases the source is never found.

The toxin was absorbed into her blood and taken up by axon terminals that release acetylcholine. It does not reach the brain in meaningful amounts, which is why she stayed alert and could feel. Inside the terminal, the toxin is an enzyme that cuts SNARE

proteins. Action potentials still arrived, Ca^{2+} channels still opened and Ca^{2+} still entered. The vesicles could not fuse, so no acetylcholine was released. Step 5 failed.

Every synapse that releases acetylcholine went quiet. The gut slowed first, which was the constipation. Then skeletal muscles stopped receiving a signal: the weak suck and cry, the drooping eyelids, the floppy neck, and last the diaphragm.

Treat it as an experiment. The variable that failed is the acetylcholine released per action potential. An antibody against the toxin binds toxin that is still in the blood, so it cannot enter any more terminals. It cannot reach toxin that is already inside terminals, so it stops the illness from progressing and does not reverse the weakness that is already there. A ventilator does the work of the diaphragm in the meantime. Recovery depends on the neurons themselves: they make new SNARE proteins and grow new terminal branches, which takes weeks to months. Infants who are supported through that time usually recover fully.

Compare this with organophosphate poisoning, where the opposite step fails. There, removal is blocked, acetylcholine stays in the cleft, and the signs at glands, gut, pupils and heart are the opposite ones.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture.

CHEMISTRY

Proteins change shape when something binds to them. That is how a receptor or a channel opens.

An enzyme speeds up one reaction and is not used up by it.

Diffusion moves molecules from higher to lower concentration, and it is fast only over very short distances.

Chemistry review page

FROM EARLIER LECTURES

Resting potential lecture: E_{Ca} is about +123 mV, free Ca^{2+} is about 1 mM outside and 0.0001 mM inside, and ligand gated channels open when a molecule binds.

Neurons and glia lecture: the axon terminal is the output zone, fast axonal transport carries vesicles from the cell body, and astrocytes take up transmitter.

Action potential lectures: voltage gated channels, inactivation of sodium channels, graded potentials, and the frequency code.

Week 3: exocytosis, endocytosis and secondary active transport.

No new anatomy is needed for this lecture, so there is no anatomy list this time.

MATH

The Nernst equation with $z = +2$, which needs a calculator with a log button.

Time = distance ÷ speed, keeping the units the same.

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

21

Sequence at a chemical synapse

Order the events of chemical synaptic transmission from action potential arrival through calcium entry and vesicle fusion to receptor binding and postsynaptic response.

Do one of the two. Head your paper with the number.

21-1

Draw one axon terminal big enough to fill half your space, with the cleft and a patch of postsynaptic membrane. Number the events in order from the action potential arriving to the postsynaptic channel opening, one arrow per step, and show where calcium enters and what it does once inside. Then redraw the terminal sitting in a calcium free bath. One line: name the step where the chain stops, and say what still happens normally upstream of it.

21-2

Draw two traces stacked on one shared time axis in milliseconds, the presynaptic action potential on top and the postsynaptic membrane potential below. Mark the synaptic delay between them, about 0.5 ms. In the gap under that delay, draw the events that are happening while nothing shows on either trace. One line: which single step you think eats most of that delay, and why.

22**Neurotransmitter classes**

Group neurocrines into their seven chemical classes, acetylcholine, amines, amino acids, peptides, purines, gases, and lipids, and match acetylcholine, the catecholamines, serotonin, glutamate, GABA, glycine, and the neuropeptides to their usual excitatory or inhibitory effect and to a site of action.

Do one of the two. Head your paper with the number.

22-1

Draw a rough brain, a spinal cord, and one peripheral effector, side by side. Place acetylcholine, a catecholamine, serotonin, glutamate, GABA, glycine, and a neuropeptide at a real site of action on your drawing. Flag each one excitatory or inhibitory with an up or down arrow drawn on the postsynaptic membrane. One line: name the one you had to flag both ways, and say what forced that.

22-2

Draw acetylcholine arriving at two postsynaptic membranes side by side, a skeletal muscle fiber with nicotinic receptors and a cardiac pacemaker cell with muscarinic receptors. Show the ion moving at each and draw the membrane potential trace under each. Same transmitter, opposite result. One line: given both drawings, state what actually decides whether a synapse is excitatory or inhibitory.

23**Neurotransmitter removal**

Name the three routes that clear a neurotransmitter from the synaptic cleft, degradation, reuptake, and diffusion, and give a drug that blocks one of them.

Do one of the two. Head your paper with the number.

23-1

Draw a synaptic cleft with transmitter still in it, then draw all three exit routes off that cleft, enzymatic breakdown, reuptake, and simple diffusion. Show where each one physically happens, on the postsynaptic side, on the presynaptic side, or out the edges. Name the transmitter you picked and put a real drug on the route it blocks. One line: what the postsynaptic cell experiences once that route is shut.

23-2

Draw the postsynaptic potential from one normal release event, membrane potential on the y axis and time on the x. On the same axes draw that trace again under two drugs, one that blocks degradation and one that blocks reuptake. Show what changes, the height, the width, or both. One line: which drug you expect to change the trace more, and what you would need to measure to be sure.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · The chemical synapse. Concentrations and voltages are typical teaching values. All figures are original drawings for this course.

Integration at the synapse

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it afterward to check what you remember. The three worked problems from the slides are written out in full here. Silverthorn chapter 9.

COMPETENCIES 24 TO 28 OF WEEK 4

23 SLIDES · 3 WORKED PROBLEMS · 1 MOVING MODEL

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each problem yourself before you read the answer.

The problem for today

SLIDES 1 TO 2 OF THE DECK

A single neuron in your spinal cord receives thousands of synapses. Some make it more likely to fire and some make it less likely, and at every moment the cell has one decision to make: send an action potential down its axon or stay silent. This lecture is about how that decision is made. We start with the two kinds of small potential a synapse can produce, the EPSP and the IPSP. Then we add them up at the trigger zone, with numbers. Last, we look at three ways the rules can be changed: inhibition of a single terminal, synapses that strengthen with use, and electrical synapses that use no transmitter at all.

Here is the problem we solve by the end. A gardener steps on a rusty nail. A week later his jaw is clamped shut, and the slightest noise or touch throws his whole body into painful spasms. He is awake through all of it. His muscles, and the motor neurons that drive them, are structurally normal. A bacterial toxin from the wound has reached his spinal cord.

Look at what the examination finds. He feels touch, sound and pain normally. He is awake and thinking clearly. His motor neurons carry action potentials, and when a motor neuron fires, its muscle contracts forcefully. Every part that excites a muscle is working. The one thing he has lost is the ability to keep muscles relaxed: a small stimulus makes opposing muscles contract together. Before you read further, write down a prediction. Is the toxin exciting his motor neurons, or is it removing something that normally holds them back?

See it move: will it fire?

SLIDE 13 OF THE DECK

This is a working model of the neuron in this lecture. It has four excitatory inputs, E1 to E4, at different distances along its

dendrites, one inhibitory input on a dendrite, I1, and one inhibitory input on the cell body, I2. The tracing on the right is the membrane potential at the trigger zone, with dashed lines at rest, -70 mV, and at threshold, -55 mV. Press the numbered cases in order, or choose which inputs fire yourself. The readouts beside the model give the highest point the trigger zone reached, how far that was from threshold, and whether the cell fired, and the sentence under the model explains why. Keep it open as you read and come back to it after each section.

Moving model: run it on the course site, [drsrennie-stack.github.io/human-physiology-Fa26/biol005-w04-synaptic-integration-notes.html](https://stack.github.io/human-physiology-Fa26/biol005-w04-synaptic-integration-notes.html)

THREE THINGS TO TRY

- 1 Press case 2, temporal summation. Then move the slider for the time between action potentials from 5 ms to 10 ms and to 25 ms. Write down the highest point each time and compare your numbers with worked problem 2. The model runs slightly higher than the hand calculation, because a real EPSP takes a moment to rise.
- 2 Press case 3, spatial summation, and then slide E4 from 400 out to 800 micrometers. The cell stops firing, and nothing changed except where one synapse sits. That is case C of worked problem 3.
- 3 Press case 4, which adds the inhibitory synapse on the cell body. Read the highest point. Then press Remove all inhibition and watch the same four EPSPs reach threshold. That is the patient in this lecture. Last, turn all the excitatory inputs off, leave I2 on, and move the chloride slider from 10 mM down to 5 mM to see the same synapse change from a small depolarization to a hyperpolarization.

Two kinds of postsynaptic potential

SLIDES 3 TO 5 OF THE DECK

The last lecture ended with one synapse: transmitter binds a receptor, a channel opens, and the postsynaptic membrane potential changes a little. That small change is a graded potential. From the graded potential lecture you know two things about it: it gets smaller as it spreads, and graded potentials can add. You also know that a neuron fires only if the membrane at its trigger zone, where the axon leaves the cell body, reaches threshold, about -55 mV. Synaptic integration is the name for the whole process in between: the neuron adds all of its excitatory and inhibitory inputs and gives one answer.

An excitatory postsynaptic potential, or EPSP, is a small depolarization. At most fast excitatory synapses in the brain, glutamate opens the AMPA receptor, which is a channel for small positive ions. Both Na^+ and K^+ can pass. At -70 mV the force on Na^+ is 131 mV inward, because E_{Na} is $+61$ mV, and the force on K^+ is only 25 mV outward, because E_{K} is -95 mV. So far more Na^+ enters than K^+ leaves, net positive charge comes in, and the membrane depolarizes. If the channel stayed open, the

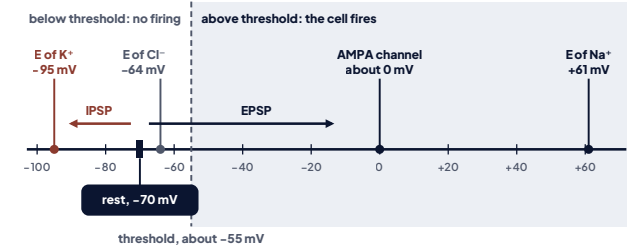
membrane would settle near 0 mV, which is well above threshold. That is what makes this synapse excitatory.

An inhibitory postsynaptic potential, or IPSP, makes the cell less likely to fire. There are two common kinds. One opens K^+ channels. K^+ leaves, and the membrane hyperpolarizes toward E_K , -95 mV. The GABA-B receptor does this through a G protein, so it is a metabotropic receptor from the last lecture, and its IPSP is slow. The other kind opens Cl^- channels. The GABA-A receptor and the glycine receptor are both Cl^- channels that open directly when the transmitter binds, so they are ionotropic and fast. Note that the transmitter does not decide whether a synapse excites or inhibits. The receptor and its channel decide. Acetylcholine excites skeletal muscle through one receptor and slows the heart through another.

Now the part that needs care. Use the rule from the resting potential lecture: opening channels for an ion moves the membrane toward that ion's equilibrium potential. With our concentrations, 110 mM chloride outside and 10 mM inside, E_{Cl} is -64 mV, which is a little above the resting potential of -70 mV. So opening Cl^- channels in our cell moves the membrane only slightly, and it moves it up, toward -64 mV. It is still inhibition, for two reasons. First, the open channels hold the membrane near -64 mV, which is 9 mV below threshold, and the membrane cannot be carried past -64 mV by chloride. Second, when an EPSP lifts the membrane above -64 mV, Cl^- enters through the open channels and cancels part of the positive charge that the EPSP brought in, so the EPSP is smaller than it would have been. This second effect is called shunting inhibition, because the open channels give the excitatory current a path out of the cell.

Cells and books vary on this point, and you should know why. Many adult neurons have a transporter that keeps Cl^- inside nearer 5 to 7 mM. With 110 mM outside, the Nernst equation then gives an E_{Cl} of about -83 to -74 mV, which is below rest, and opening Cl^- channels lets Cl^- in and hyperpolarizes the cell. Silverthorn describes the IPSP that way. Both descriptions are correct for the cells they describe. The general rule covers every case: a synapse is excitatory if its channel moves the membrane toward a value above threshold, and inhibitory if that value is below threshold.

Where each kind of channel moves the membrane, compared with threshold



A synapse is excitatory if its channel moves the membrane toward a value above threshold. It is inhibitory if that value is below threshold.

Figure 2. The equilibrium potentials from the first lecture this week, with threshold added.

Synaptic integration

The process by which a neuron adds all of its excitatory and inhibitory inputs and produces one output: fire or do not fire.

EPSP

Excitatory postsynaptic potential. A small, graded depolarization, usually from a cation channel such as the AMPA receptor letting Na^+ in.

IPSP

Inhibitory postsynaptic potential. A graded potential that makes firing less likely, from opening K^+ channels (hyperpolarizes toward -95 mV) or Cl^- channels (holds the membrane near E_{Cl}).

Shunting inhibition

Inhibition by open channels that let excitatory current leave the cell, so EPSPs are smaller even though the membrane potential itself hardly moved.

GABA-A and glycine receptors

Ionotropic receptors that are Cl^- channels. GABA is the main fast inhibitory transmitter in the brain and glycine in the spinal cord.

GABA-B receptor

A metabotropic receptor that opens K^+ channels through a G protein and gives a slow IPSP.

Working backwards from a recording

SLIDE 6 OF THE DECK

This problem is the reverse of the last section. You are given the recording and asked for the channel. It is worth doing carefully, because it is exactly what you will do in lab with the synapse simulation.

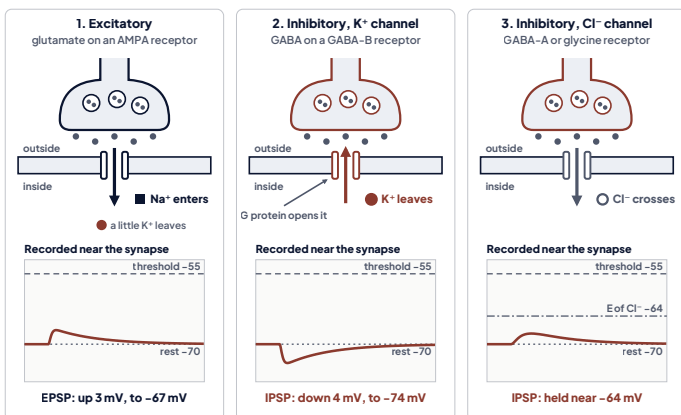


Figure 1. Three synapses and the potential each one produces. Sizes are illustrative. One synapse alone usually gives less than 1 mV.

WORKED PROBLEM 1

An electrode in a neuron records three traces on different trials. Each starts at -70 mV. The first rises 3 mV, the second falls 4 mV, and the third stays flat. For each, name the channel that opened and the ion that moved. Give the flat trace two different explanations, and say what second recording would tell them apart.

You need $E_K = -95$ mV, $E_{Cl} = -64$ mV with our numbers and lower in many neurons, the AMPA channel at about 0 mV, and $E_{Na} = +61$ mV.

THE WORK

- 1 The trace that rises 3 mV.** The membrane moved toward a value above -70 mV, and the channel that does this at a fast excitatory synapse is a cation channel such as the AMPA receptor. It passes Na^+ and K^+ . At -70 mV the force on Na^+ is 131 mV inward and the force on K^+ is only 25 mV outward, so far more Na^+ enters than K^+ leaves. Net positive charge enters, and the trace rises. This is an EPSP.
- 2 The trace that falls 4 mV.** The membrane moved toward a value below -70 mV. With our numbers only E_K , at -95 mV, is below rest, so a K^+ channel opened and K^+ left the cell. A GABA-B receptor does this through a G protein. In a neuron that keeps less chloride inside, a Cl^- channel could give the same trace, with Cl^- entering. Either way it is an IPSP.
- 3 The flat trace, explanation one.** Nothing happened at the synapse. The presynaptic neuron did not fire, or it fired and released no transmitter, so no channel opened.
- 4 The flat trace, explanation two.** A Cl^- channel opened in a cell whose chloride equilibrium potential is equal to its resting potential. There is no net force on Cl^- at that voltage, so no net charge crosses and the voltage does not move, even though the channels are open.
- 5 The second recording that separates them.** Fire a nearby excitatory input at the same moment and compare the EPSP with the EPSP you get from that input alone. If the quiet synapse opened no channels, the EPSP is its usual size. If it opened Cl^- channels, the EPSP is smaller, because as soon as the membrane rises above E_{Cl} , Cl^- enters and cancels part of the positive charge. That is how an input can be inhibitory while the membrane potential barely moves.

about 15 mV above rest. If nothing were lost on the way, it would take about 30 of those EPSPs arriving together to fire the cell. This is very different from the neuromuscular junction, where one action potential in the motor neuron always fires the muscle fiber. In the central nervous system no single synapse decides anything. A spinal motor neuron carries on the order of 10,000 synapses, and its output depends on many of them at once.

The arithmetic of integration can be written in one line. The potential at the trigger zone equals -70 mV, plus the EPSPs arriving, minus the IPSPs arriving, and the cell fires if that value reaches about -55 mV. Arriving means the size of each potential when it reaches the trigger zone, after it has shrunk with distance and with time. Adding potentials as plain numbers is a simplification that works well for small potentials. Real neurons add a little less than the arithmetic says, because each EPSP reduces the force that drives the next one.

There are two ways to add. In temporal summation, one presynaptic neuron fires several times in quick succession. Each EPSP begins before the one before it has decayed, so each starts from a higher level. In spatial summation, several presynaptic neurons fire at about the same moment and their EPSPs arrive at the trigger zone together. Real neurons use both at once: many inputs, each firing a train of action potentials. An EPSP and an IPSP that arrive together cancel, in whole or in part, in the same sum.

Temporal summation depends entirely on timing. After its peak, an EPSP falls by half roughly every 10 ms in the teaching values used here, because charge leaks back out through the membrane. If the next action potential arrives after 25 ms, less than a fifth of the first EPSP is left and almost nothing is added. If it arrives after 5 ms, about 71 percent is left. This is how the frequency code from the lecture on action potentials is read by the next cell. A faster train of action potentials in one axon produces a larger depolarization at the trigger zone of the cell it synapses on.

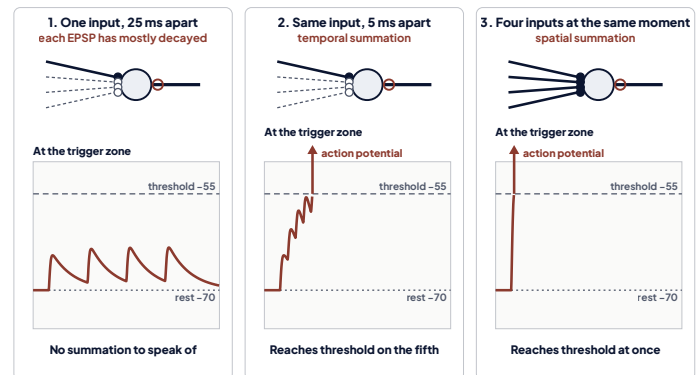


Figure 3. Tracings calculated with the same values as the moving model. Each EPSP is about 5 mV at the trigger zone.

One EPSP is small, so EPSPs have to add

SLIDES 7 TO 9 OF THE DECK

A single excitatory synapse on a neuron in the central nervous system typically produces an EPSP of roughly 0.5 mV at the cell body, and the value varies from synapse to synapse. Threshold is

WORKED PROBLEM 2

One excitatory input gives an EPSP of 5.4 mV at the trigger zone. After its peak, an EPSP falls by half about every 10 ms. The neuron rests at -70 mV and its threshold is -55 mV. Does the cell fire if this one input fires every 25 ms? Every 10 ms? Every 5 ms?

The peak after each new EPSP is 5.4 mV plus whatever is left of the earlier ones. What is left is one half after 10 ms, about 0.71 after 5 ms and about 0.18 after 25 ms.

THE WORK

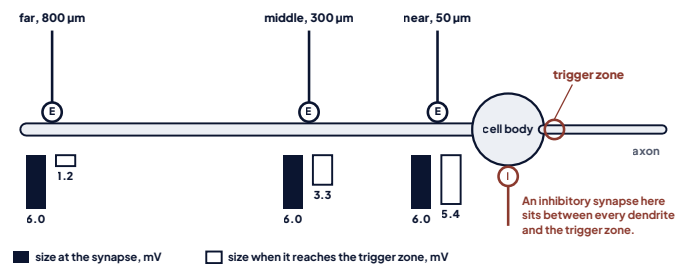
- 1 How far is threshold?** From -70 mV to -55 mV is 15 mV. One EPSP of 5.4 mV reaches -64.6 mV, which is 9.6 mV short.
- 2 Every 25 ms.** After 25 ms an EPSP has been halved two and a half times, so less than one fifth of it is left, about 1 mV. The second EPSP peaks near 6.4 mV above rest, and the later ones go no higher than about 6.6 mV. The cell does not fire however long the train continues.
- 3 Every 10 ms.** Exactly half is left each time. The peaks are 5.4, then $5.4 + 2.7 = 8.1$, then $5.4 + 4.05 = 9.5$, then 10.1 mV, and they level off below $5.4 \times 2 = 10.8$ mV. That is still 4 mV short of threshold.
- 4 Every 5 ms.** After 5 ms about 71 percent is left, because $0.71 \times 0.71 = 0.5$. The peaks are 5.4, 9.2, 11.9, 13.8 and then 15.2 mV. The fifth EPSP carries the trigger zone past 15 mV, and the cell fires.
- 5 What has to be true about timing.** The next EPSP has to arrive before the last one has decayed. The membrane potential takes roughly 10 ms to fall by half, so action potentials more than a few tens of milliseconds apart do not add, and action potentials a few milliseconds apart do. This is how the frequency code from the last lecture is read by the next cell: a faster train in one axon produces a larger depolarization at the trigger zone.

dendrite therefore has less effect at the trigger zone than the same synapse close to the cell body. In this lecture we use one teaching rule for that loss: the fraction that arrives is e raised to the power of minus the distance divided by 500 micrometers. With that rule, a 6 mV EPSP arrives as 5.4 mV from 50 micrometers, 3.3 mV from 300 micrometers and 1.2 mV from 800 micrometers. Real dendrites differ from one neuron to the next, and some have voltage gated channels that boost distant inputs, so treat the rule as an illustration of the direction of the effect.

Location matters for inhibition too. Inhibitory synapses are often found on the cell body and on the first part of the axon. Every EPSP from every dendrite has to pass that point on its way to the trigger zone, so one well placed inhibitory synapse can cancel the combined effect of many excitatory ones.

Two wiring patterns appear all through the nervous system. In divergence, the axon of one neuron branches and synapses on many target cells, so one signal reaches many places. In convergence, many presynaptic neurons synapse on one target cell, which integrates all of them. A spinal motor neuron is a good example of convergence: it receives sensory input, input from the brain and input from local inhibitory interneurons, and its one output decides whether a muscle contracts. The largest example in the body is the Purkinje cell of the cerebellum, which receives on the order of 100,000 to 200,000 synapses. That number is approximate.

The same 6 mV EPSP, started at three different distances from the cell body



Sizes use the teaching rule in this lecture: a graded potential shrinks by the factor $e^{-(\text{distance} / 500 \mu\text{m})}$.

Figure 4. Three identical synapses at different distances, and one inhibitory synapse on the cell body.

FOUR EXCITATORY INPUTS, EACH 6 MV AT ITS OWN SYNAPSE, AND ONE INHIBITORY INPUT

INPUT	DISTANCE FROM THE CELL BODY	FRACTION THAT ARRIVES	SIZE AT THE TRIGGER ZONE
E1	50 μm	0.90	+5.4 mV
E2	150 μm	0.74	+4.4 mV
E3	350 μm	0.50	+3.0 mV
E4	400 μm	0.45	+2.7 mV
I1 (-5 mV at its synapse)	250 μm	0.61	-3.0 mV

Temporal summation

Adding of graded potentials that arrive one after another from the same input, close enough in time that each begins before the last has decayed.

Spatial summation

Adding of graded potentials that arrive at the trigger zone at the same time from different synapses.

Trigger zone

The region where the axon leaves the cell body, rich in voltage gated Na^+ channels. It is where the sum of all inputs is compared with threshold.

Location, and a full worked problem

SLIDES 10 TO 12 OF THE DECK

A graded potential loses size as it spreads, because charge leaks out through the membrane along the way. A synapse far out on a

WORKED PROBLEM 3

Using the table above, with rest at -70 mV and threshold at -55 mV, decide whether the cell fires in each case. A: E1 alone. B: all four excitatory inputs together. C: the same four, with E4 moved out to 800 micrometers, where the fraction that arrives is 0.20. D: case B plus I1. E: case D, but the inhibitory terminal cannot release its transmitter.

Add the sizes at the trigger zone to -70 mV and compare the result with -55 mV.

THE WORK

- 1 **Case A, E1 alone.** $-70 + 5.4 = -64.6$ mV. That is 9.6 mV short of threshold, so the cell does not fire.
- 2 **Case B, all four excitatory inputs at the same moment.** $5.4 + 4.4 + 3.0 + 2.7 = 15.5$ mV. $-70 + 15.5 = -54.5$ mV. That is above -55 mV, so the cell fires. This is spatial summation.
- 3 **Case C, the same four, with E4 moved out to 800 μm .** E4 now arrives as $6 \times 0.20 = 1.2$ mV. The sum is $5.4 + 4.4 + 3.0 + 1.2 = 14.0$ mV, which gives -56.0 mV. The cell does not fire. Nothing about the synapse changed except where it sits.
- 4 **Case D, all four as in case B, plus I1.** I1 makes -5 mV at 250 μm and arrives as $-5 \times 0.61 = -3.0$ mV. The sum is $15.5 - 3.0 = 12.5$ mV, which gives -57.5 mV. The cell does not fire. One inhibitory input has canceled the effect of four excitatory ones.
- 5 **Case E, case D with the inhibitory terminal unable to release.** The -3.0 mV is gone, the sum is back to 15.5 mV, and the cell fires again. Keep this case in mind for the patient.
- 6 **The simplification.** Adding potentials as plain numbers is close to right for small potentials and is how these problems are worked in this course. In a real neuron the sum comes out a little smaller than the arithmetic, and an inhibitory synapse that opens Cl^- channels on the cell body removes more than its own size suggests. The model on slide 13 shows that case.

ONE MORE TO TRY ON YOUR OWN

A neuron has three excitatory inputs worth about 4 mV each at the trigger zone and two inhibitory inputs worth about 5 mV each. Can it ever reach threshold?

All three excitatory inputs fired once, at the same moment, give 12 mV, which is 3 mV short. So single volleys cannot do it, even with no inhibition. Temporal summation can. If all three fire twice, 5 ms apart, the second volley adds 12 mV to the $0.71 \times 12 = 8.5$ mV that is left, which is 20.5 mV and well past threshold. Add one inhibitory input at the time of the second volley and the total falls to 15.5 mV, which still fires. Add both and it falls to 10.5 mV, which does not.

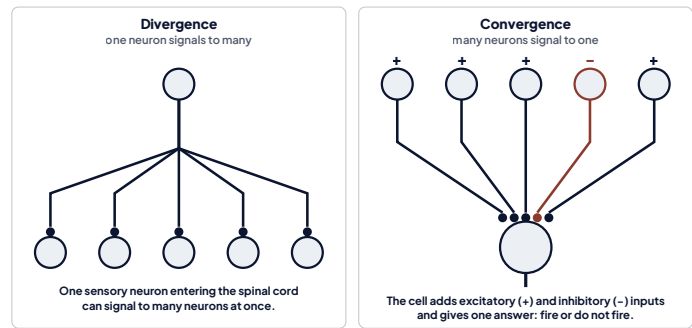


Figure 5. Divergence and convergence.

Divergence

One neuron synapsing on many target cells.

Convergence

Many neurons synapsing on one target cell, which integrates them.

Presynaptic inhibition and facilitation

SLIDES 14 TO 15 OF THE DECK

Everything so far has acted on the receiving cell. There is another place to act. In presynaptic inhibition a modulatory neuron ends on the axon terminal of another neuron, which is called an axoaxonic synapse. Its transmitter reduces the amount of Ca^{2+} that enters that terminal when an action potential arrives. From the last lecture you know that Ca^{2+} entry is what makes vesicles fuse. Less Ca^{2+} entering means fewer vesicles fuse, less transmitter is released, and the EPSP from that one input is smaller. The postsynaptic cell has not changed at all, so it responds to its other inputs exactly as before.

Compare that with postsynaptic inhibition, which is the IPSP from the first section. It acts on the receiving cell, so it reduces the effect of every input to that cell. The advantage of presynaptic control is selectivity: one input can be turned down while the others are left alone. If you needed to cut the effect of input A by half and leave inputs B and C untouched, an inhibitory synapse on the terminal of A would do it. An inhibitory synapse on the cell body would fail the requirement, because it would lower B and C by just as much.

Presynaptic facilitation is the opposite change at a terminal. The modulatory transmitter keeps the terminal depolarized a little longer when an action potential arrives, more Ca^{2+} enters, and more transmitter is released. It was worked out in the sea slug *Aplysia*, where serotonin closes K^+ channels in a sensory terminal.

The standard example of presynaptic inhibition in the human body is pain. Pain fibers make their first synapse in the dorsal horn of the spinal cord. Interneurons there release enkephalins, the body's own opioids, onto the terminals of the pain fibers. Opioid receptors on those terminals reduce Ca^{2+} entry, less transmitter is released, and less of the pain signal reaches the next neuron. Morphine and the other opioid drugs act at the same receptors. Opioids also act at other sites, including on the postsynaptic cell and in the brain, so this is one part of how they work and not the whole of it.

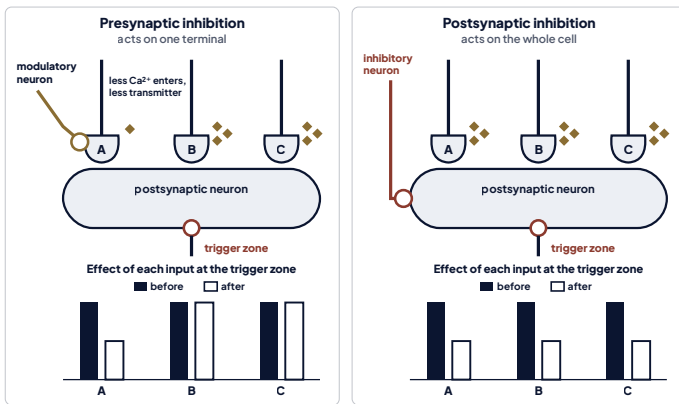


Figure 6. The same three inputs, inhibited in two different places.

Axooaxonic synapse

A synapse made by one axon terminal onto another axon terminal.

Presynaptic inhibition

A decrease in transmitter release from one terminal, caused by a modulatory input that reduces Ca^{2+} entry into that terminal. It is selective for that one input.

Presynaptic facilitation

An increase in transmitter release from one terminal, caused by a modulatory input that increases Ca^{2+} entry.

Postsynaptic inhibition

An IPSP in the receiving cell. It reduces the effect of all inputs to that cell.

Synaptic plasticity

SLIDES 16 TO 18 OF THE DECK

Synaptic plasticity means that the strength of a synapse depends on its history. The same presynaptic action potential can produce a larger or a smaller EPSP than it did before. Some of these changes are brief. In facilitation, each EPSP in a short rapid train is larger than the one before, because Ca^{2+} from one action potential has not been cleared from the terminal when the next arrives, so more vesicles fuse. In depression, the EPSPs get smaller during a long rapid train, because the terminal is using vesicles faster than it can refill them. Both changes are in the presynaptic terminal, and both fade within seconds to minutes after the train stops. A memory lasts far longer than that, so there must also be changes that last.

Long term potentiation, or LTP, is an increase in synaptic strength that lasts hours or longer. It was discovered in the hippocampus, a part of the brain needed for forming new memories, and it is best understood at glutamate synapses. Take it one step at a time. Glutamate binds two receptors on the postsynaptic membrane. The AMPA receptor opens at once and produces the ordinary EPSP. The NMDA receptor also binds glutamate, but near the resting potential its channel is plugged by a Mg^{2+} ion from the outside fluid, which the negative interior holds in the pore. So after one action potential, the NMDA receptor has glutamate on it and passes almost nothing.

A rapid burst of input changes that. Many EPSPs add, by the temporal and spatial summation you have already worked through, and the membrane depolarizes strongly. A less negative interior no longer holds the positive Mg^{2+} ion, so the plug leaves

the pore. With glutamate still bound, the NMDA channel is now open, and it lets Ca^{2+} in as well as Na^{+} . Ca^{2+} acts as a second messenger. It activates protein kinases in the postsynaptic cell, and these add more AMPA receptors to the membrane and make the existing ones pass more current. The postsynaptic cell may also send a signal back across the cleft so that the terminal releases more glutamate. Changes that last for days also need new proteins to be made.

The result is that the same single action potential now produces a larger EPSP. Notice why one action potential alone cannot do this. The NMDA receptor opens only when two things are true together: glutamate is bound, which says the presynaptic cell fired, and the membrane is already strongly depolarized, which says the postsynaptic cell was strongly driven at that moment. One EPSP of half a millivolt does not remove the Mg^{2+} .

In the standard experiment, the EPSP is tested with a single action potential once a minute. After one burst of rapid stimulation it jumps, settles at a larger size than before and stays there for as long as the recording lasts. If NMDA receptors are blocked with a drug, ordinary transmission is normal, because AMPA receptors carry it. The burst still causes a brief rise, which is short term plasticity in the terminal, but the lasting increase does not happen. So the AMPA receptor carries the ordinary, moment to moment signal, and the NMDA receptor carries the signal that something should change.

Long term depression, or LTD, is the reverse. Prolonged slow stimulation lets in a small amount of Ca^{2+} , which activates a different set of enzymes. AMPA receptors are removed from the membrane, and the EPSP becomes smaller for a long time. Most neuroscientists think changes like these are part of how learning and memory work, and blocking NMDA receptors in the hippocampus does impair some kinds of learning in animals. How a particular memory is stored is still being worked out, so treat LTP as a well supported mechanism for strengthening a synapse and not as a full explanation of memory.

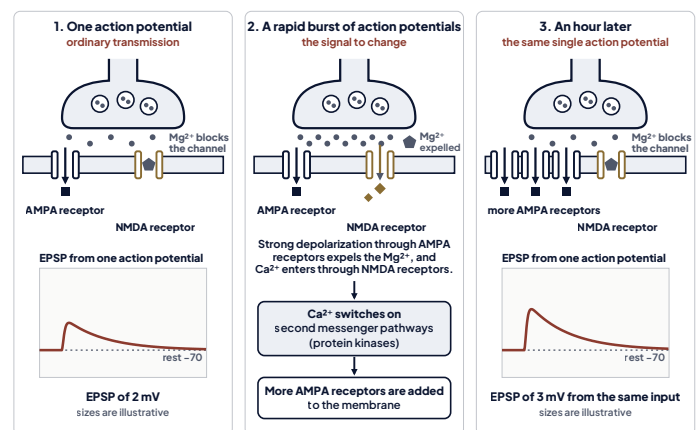


Figure 7. One glutamate synapse before, during and after a rapid burst of activity.

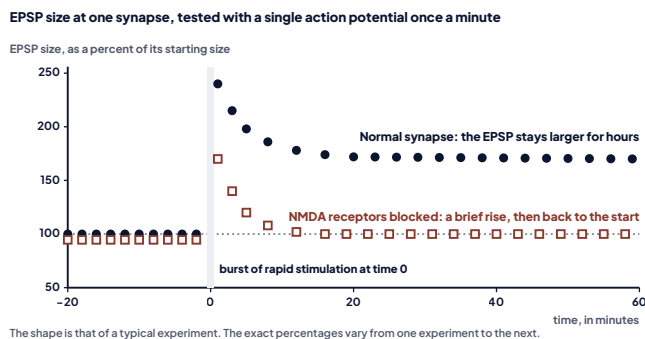


Figure 8. The size of the EPSP before and after one burst of rapid stimulation, with and without working NMDA receptors.

Synaptic plasticity

A change in the strength of a synapse that depends on its earlier activity.

Facilitation and depression

Short term increases and decreases in transmitter release during a train of action potentials, lasting milliseconds to minutes.

AMPA receptor

An ionotropic glutamate receptor that opens at once and produces the ordinary fast EPSP.

NMDA receptor

An ionotropic glutamate receptor whose channel is blocked by Mg^{2+} near rest. It opens only when glutamate is bound and the membrane is strongly depolarized, and it lets Ca^{2+} in.

Long term potentiation (LTP)

A long lasting increase in the size of the EPSP at a synapse after strong, rapid activity. It depends on Ca^{2+} entry through NMDA receptors and on adding AMPA receptors.

Long term depression (LTD)

A long lasting decrease in the size of the EPSP, with AMPA receptors removed from the membrane.

Electrical synapses

SLIDES 19 TO 20 OF THE DECK

Not every synapse uses a transmitter. At an electrical synapse two cells are joined by gap junctions. Each gap junction channel is built from proteins called connexins. Six connexins form a half channel in one membrane, six more form a half channel in the other, and the two line up to make one pore from the cytoplasm of one cell to the cytoplasm of the next. Where they occur, the two membranes lie only about 3.5 nm apart, compared with a cleft of about 20 to 40 nm at a chemical synapse.

Ions carry current straight through the pore, so a depolarization in one cell depolarizes its neighbor with almost no delay. A chemical synapse takes about 0.5 to 1 ms, and sometimes longer, because Ca^{2+} has to enter, vesicles have to fuse, and transmitter has to cross and bind. Current can usually pass through a gap junction in either direction, so a group of cells joined this way tends to fire together. You find electrical synapses in heart muscle, in the smooth muscle of the gut and the uterus, between glial cells, and between some neurons in the brain.

What an electrical synapse gives up is flexibility. The signal keeps its sign, so a depolarization can only pass on as a smaller

depolarization. It cannot be turned into inhibition, it cannot be amplified, and there is no terminal to modulate and no receptor number to change. Gap junctions can open and close, but that is a much coarser control. So the two kinds of synapse suit different jobs. When every muscle cell in the wall of the heart must contract at nearly the same moment, gap junctions are the right choice, because delays at millions of chemical synapses would add up and the contraction would not be coordinated. When a spinal motor neuron must be able to ignore a sensory signal on some occasions and respond to it on others, only a chemical synapse will do. The delay of a chemical synapse is the cost of being able to integrate.

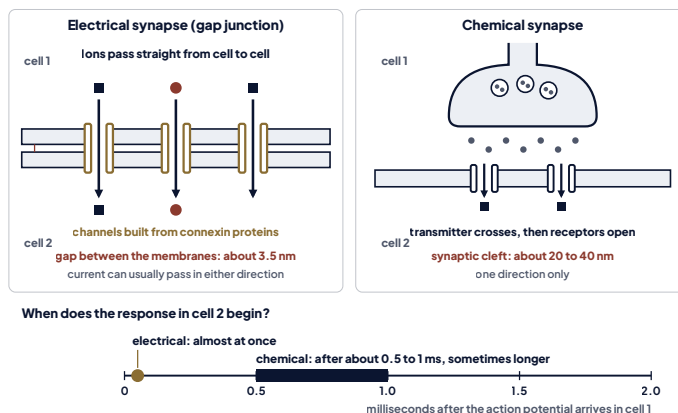


Figure 9. The two kinds of synapse, with the time each one takes.

TWO WAYS FOR ONE CELL TO SIGNAL TO THE NEXT

	CHEMICAL SYNAPSE	ELECTRICAL SYNAPSE
What crosses	Neurotransmitter, across a cleft of about 20 to 40 nm	Ions, through connexin channels across a gap of about 3.5 nm
Delay	About 0.5 to 1 ms, sometimes longer	Almost none
Direction	One way	Usually both ways
Sign of the signal	Excitatory or inhibitory, depending on the receptor	The same sign in both cells
Amplification	Yes	No. The signal can only get smaller
Adjustable	Highly: presynaptic control, receptor number, plasticity	Much less
Best suited for	Integration and decisions	Speed and firing together

Electrical synapse

A junction where current passes directly from one cell to the next through gap junction channels.

Gap junction

A cluster of channels that join the cytoplasm of two cells. Ions and small molecules pass through.

Connexin

The protein that gap junction channels are built from, six in each membrane.

The patient, explained

SLIDES 21 TO 22 OF THE DECK

Start with a normal spinal motor neuron. It receives excitatory input all the time, from sensory neurons and from the brain. Inhibitory interneurons that release glycine or GABA keep it below threshold unless the excitation is strong. Put illustrative numbers on it. An excitatory input worth 18 mV and an inhibitory input worth -8 mV arrive together, and $-70 + 18 - 8 = -60$ mV, so the cell does not fire. Now block release from the inhibitory terminal. $-70 + 18 = -52$ mV, which is above threshold. The same input that did nothing a moment ago now fires the motor neuron.

That is what happened to the gardener. The bacterium *Clostridium tetani* grew in the deep wound and released tetanus toxin. Motor axon terminals took the toxin up, and retrograde axonal transport, from the neuron lecture, carried it back to the cell bodies in the spinal cord and brainstem. That trip is part of why a week went by. In the spinal cord the toxin passed from the motor neuron into the terminals of the inhibitory interneurons that synapse on it. There it cut one of the proteins that a vesicle needs in order to fuse with the membrane, so those terminals could no longer release glycine or GABA.

Nothing was added to the excitatory side. His motor neurons still receive the same EPSPs from every touch and sound, but the inhibitory term is gone from the sum, so ordinary input now reaches threshold. Normally, when one muscle contracts, inhibitory interneurons keep the opposing muscle quiet. Without them, both contract at once and with full force. The strong jaw closing muscles clamp the jaw shut, which gives the disease its old name, lockjaw, and any stimulus sets off a spasm of the whole body.

Treat it as an experiment. The variable that failed is inhibitory transmitter release onto motor neurons, and through it the sum at the trigger zone. Three physiological targets follow from that. The GABA-A receptors on the motor neuron are intact, so a drug that makes those channels open more readily, such as a benzodiazepine, increases whatever inhibition remains and puts a negative term back into the sum. Every sound and touch adds EPSPs, so a quiet, dark room lowers the excitatory side of the same sum. And antibody against the toxin binds whatever toxin has not yet entered neurons. Toxin already inside the terminals cannot be reached, so recovery waits for those terminals to be repaired or replaced, which takes weeks. Vaccination works by having that antibody present before any toxin arrives.

Two comparisons test whether you have the mechanism. Strychnine blocks the glycine receptor on the motor neuron. Glycine is still released, but its receptor cannot open its Cl^- channel, so there is no IPSP, and the result is convulsions set off by small stimuli, very much like tetanus. One agent acts before the cleft and the other after it, and both remove the same term from the sum. Botulinum toxin is a close relative of tetanus toxin and cuts the same family of vesicle proteins, but it stays in the motor neuron terminal at the muscle, where it blocks acetylcholine release. The muscle cannot be excited at all, and the result is limp paralysis. The mechanism is the same, the synapse is different, and the effect is the opposite.

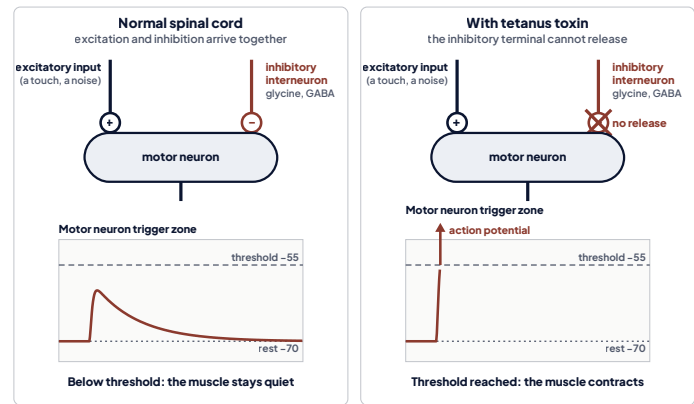


Figure 10. A spinal motor neuron with the same excitatory input in both panels.

What this lecture assumes you already have

If any line here is shaky, fix it before you work the lecture.

CHEMISTRY

Ions and charge: Na^+ , K^+ , Cl^- , Ca^{2+} and Mg^{2+} , and that like charges repel.

Proteins as channels, receptors and enzymes, and what a protein kinase does.

Chemistry review page

FROM EARLIER LECTURES

The resting membrane potential: $E_{\text{K}} -95$ mV, $E_{\text{Na}} +61$ mV, $E_{\text{Cl}} -64$ mV, rest -70 mV, and the rule that opening channels for an ion moves the membrane toward that ion's equilibrium potential.

Neurons and glia: dendrites and cell body as the input zone, the trigger zone, and retrograde axonal transport.

Graded potentials: they shrink with distance and can add, and threshold is about -55 mV.

How action potentials carry information: the frequency code.

The chemical synapse: Ca^{2+} entry and vesicle fusion, ionotropic and metabotropic receptors, glutamate, GABA and glycine.

The only anatomy used is the dorsal horn of the spinal cord, where pain fibers make their first synapse, and the hippocampus and cerebellum, which are named and not described.

MATH

Adding and subtracting signed numbers.

Multiplying by a fraction, for example 6×0.45 .

No logarithms are needed in this lecture. The one Nernst value, E_{Cl} , is given.

Close the page and produce it

These are the Week 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one. Competency 29, the synapse simulation, is done in lab. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

24

Excitatory and inhibitory postsynaptic potentials

Distinguish an EPSP from an IPSP by the ion channel opened and the direction of the membrane potential change, and contrast fast responses through ionotropic receptor-channels with slow responses through metabotropic G protein coupled receptors, including how a neuromodulator differs from a neurotransmitter.

Do one of the two. Head your paper with the number.

24-1

Draw a postsynaptic neuron with two synapses on one dendrite, one excitatory and one inhibitory. At each, draw the channel that opened and the ion moving through it with an arrow for direction. Under each synapse draw its own trace starting from about -70 mV and label the peak value. One line: how an inhibitory input can still be inhibitory even when the membrane potential barely moves.

24-2

Draw three recorded traces from the same neuron on shared axes, all starting near -70 mV, one rising 3 mV, one falling 4 mV, and one staying flat. For each trace, work backwards and draw the channel and the ion that would produce it. Give the flat trace two different explanations. One line: which explanation a second recording could rule out, and what that recording would be.

25

Summation and integration

Distinguish temporal from spatial summation and determine whether a stated combination of EPSPs and IPSPs will bring the trigger zone to threshold.

Do one of the two. Head your paper with the number.

25-1

Draw the same neuron twice. On the first, one presynaptic neuron fires four times in quick succession. On the second, four separate presynaptic neurons fire at the same moment. Under each, draw the trace at the trigger zone with a resting line near -70 mV and a dashed threshold line near -55 mV. Label which is temporal and which is spatial. One line: what has to be true about timing for the first one to work at all.

25-2

Draw a neuron with five inputs, three excitatory worth about 4 mV each and two inhibitory worth about 5 mV each. Choose which fire and in what order, then draw the trace at the trigger zone against a threshold line near -55 mV, writing your running total along the trace. Now change one input, its timing or its sign, so the outcome flips. Write which change you made and why it was enough.

26

Presynaptic modulation

Compare presynaptic inhibition and facilitation with postsynaptic modulation and state the advantage of presynaptic control at a single input, which is selectivity.

Do one of the two. Head your paper with the number.

26-1

Draw a neuron with three excitatory inputs converging on it. Add a modulatory neuron that synapses on one input's terminal, not on the cell body. Show what it does to calcium entry and to transmitter release at that terminal only. Draw the postsynaptic response to all three inputs before and after. One line: what postsynaptic inhibition would have done to the other two inputs.

26-2

You need to cut one input's effect by half and leave the other inputs untouched. Draw two attempts side by side, one acting on that input's presynaptic terminal and one acting on the postsynaptic membrane. On each drawing, shade every synapse the attempt affected. Mark the attempt that failed the requirement. One line: name the property that makes presynaptic control the right tool for this job.

27

Synaptic plasticity

Explain long term potentiation as a mechanism of learning, tracing how glutamate acts on AMPA and NMDA receptors, how strong depolarization removes the magnesium block, how calcium entry starts second messenger pathways, and how repeated stimulation leads to insertion of new AMPA receptors.

Do one of the two. Head your paper with the number.

27-1

Draw a glutamate synapse with both AMPA and NMDA receptors in the postsynaptic membrane. Show the magnesium sitting in the NMDA channel at rest. In a second panel, draw the same synapse during repeated high frequency stimulation with the magnesium displaced and calcium entering. In a third, draw what the membrane looks like an hour later. One line: why one presynaptic action potential on its own cannot do this.

27-2

Draw a graph of postsynaptic response size against time, with a burst of high frequency stimulation marked on the time axis, and plot the response before, during, and long after the burst. On the same axes, draw a second curve for the identical experiment with NMDA receptors blocked. One line: which receptor carries the ordinary moment to moment signal, and which one carries the signal that something should change.

28

Electrical synapses

Compare an electrical synapse at a gap junction with a chemical synapse by speed and by capacity for modulation.

Do one of the two. Head your paper with the number.

28-1

Draw two synapses side by side in cross section at the same scale, an electrical one at a gap junction and a chemical one. Show what physically crosses at each and make clear which gap is wider. Underneath both, draw one shared time axis and mark when the postsynaptic response begins for each. One line: what the chemical synapse buys with the delay it pays.

28-2

Pick two real jobs, one where many cells must fire as a single unit and one where a signal must be turned up, turned down, or blocked outright. Draw the synapse you would build for each and label it electrical or chemical. On each drawing, circle the feature that made it right for that job. One line: what fails if you swap the two.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · Integration at the synapse. Potential sizes and distances are teaching values. All figures are original drawings for this course.